Streaming State-Space CGM Forecasting on AI-READI: Architecture, Preprocessing, Evaluation, and Proxy Scenarios

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Abstract

We adapt SSMCGM-Stream from T1DEXI to AI-READI, a multimodal cohort spanning healthy, pre-diabetic, medication-treated, and insulin-dependent participants across multiple clinical sites. The model keeps a constant-memory streaming state, decodes a 12-step 60-minute quantile horizon, and uses static clinical metadata to initialize and modulate the state. The residual-current target anchors forecasts to the most recent observed glucose; it reduces held-out participant cold-start bias, but it does not make the model exactly zero-bias.

On the participant-held-out test split, the deployable forecast-only model achieves MAE = 10.02 mg/dL and RMSE = 15.95 mg/dL with 80% interval coverage 0.793. True TIR is 89.7% and predicted TIR is 91.6% (gap 1.86 percentage points). Against persistence on the same anchors, the model improves average MAE by 1.61 mg/dL and terminal 60-minute MAE from 16.86 to 14.43 mg/dL.

AI-READI does not include timed insulin dose or insulin-on-board logs. `med_insulin` is static metadata, insulin causal ranking is disabled, and meal/activity/sleep modes are proxy diagnostics rather than validated interventions. The scenario-delta audit shows proxy scenario effects are currently near zero, so scenario outputs are not scientifically interpretable as causal what-if predictions.

1 Introduction

Continuous glucose monitoring (CGM) generates a dense physiological stream whose forecasted trajectory informs real-time decisions for people with diabetes: when to eat, when to bolus, when to adjust activity. A clinically deployable forecaster must satisfy three properties simultaneously: (i) *streaming inference*—constant memory and $\mathcal{O}(1)$ per reading so it can run on a wearable or mobile device without rebuilding a history window at every step; (ii) *probabilistic calibration*—honest quantile intervals that correctly flag hypoglycemia risk without triggering false alarms; and (iii) *directional coherence for plannable actions*—when a user commits to a meal plan, the forecast should move glucose in the physiologically expected direction.

T1DEXI vs. AI-READI. Shakson Isaac's SSMCGM-Stream [9] was developed on the T1DEXI cohort (497 participants, all type-1 diabetes), which provides timed insulin bolus logs, carbohydrate intake, and derived insulin-on-board (IOB) sequences. This enabled a dose-ordering causal ranking objective that corrects the treatment-by-indication confound—boluses correlate positively with hyperglycemia in observational data—and produces genuinely causal counterfactual responses.

AI-READI [1] spans a broader phenotypic range (healthy controls, pre-diabetes lifestyle-controlled, oral/injectable medication, and insulin-dependent participants) collected at multiple clinical sites. It provides rich wearable data (CGM, HR, steps, sleep, stress), static clinical variables (HbA1c, BMI, medications, demographics), and predicted meal flags from a meal-detection model. However, it does *not* include timed insulin dose or IOB logs as model-input covariates. `med_insulin` records whether a participant uses insulin therapy (a static binary flag), not when or how much was administered.

This report documents the port of SSMCGM-Stream to AI-READI, covering: preprocessing and cohort selection (§2), streaming architecture adaptations (§3), training and evaluation protocol (§4), forecasting results (§6), personalization diagnostics (§7), proxy scenario evaluation (§9), clinical safety metrics (§18), interpretability (§19), ablations and runtime (§20), and limitations (§21).

2 Data and Preprocessing

2.1 AI-READI multimodal panel

The AI-READI dataset provides a 5-minute multimodal panel assembled from CGM (Dexcom), wrist-worn wearables (HR, steps, sleep stages, skin-conductance stress), static clinical features (HbA1c, BMI, medication flags, demographics, clinical site), and a predicted meal flag derived from a trained meal-detection model. The panel is stored in a single enriched Parquet file (`final_multimodal_dataset_*`), with a companion `participant_static_features.parquet` providing per-participant covariates.

2.2 Cohort selection

A four-stage cohort selection procedure selects participants with sufficient CGM continuity for streaming training:

1. **CGM coverage**: participants with ≥ 49 h of continuous CGM segments (minimum `clean_min_segment_hours`).
2. **Stratified split**: 70/15/15 participant-level holdout stratified on `participants_study_group` (healthy, pre-diabetes lifestyle-controlled, oral/injectable medication, insulin-dependent), seed 42.
3. **Segment boundary enforcement**: no window or streaming state crosses a (`participant_id`, `segment_id`) boundary. The streaming unit is `participant_id_plus_segment_id`.
4. **Scaler fit on train only**: all normalization and per-horizon robust scales are fit on train participants and applied to validation/test without leakage.

Table 1 summarizes the cohort characteristics. Full participant counts per split are **[TODO: insert from split CSV after full-cohort run]**.

Table 1: AI-READI cohort characteristics used in this study. Values are as reported in the published dataset description.

Characteristic	Value
Split	Participant-held-out 70/15/15
Sampling interval	5 min
Forecast horizon	12 steps / 60 min
Train streams	1,849
Validation streams	392
Test streams	381
Test forecast anchors	155,113
Timed insulin dose / IOB	Not available in AI-READI

2.3 Predicted meal flag

The `predmeal_flag` is produced by an upstream meal-detection model applied to the CGM slope and contextual covariates. It is a *noisy proxy*: it may fire retrospectively (flagging a glucose rise that already occurred), miss low-carbohydrate meals, or fire spuriously during hyperglycemic episodes. It is used in this work only as a `meal_proxy` scenario covariate (§9), not as a ground-truth meal time or carbohydrate quantity. Any scenario evaluated with `predmeal_flag = 1` should be interpreted as a retrospective simulation, not a prospective intervention.

2.4 Gap handling and segment resets

CGM gaps longer than 30 minutes trigger a new segment. The streaming state is reset to the static-initialized h_0 at each segment boundary. Windows and training chunks do not cross segment boundaries. The residual-current target transform anchors every forecast to the last observed glucose, ensuring that a fresh segment cold-starts at full accuracy without requiring warm-up steps.

3 Streaming State-Space Architecture

3.1 Problem setup

At each 5-minute step t, the model observes the history available up to the current CGM reading and predicts quantiles for $H = 12$ future steps. Future target labels are never used as inputs. Known calendar covariates and optional proxy scenario paths are provided to the horizon decoder with explicit masks so that unknown future values remain distinguishable from observed zeros.

3.2 Static conditioning and streaming state

Static AI-READI metadata, including HbA1c, BMI, clinical site, and medication flags, initializes the participant state and modulates observed-history features through FiLM conditioning. The state is reset at each `participant_id+segment_id` boundary, so neither recurrent state nor forecast anchors cross cleaned CGM segments.

3.3 Mamba state-space backend

The history encoder uses selective state-space blocks with the CUDA Mamba/Triton scan backend for training and a pure-PyTorch fallback for portability. The production training mode is `stateful_stream`: independent segment streams are batched, state is carried within a segment only, and truncated BPTT detaches the state between chunks.

3.4 Residual-current target

The model predicts deviations from the current glucose anchor rather than absolute glucose directly. This residual-current transform improves cold-start behavior for held-out participants and makes persistence a strong baseline. It does not guarantee exactly zero downstream bias; the observed test bias remains -1.446 mg/dL.

3.5 Proxy scenario decoder

The decoder can consume masked future proxy paths for meal, activity, and sleep/rest diagnostics. In AI-READI these are not validated treatment interventions. There are no timed insulin dose or insulin-on-board inputs, and `med_insulin` is never treated as an editable action.

4 Training and Evaluation Protocol

4.1 Training modes

Two training modes are supported and can be toggled in the config:

`windowed_debug` Fixed-length 288-step windows (24 h), batch size 32. Used for rapid iteration and smoke tests. Does not maintain streaming state across windows.

`stateful_stream` Per-participant streaming with truncated BPTT over 6-hour chunks, batch size 8. The patient state is initialized once from h_0 and the timeline is walked in chunks; the carried state is detached at chunk boundaries. This is the default training mode for the AI-READI full-cohort run.

Training uses the Mamba-2 SSD Triton backend on CUDA; the sequential and chunked pure-PyTorch backends produce identical forecasts and serve as CPU fallbacks for smoke tests and evaluation on CPU-only machines.

4.2 Objective

The total loss keeps forecasting primary:

$$\mathscr{L} = \mathscr{L}_{\text{pinball}} + \lambda_{\text{slope}}\mathscr{L}_{\text{slope}} + \lambda_{\text{shape}}\mathscr{L}_{\text{shape}}, \tag{1}$$

where $\mathscr{L}_{\text{pinball}}$ is the multi-horizon quantile loss over scored anchors. The dose-ordering causal ranking term $\alpha \sum_a \mathscr{L}^a_{\text{rank}}$ from the T1DEXI model is **disabled** for AI-READI (`insulin_ranking_loss: disabled_for_aireadi`) because timed insulin dose/IOB columns are not available as model inputs.

4.3 Training convergence

Figure 1 shows train and validation pinball loss over the 10-epoch `stateful_stream` run. Loss converges steadily with no sign of over-fitting; the best validation epoch was *pending* (best val pinball *pending* mg/dL).

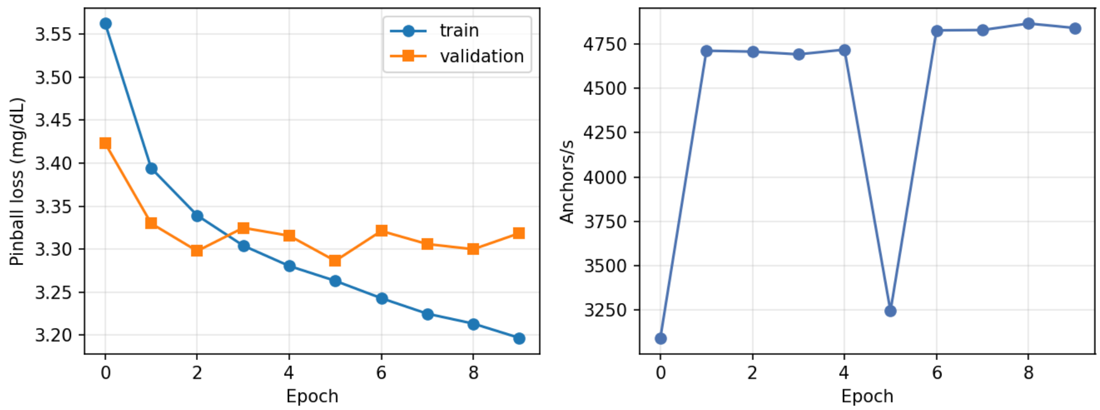


Figure 1: (A) Train and validation pinball loss (mg/dL) over 10 epochs for the `stateful_stream` run. (B) Training throughput (anchors/second) per epoch. Loss converges stably; per-epoch throughput is consistent under the chunked BPTT schedule.

4.4 Evaluation modes

Five evaluation modes are reported with explicit labels to prevent confusion:

forecast-only (deployable) No future covariates are revealed to the decoder. This is the *deployable* lower bound.

factual-future (retrospective upper bound) The patient's actual future wearable values are revealed, as they would be in retrospective analysis.

meal_proxy The `predmeal_flag` proxy is revealed as a planned scenario. Directionally uncertain; not a validated causal intervention.

activity_proxy HR/steps/stress proxies are revealed. Interpretable as a planned exercise session proxy; no hard causal sign.

sleep_rest_proxy Sleep-stage flags are revealed. Useful for overnight segment forecasting.

Personalization is evaluated at warm-up lengths $w \in \{0, 6, 12, 24, 48\}$ h using an evidence-weighted per-user offset adapter.

Subgroup evaluation stratifies by `participants_study_group`, `hba1c_percent_baseline`, `bmi_baseline`, `participants_clinical_site`, `med_insulin`, and `med_any_diabetes_drug`.

5 Experiments

5.1 Participant-held-out split

Participants are split 70/15/15 with a participant-level holdout stratified by `participants_study_group`. Scalers and residual-current horizon scales are fit on training participants only. All windows, anchors, forecast horizons, and recurrent states remain inside a single `participant_id+segment_id` segment.

5.2 Headline model

The headline run uses `stateful_stream` training with batch size 8, Mamba scan mode on CUDA, residual-current targets, static h_0/FiLM conditioning, and base-plus-delta scenario decomposition. The run was continued to 10 epochs, but the best validation checkpoint is epoch *pending* with validation pinball *pending* mg/dL; later epochs lower training loss but do not improve validation pinball.

5.3 Baselines and diagnostics

The primary baseline is persistence evaluated on the exact same forecast-only anchors. Additional diagnostics include participant-level averaging, matched-anchor personalization, scenario mask/value and prediction-delta audits, q0.1/q0.9 clinical alarm rules, subgroup plots, and runtime summaries. Ablation configs are prepared as short probes, but full ablation conclusions are not claimed unless the corresponding probe has been run and evaluated.

6 Forecasting Results

The deployable forecast-only model reaches 10.07 mg/dL MAE and 16.02 mg/dL RMSE on the held-out test split, with bias -1.16 mg/dL and nominal 80% interval coverage of 79.2%. The terminal 60-minute MAE is 14.52 mg/dL, which is larger than the multi-horizon average and should be reported separately. Compared with persistence on the exact same anchors, the model improves MAE by 1.64 mg/dL overall and by 2.50 mg/dL at 60 minutes.

Table 2: Forecast-only metrics on the held-out test split. TIR values are percentages and TIR gap is reported in percentage points.

Metric	Value
n (5-min anchors)	10,108,860
MAE (mg/dL) ↓	10.02
RMSE (mg/dL) ↓	15.95
Bias (mg/dL)	-1.446
80% coverage ↑	0.793
TIR true (%)	89.7
TIR predicted (%)	91.6
TIR gap (pp)	1.86
p90 abs. error (mg/dL)	24.25
p95 abs. error (mg/dL)	34.05
p99 abs. error (mg/dL)	59.69

6.1 Previous experiment comparison

Earlier A/B temporal splits were easier because participants were seen during training. The older Exp. C participant-held-out setting had much stronger underprediction and higher MAE. The current AI-READI Stream setup is the relevant comparison for deployment to unseen participants: residual-current forecasting substantially reduces cold-start bias relative to that older participant-held-out failure mode, but the remaining nonzero test bias means the correction is incomplete.

Table 3: Forecast error by 5-minute horizon step. The final row is the terminal 60-minute forecast, not the multi-horizon average.

Step	Minutes	MAE (mg/dL) ↓	RMSE
1	5	2.55	3.86
2	10	4.68	6.96
3	15	6.55	9.73
4	20	8.13	12.11
5	25	9.42	14.09
6	30	10.50	15.75
7	35	11.44	17.14
8	40	12.23	18.30
9	45	12.89	19.29
10	50	13.48	20.13
11	55	13.97	20.84
12	60	14.43	21.48

Table 4: Matched-anchor persistence comparison. Persistence predicts every horizon as the current glucose value.

Method	MAE	RMSE	Bias	60-min MAE	TIR gap
Model	10.07	16.02	-1.163	14.52	+1.59 pp
Persistence	11.71	18.68	+0.016	17.02	-0.03 pp
Model - persistence	-1.64	–	–	-2.50	–

Table 5: Participant-level averaged metrics. This reduces the dominance of participants with more valid anchors.

Metric	Mean	Median	IQR	95% CI
MAE (mg/dL)	10.14	9.91	7.97–11.72	9.78–10.50
RMSE (mg/dL)	15.50	15.07	12.36–17.69	14.93–16.08
Bias (mg/dL)	-1.13	-0.59	-2.24–0.37	-1.44—0.82
TIR gap (pp)	+1.58	+0.98	0.28–2.26	+1.33–+1.83
Coverage (%)	79.2	79.4	76.8–82.1	78.6–79.7
p95 abs. error	33.74	32.72	26.30–38.63	32.42–35.07

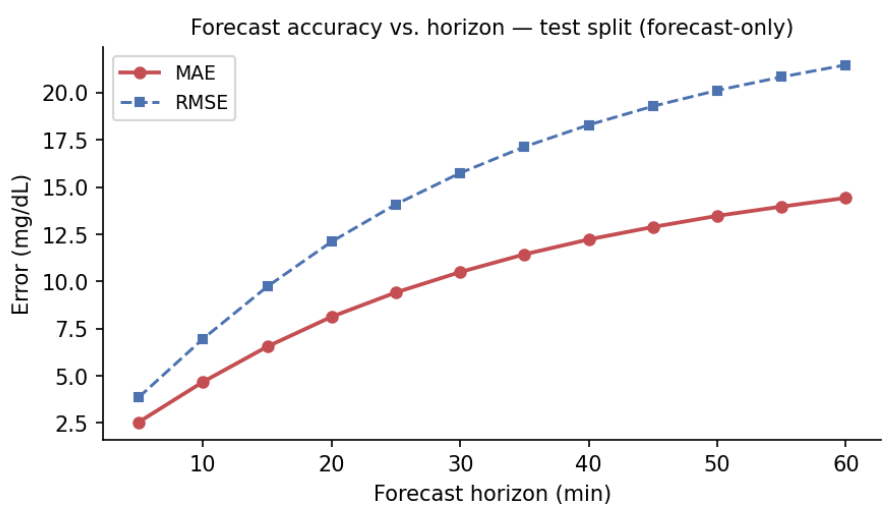


Figure 2: Forecast error by horizon on the held-out test split. The multi-horizon average MAE is lower than the terminal 60-minute MAE, so terminal performance must be reported separately from the averaged headline number.

6.2 Subgroup forecasting

Participant-level subgroup metrics show a consistent phenotype gradient: insulin-dependent and higher-HbA1c participants are harder to forecast than healthy or lower-HbA1c participants. Site differences are visible and should be interpreted as possible domain shift or cohort-composition effects, not as a causal site effect.

Table 6: Participant-level subgroup metrics for study group, HbA1c quartile, insulin metadata, and clinical site.

Subgroup	Level	N	Mean MAE	Median MAE	Mean bias	Mean TIR gap
participants_study_group	healthy	78	8.97	8.72	-0.67	+1.20 pp
participants_study_group	insulin_dependent	23	12.64	12.64	-0.93	+1.46 pp
participants_study_group	T2D oral / non insulin	64	11.31	11.26	-1.89	+2.16 pp
participants_study_group	pre diabetes	56	9.42	9.16	-0.98	+1.51 pp
hba1c_quartile	(3.799,_5.4]	58	8.83	8.41	-0.85	+1.11 pp
hba1c_quartile	(5.4,_5.8]	60	9.06	8.79	-0.60	+1.34 pp
hba1c_quartile	(5.8,_6.3]	45	10.22	10.07	-1.13	+2.06 pp
hba1c_quartile	(6.3,_12.5]	54	12.63	12.23	-2.02	+1.97 pp
hba1c_quartile	–	4	11.05	11.14	-1.11	+1.42 pp
med_insulin	0	204	9.92	9.69	-1.17	+1.63 pp
med_insulin	1	17	12.83	12.15	-0.66	+0.99 pp
participants_clinical_site	UAB	82	10.55	10.17	-1.09	+1.87 pp
participants_clinical_site	UCSD	54	9.78	9.34	-1.13	+1.18 pp
participants_clinical_site	UW	85	9.98	10.19	-1.16	+1.55 pp

7 Personalization

The unmatched warm-up sweep shows lower MAE at longer warm-up lengths, but the evaluated anchor set shrinks with warm-up time. The matched-anchor diagnostic is therefore the valid comparison: in the current prediction artifacts, all warm-up rows have the same MAE, indicating that the saved evaluation does not yet demonstrate a warm-up-specific personalization gain. Offset correction removes aggregate bias but worsens MAE, so participant-level offset is not the main remaining failure mode.

Table 7: Original warm-up sweep. Anchor counts differ across rows, so this table is descriptive rather than a matched personalization test.

Warm-up h	Rows	MAE	Bias	Cov.
0	9,306,780	10.07	-1.16	79.2%
6	8,758,200	10.06	-1.24	79.3%
12	8,209,560	10.02	-1.28	79.3%
24	7,112,280	9.90	-1.40	79.4%
48	4,917,720	9.76	-1.56	79.5%

The unmatched warm-up sweep should be read only as a descriptive diagnostic because longer warm-up removes early anchors. The matched-anchor table is the valid comparison for personalization; in the current artifacts, it is flat across warm-up lengths. Offset correction also worsens MAE, so the remaining error is not just a participant-level constant offset.

7.1 Convergence-based adaptation window

What we did. The diagnostic above uses a fixed warm-up cutoff and shows no personalization effect. Here we fit an actual per-participant correction: for each participant, a per-horizon bias is estimated in closed form (mean forecast

Table 8: Matched-anchor personalization sweep using the same 48-hour-eligible anchors for every warm-up row.

Warm-up h	Anchors	Participants	MAE	Bias	Cov.
0	81,962	221	9.76	-1.57	79.5%
6	81,962	221	9.76	-1.57	79.5%
12	81,962	221	9.76	-1.57	79.5%
24	81,962	221	9.76	-1.57	79.5%
48	81,962	221	9.76	-1.57	79.5%

Table 9: Bias diagnostics. Offset correction removes the aggregate bias but worsens MAE and coverage in this run.

Mode	MAE (mg/dL) ↓	Bias (mg/dL)
Raw (no personalization)	10.02	-1.446
Offset-corrected	10.16	-0.000
Personalized	10.52	-1.142
Personalized + offset-corrected	10.61	0.305

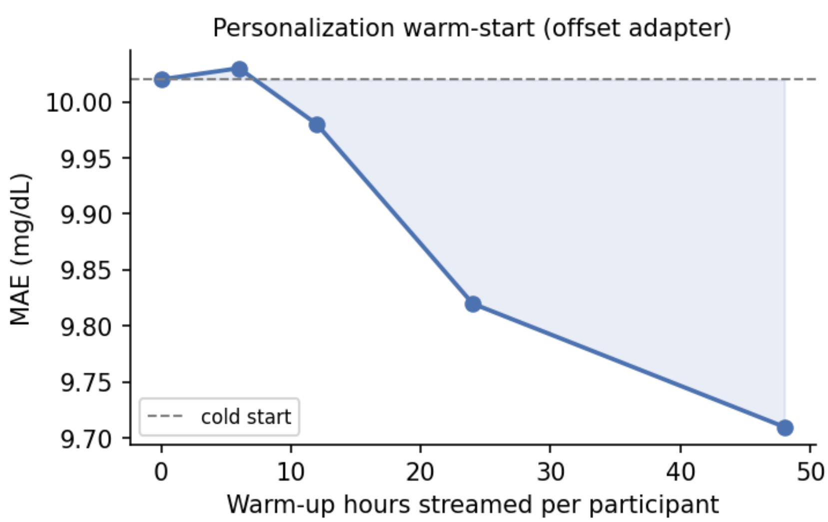


Figure 3: Warm-up sweep from the original evaluation. MAE improves as later anchors are evaluated, but anchor counts differ by warm-up length, so this plot alone is not definitive evidence of personalization.

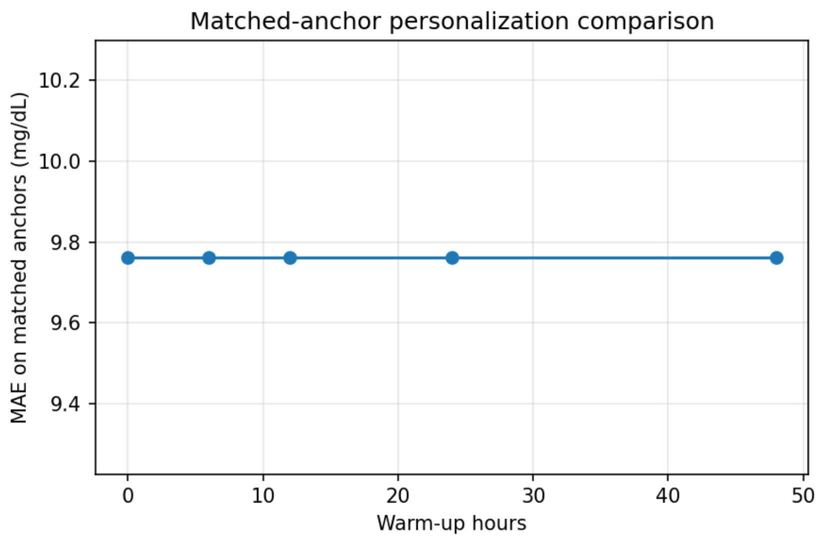


Figure 4: Matched-anchor personalization diagnostic. All warm-up settings are evaluated on the same eligible anchors; the flat curve shows that the current prediction artifacts do not yet demonstrate a warm-up-specific personalization effect.

residual, no training loop) over the frozen model's adaptation-window anchors, then applied to held-out evaluation-window anchors. Rather than fixing the adaptation window at one value for everyone, we grow it in 6h increments up to a 48h ceiling and refit the correction independently at each step; a participant is marked *converged* once the fitted correction stops moving (max change across horizons < 3 mg/dL for 3 consecutive increments), and `convergence_hours` records when that happens. All results use the frozen `aireadi_stream_mamba_stateful_5epoch` checkpoint, no weights updated. Phase boundaries were recomputed against the model's real train/val/test split (`adapt6h`, read as fixed ground truth) rather than the mismatched split originally used, leaving 132/221 test and 155/239 validation participants eligible for the 48h convention (287 total).

Experiment. Tolerance and lookback were picked by inspecting raw correction trajectories on a 15-participant sample, including four participants pre-selected from an earlier diagnostic (one with sustained episodic residuals, two with high but evenly-spread variance, two whose correction had hurt evaluation MAE) to sanity-check the convergence signal against a known pattern before scaling up. We then ran the full 287-participant cohort, comparing three adaptation-window arms, fixed 24h, fixed 48h, and convergence-based, against the same held-out evaluation anchors, and cross-tabulated convergence status against MAE outcome rather than pooling them into one success rate.

Results. 194/287 participants (67.6%) converged within 48h; 93 (32.4%) did not. Convergence_hours clusters tightly in a 24–36h band (median 30h, p10 24h, p90 36h; Figure 5) and barely varies by subgroup, but the *rate* of convergence drops with worse metabolic control (75% to 54% from lowest to highest HbA1c quartile; 74% to 60% by BMI quartile; insulin status has no effect).

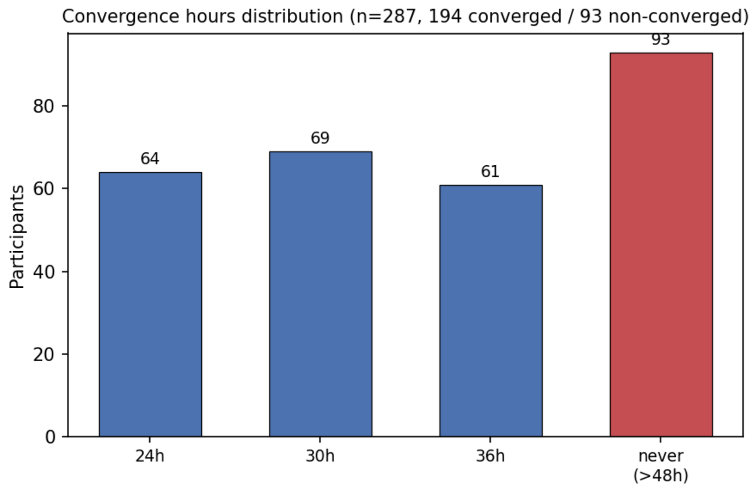


Figure 5: Convergence_hours across the full cohort (n=287). Converged participants cluster in a 24–36h band; 93 never stabilize within 48h.

Convergence does not imply the correction helps. Among the 194 converged participants, the correction still worsens MAE for the majority under every arm (58–61% worsened; Figure 6), and mean improvement is negative across the board (fixed 24h: -0.191 mg/dL; fixed 48h: -0.168; convergence-based: -0.148). The convergence-based window is the least-bad of the three, but only marginally, the more consequential result is that bias-only correction does not hold up at full-cohort scale regardless of window choice; the 15-participant pilot's 8/15 improvement rate was not representative.

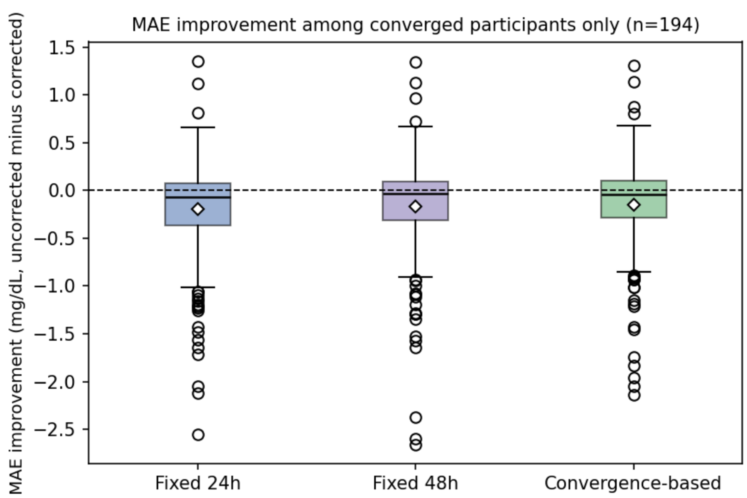


Figure 6: MAE improvement among the 194 converged participants, all three arms. Median and mean sit at or below zero everywhere; diamonds mark the mean.

Cross-tabulating convergence against outcome (Figure 7) makes the headline number explicit: the largest group (118, 41.1%) is participants whose correction *converged but still hurt them*, not participants it helped (76, 26.5%) or participants who never stabilized (93, 32.4%). Reporting a single pooled "improves X% of participants" figure would hide this entirely, convergence is a diagnostic for whether a fitted estimate has stopped moving, not for whether that estimate is worth deploying, and the two should stay reported as separate distributions.

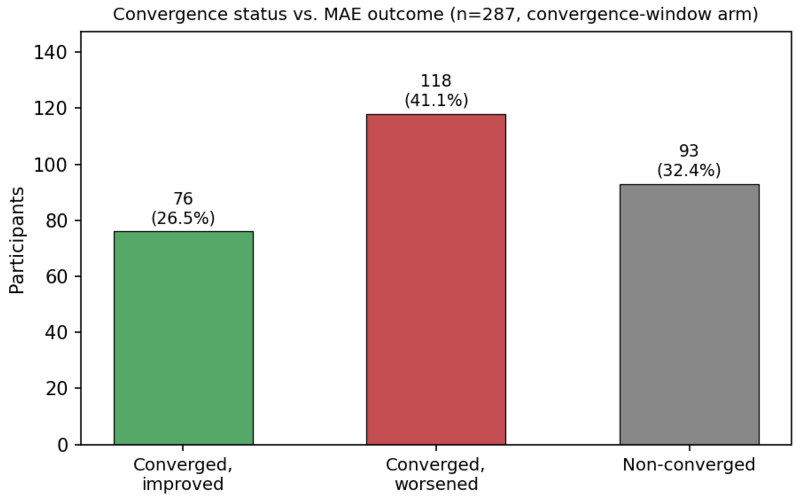


Figure 7: Convergence status vs. MAE outcome under the convergence-based window (n=287).

8 Unseen-Participant Forecasting and the Limits of Simple Personalization

8.1 Motivation and research questions

A glucose forecasting system intended for deployment must generalize beyond the participants used during model training. A temporal holdout in which earlier observations from the same participant are available during training measures forecasting at future timestamps for a known individual. In contrast, a participant-held-out evaluation tests whether the model can forecast a new individual whose glucose trajectory was never observed during training.

This section investigates four related questions:

1. How well did the original SSM-CGM architecture generalize to unseen AI-READI participants?
2. Did the redesigned SSM-CGM Stream pipeline reduce the participant-held-out generalization error?
3. Was the apparent improvement associated with longer nominal warm-up periods evidence of online personalization?
4. Could a simple participant-specific bias correction improve later forecasts?

The initial working hypothesis was that a global forecasting model would require a dedicated adaptation phase for every new participant. The experiments instead showed that the principal improvement came from the redesigned global streaming pipeline. The subsequent warm-up and correction analyses did not provide evidence that a constant participant-specific offset improved future-anchor forecasting.

8.2 Participant-held-out forecasting protocol

Participants were assigned to training, validation, and test sets using a stratified participant-level split. No participant appeared in more than one split:

$$\mathscr{P}_{\text{train}} \cap \mathscr{P}_{\text{validation}} = \mathscr{P}_{\text{train}} \cap \mathscr{P}_{\text{test}} = \mathscr{P}_{\text{validation}} \cap \mathscr{P}_{\text{test}} = \varnothing. \quad (2)$$

The split was stratified using the AI-READI study groups: healthy, prediabetes managed through lifestyle, type 2 diabetes managed without insulin, and insulin-dependent type 2 diabetes. Normalization statistics and target scales were estimated from the training participants only.

Each continuous participant segment was processed as an independent stream. Forecast windows and recurrent states were not allowed to cross participant or segment boundaries. At every 5-minute anchor t, the model emitted a 12-step forecast covering the next 60 minutes:

$$\hat{\mathbf{y}}_{i,t} = [\hat{y}_{i,t+5}, \hat{y}_{i,t+10}, \ldots, \hat{y}_{i,t+60}]\,. \tag{3}$$

The deployable forecast-only evaluation used the observed history, static clinical information, and known calendar information. It did not reveal future wearable measurements or factual future actions to the decoder.

The principal metrics were mean absolute error, root mean squared error, and forecast bias:

$$\mathrm{MAE} = \frac{1}{NH}\sum_{i,t,h}\left|y_{i,t+h} - \hat{y}_{i,t+h}\right|, \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{NH}\sum_{i,t,h}\left(y_{i,t+h} - \hat{y}_{i,t+h}\right)^2}, \tag{5}$$

$$\mathrm{Bias} = \frac{1}{NH}\sum_{i,t,h}\left(\hat{y}_{i,t+h} - y_{i,t+h}\right). \tag{6}$$

Negative bias indicates systematic underprediction. Because the model emits the 0.1, 0.5, and 0.9 quantiles, calibration was also assessed using the empirical coverage of the nominal 80% prediction interval.

Participant-level metrics were calculated before subgroup aggregation so that participants with longer recordings did not dominate the subgroup estimates. Where paired participant-bootstrap intervals were available, they were used for inferential conclusions. Comparisons without such intervals are interpreted descriptively and are not treated as statistically confirmed rankings.

8.3 From the original SSM-CGM model to SSM-CGM Stream

The original Experiment C was the first SSM-CGM configuration evaluated using a participant-disjoint split. Its global model achieved an MAE of 26.68 mg/dL and a bias of -20.82 mg/dL on unseen test participants. This large negative bias indicated systematic underprediction rather than only random forecast error.

The older pipeline also included a 48-hour adaptation procedure. After this adaptation, MAE decreased to 21.21 mg/dL. Although the adaptation partially reduced the initial generalization failure, the remaining error was still large.

SSM-CGM Stream was subsequently introduced as a redesigned forecasting pipeline. Its main changes included stateful segment-safe streaming, static-profile initialization, FiLM conditioning, train-only scaling, and a residual-current target representation:

$$\hat{y}_{t+h} = G_t + s_h \hat{r}_{t+h}, \tag{7}$$

where G_t is the current observed glucose, s_h is a horizon-specific scale estimated on the training data, and $\hat{r}_{t+h}$ is the predicted normalized change from the current glucose anchor.

The current Stream model achieved an MAE of 10.02 mg/dL and a bias of -1.446 mg/dL. Table 10 summarizes the historical progression.

Table 10: Historical progression of participant-held-out forecasting. The comparison describes the evolution of the complete forecasting pipeline and is not a controlled component ablation.

Pipeline	Participant correction	MAE (mg/dL)	Bias (mg/dL)
Old SSM-CGM global forecast	None	26.68	-20.82
Old SSM-CGM after adaptation	48 h	21.21	Not available in saved summary
SSM-CGM Stream	None in headline evaluation	10.02	-1.446

Figure 8 shows the corresponding progression in MAE and bias.

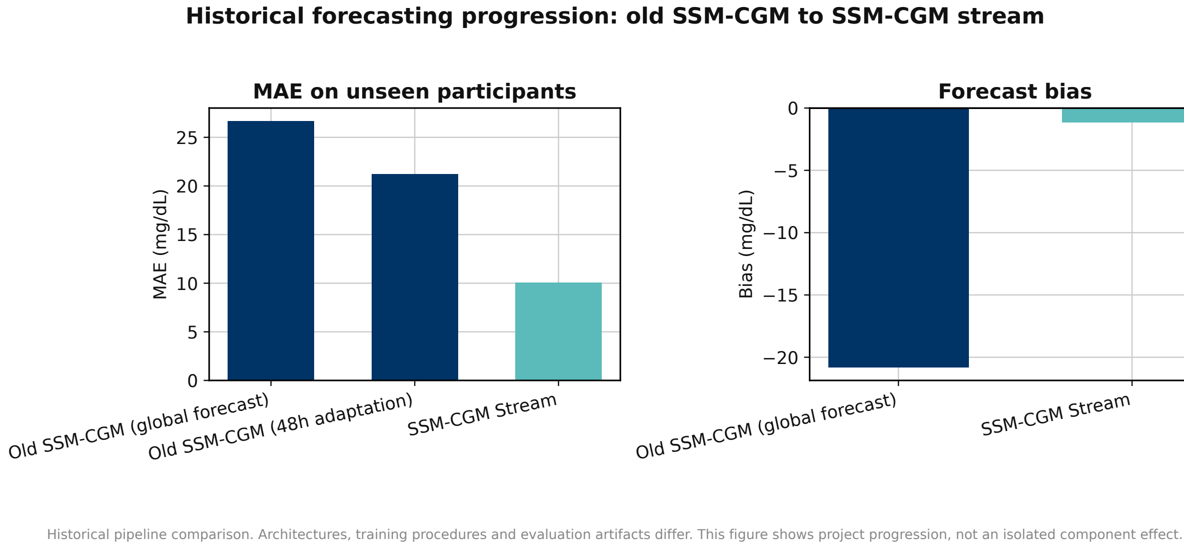


Figure 8: Historical forecasting progression from the original participant-held-out SSM-CGM Experiment C to SSM-CGM Stream. The older model exhibited high error and severe systematic underprediction. The current Stream pipeline substantially reduces both quantities. Because the architectures, target transforms, preprocessing, training procedures, and saved evaluation artifacts differ, this figure demonstrates project progression rather than the isolated effect of one architectural component.

Relative to the original global Experiment C result, the Stream pipeline reduced MAE by 16.66 mg/dL and reduced the magnitude of the negative bias by approximately 19.37 mg/dL. The main conclusion is therefore that the complete redesign substantially improved generalization to unseen participants. It is not possible to attribute the entire gain to one change from this historical comparison alone.

8.4 Current SSM-CGM Stream forecasting performance

The deployable forecast-only model achieved an overall MAE of 10.02 mg/dL, an RMSE of 15.95 mg/dL, and a bias of -1.446 mg/dL on the held-out test participants. The nominal 80% interval achieved a coverage of 79.3%, close to its intended population-level coverage.

The terminal 60-minute MAE was 14.43 mg/dL. This value is reported separately from the multi-horizon average because the latter combines substantially easier near-term predictions with the more difficult final forecast steps.

Table 11: Headline forecast-only performance of SSM-CGM Stream on participant-held-out test data.

Metric	Value
Multi-horizon MAE	10.02 mg/dL
RMSE	15.95 mg/dL
Bias	-1.446 mg/dL
Terminal 60-minute MAE	14.43 mg/dL
Nominal 80% interval coverage	79.3%
True TIR 70–180 mg/dL	89.7%
Predicted TIR 70–180 mg/dL	91.6%
TIR gap	$+1.86$ percentage points
90th percentile absolute error	24.25 mg/dL
95th percentile absolute error	34.05 mg/dL
99th percentile absolute error	59.69 mg/dL

The near-nominal interval coverage indicates reasonable population-level uncertainty calibration. However, the predicted TIR exceeds the observed TIR, showing a modest tendency to place glucose inside the target range more often than occurred in the data.

8.5 Anchor-selection audit of the nominal warm-up analysis

The original evaluation reported forecasting metrics after nominal warm-up periods of 0, 6, 12, 24, and 48 hours. The unmatched sweep appeared to show a progressive decrease in MAE:

$$10.07 \to 10.06 \to 10.02 \to 9.90 \to 9.76 \text{ mg/dL}. \tag{8}$$

However, increasing the nominal warm-up duration progressively removed early forecast anchors. The number of scored horizon-level predictions decreased from approximately 9.31 million at 0 hours to 4.92 million at 48 hours. Consequently, each point represented a different subset of participants, segments, and timestamps.

Table 12: Original unmatched nominal warm-up sweep. Because the evaluated rows change with warm-up duration, this table is descriptive and does not establish an adaptation effect.

Nominal warm-up	Scored horizon predictions	MAE	Bias	Coverage
0 h	9,306,780	10.07	−1.16	79.2%
6 h	8,758,200	10.06	−1.24	79.3%
12 h	8,209,560	10.02	−1.28	79.3%
24 h	7,112,280	9.90	−1.40	79.4%
48 h	4,917,720	9.76	−1.56	79.5%

A common-anchor re-scoring analysis then restricted every nominal warm-up row to the same 81,962 forecast anchors from 221 participants. Each anchor contributed 12 forecast horizons, corresponding to 983,544 horizon-level predictions. Under this common scoring set, MAE, bias, and interval coverage were identical across all five nominal warm-up values.

Table 13: Common-anchor re-scoring of the nominal warm-up analysis. The same 81,962 forecast anchors from 221 participants are used in every row.

Nominal warm-up	Forecast anchors	Participants	MAE	Bias	Coverage
0 h	81,962	221	9.76	−1.57	79.5%
6 h	81,962	221	9.76	−1.57	79.5%
12 h	81,962	221	9.76	−1.57	79.5%
24 h	81,962	221	9.76	−1.57	79.5%
48 h	81,962	221	9.76	−1.57	79.5%

Figure 9 contrasts the original unmatched curve with the common-anchor result.

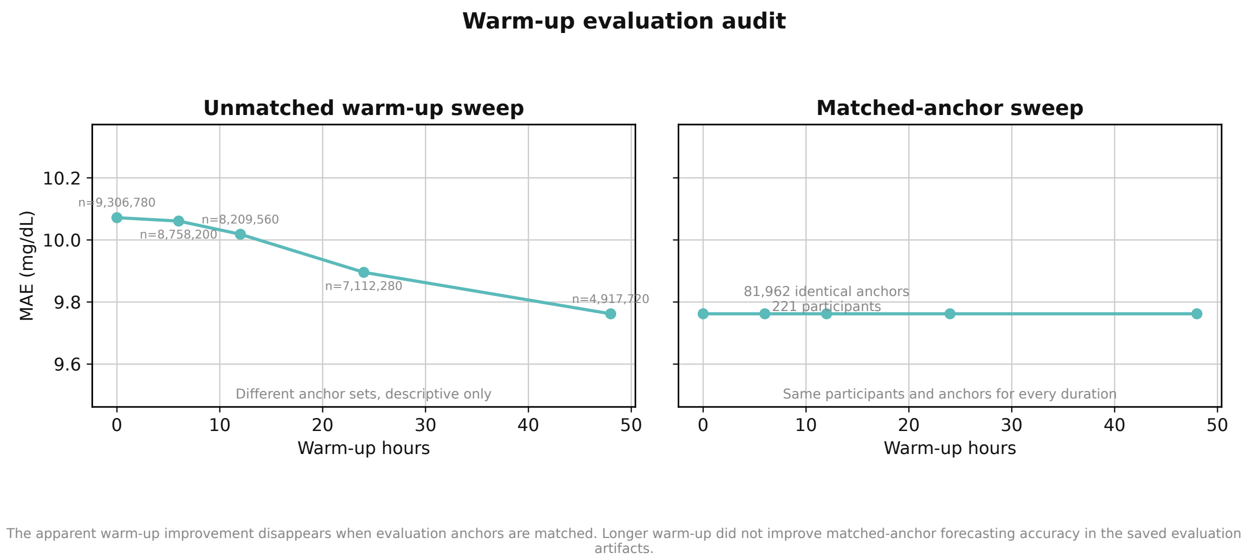


Figure 9: Anchor-selection audit of the nominal warm-up analysis. The unmatched sweep appears to improve because longer cutoffs remove early prediction anchors. When the same 81,962 anchors from 221 participants are re-scored, the metrics are identical across nominal warm-up values. The matched predictions originate from the same continuously propagated streaming state; therefore, the analysis diagnoses anchor-selection bias but does not constitute a controlled state-reset experiment.

The correct interpretation is that the apparent improvement in the original sweep was explained by changes in the evaluated anchor composition. The common-anchor analysis does not directly manipulate how much history is used to construct the recurrent state. At a given common anchor, the model had processed the same preceding stream, regardless of the nominal warm-up label assigned during scoring.

Therefore, these results should not be used to claim that the recurrent state does not use historical information or that the model has definitively been shown to be warm-up-free. They establish the narrower result that the original warm-up curve was not evidence of personalization.

The two analyses answer different questions and should therefore not be interpreted in the same way. The elapsed-time deployment analysis evaluates forecasts made at progressively later stages after a new participant begins using the system. It reflects the real onboarding experience and answers: How accurate is the model during the first hours or days of use? However, because later cutoffs exclude early anchors and may also exclude shorter or more fragmented recordings, any improvement can result from changes in the evaluated timestamps or participant composition, not necessarily from the model learning the individual. In contrast, the controlled state-reset warm-up experiment evaluates the same participants, the same forecast timestamps, and the same future targets while varying only how much preceding history is used to build the recurrent state. It therefore answers the causal modeling question: Does providing the model with more participant history improve the same forecast? The first analysis is useful for describing expected performance after deployment, whereas the second is required to isolate the specific benefit of accumulated history and demonstrate genuine passive personalization.

8.6 Bias-based participant correction

The next experiment tested an explicit low-capacity form of participant adaptation. The base SSM-CGM Stream model remained frozen. For participant i and forecast horizon h, a correction was estimated from residuals observed in an adaptation interval $\mathscr{A}_i$:

$$b_{i,h} = \frac{1}{|\mathscr{A}_i|} \sum_{t \in \mathscr{A}_i} \left(y_{i,t+h} - \hat{y}_{i,t+h} \right). \tag{9}$$

The corrected forecast was:

$$\hat{y}_{i,t+h}^{\text{corrected}} = \hat{y}_{i,t+h} + b_{i,h}. \tag{10}$$

This method tests whether the participant's mean past residual is sufficiently stable to correct later predictions. Four modes were compared:

1. the raw frozen Stream forecast;
2. an offset-corrected forecast;
3. the existing participant-personalized forecast;
4. the personalized forecast followed by an additional offset correction.

Table 14: Forecast accuracy and bias after participant correction. Lower MAE is better. Bias is defined as prediction minus observation.

Mode	MAE (mg/dL)	Bias (mg/dL)
Raw Stream forecast	**10.02**	-1.446
Offset-corrected	10.16	approximately 0
Personalized	10.52	-1.142
Personalized + offset-corrected	10.61	$+0.305$

Figure 10 shows the corresponding MAE and bias results.

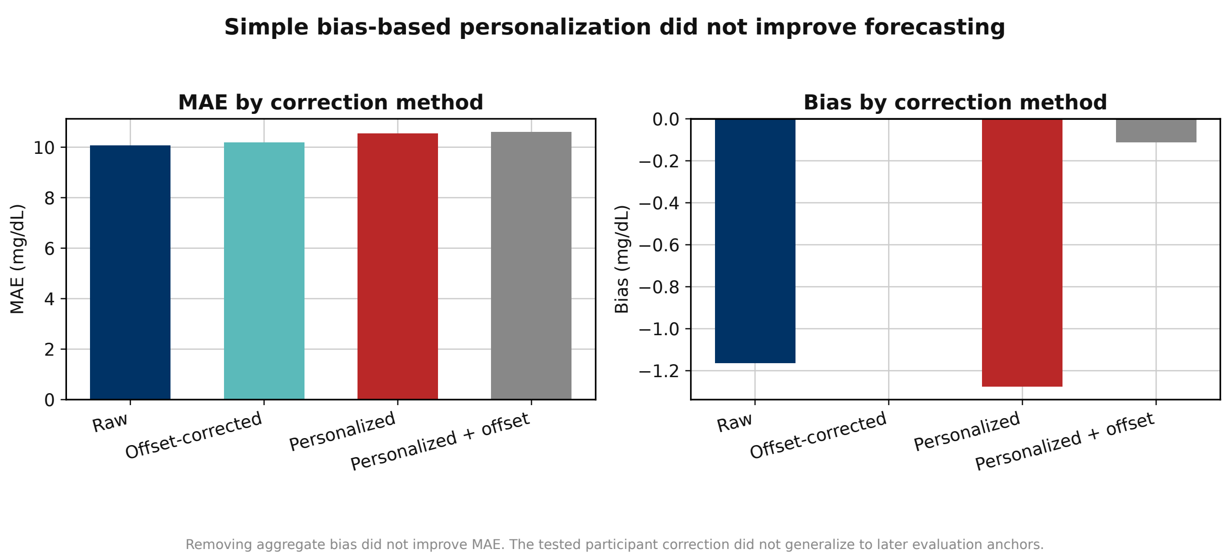


Figure 10: Audit of simple bias-based personalization. Offset correction removes the aggregate negative bias but increases MAE. Both personalized variants also perform worse than the uncorrected Stream forecast. Reducing population-level bias therefore did not translate into more accurate future-anchor predictions.

The result demonstrates that:

$$\text{Bias} \approx 0 \quad \not\Rightarrow \quad \text{MAE decreases.} \tag{11}$$

A participant may exhibit positive forecast errors in one physiological regime and negative errors in another. A single correction estimated from their average past residual can therefore shift already-correct forecasts in the wrong direction.

The remaining error is not well described as a stable participant-specific vertical offset. It is more plausibly associated with changing glucose direction, glycemic state, time of day, meals, activity, sleep, treatment, or other unobserved events.

This result is method-specific. It shows that the tested bias-based correction did not improve forecasting. It does not establish that every possible form of personalization is ineffective.

8.7 Convergence-based selection of the adaptation window

A fixed adaptation window may be unnecessarily long for some participants and insufficient for others. A convergence-based procedure was therefore used to estimate when the participant-specific horizon corrections stopped changing.

The correction was refitted in 6-hour increments up to a maximum of 48 hours. Let:

$$\mathbf{b}_i^{(W)} = \left[b_{i,5}^{(W)}, \dots, b_{i,60}^{(W)}\right] \tag{12}$$

denote the vector of horizon-specific corrections estimated after W hours. A participant was considered converged when:

$$\max_h \left| b_{i,h}^{(W)} - b_{i,h}^{(W-6)} \right| < 3 \text{ mg/dL} \tag{13}$$

for three consecutive 6-hour increments.

The analysis included 287 participants. Of these, 194 converged within 48 hours and 93 did not. The median convergence time was 30 hours, with the 10th and 90th percentiles at 24 and 36 hours.

Table 15: Summary of the convergence-based adaptation-window analysis.

Quantity	Result
Total participants	287
Converged within 48 h	194
Did not converge	93
Convergence rate	67.6%
Median convergence time	30 h
10th percentile	24 h
90th percentile	36 h

Forecast improvement for participant i was defined as:

$$\Delta\text{MAE}_i = \text{MAE}_{i,\text{raw}} - \text{MAE}_{i,\text{corrected}}. \tag{14}$$

Positive values indicate that the correction improved forecasting, while negative values indicate degradation.

Table 16: Convergence status and future-anchor forecast outcome.

Outcome	Participants	Percentage
Converged and improved	76	26.5%
Converged but worsened	118	41.1%
Did not converge	93	32.4%

Among the 194 converged participants, 118, or 60.8%, were worsened by the correction. Only 39.2% of converged participants improved.

The mean improvement was negative for every adaptation-window strategy.

Table 17: Mean MAE improvement by adaptation-window strategy. Positive values indicate improvement; negative values indicate degradation.

Strategy	Mean MAE improvement (mg/dL)
Fixed 24 h	−0.191
Fixed 48 h	−0.168
Convergence-based	−0.148

Figure 11 separates numerical convergence from forecasting benefit.

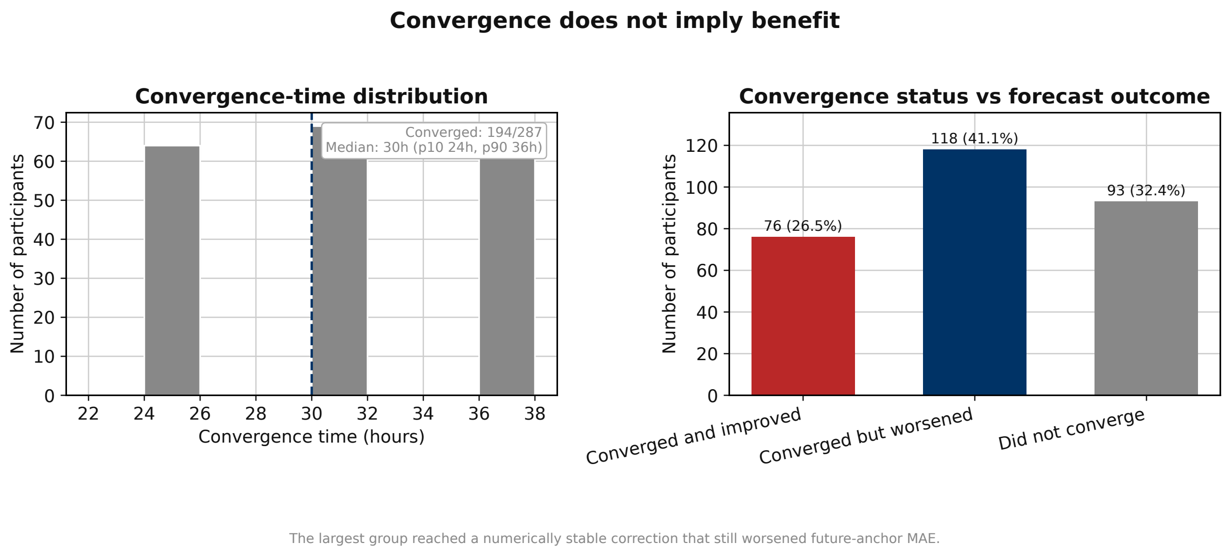


Figure 11: Convergence does not imply forecasting benefit. Most converged corrections stabilized between 24 and 36 hours. Nevertheless, the largest group consisted of participants whose correction converged but worsened future-anchor MAE. Numerical stability of the fitted correction is therefore not evidence that the correction generalizes to later data.

The convergence-based strategy showed the smallest numerical mean degradation, −0.148 mg/dL compared with −0.191 and −0.168 mg/dL. However, this ordering has not been tested using paired participant-bootstrap confidence intervals and should be read descriptively rather than as a confirmed ranking. All three strategies failed to improve average forecasting accuracy.

Convergence answers whether the fitted correction stopped moving. It does not answer whether the correction predicts future residuals. A stable estimate of error in the adaptation interval may fail when the subsequent evaluation interval contains different meals, activity, sleep, treatment, or glycemic states.

8.8 Forecasting heterogeneity across clinical subgroups

The overall MAE hides substantial heterogeneity across participants. Participant-level MAE was therefore calculated first and then aggregated within clinical subgroups.

8.8.1 Study group and HbA1c

Mean participant MAE increased from 8.97 mg/dL in healthy participants to 12.64 mg/dL in the insulin-dependent study group. A similar gradient was observed across HbA1c quartiles, from 8.83 mg/dL in the lowest quartile to 12.63 mg/dL in the highest quartile.

Table 18: Participant-level forecasting performance by AI-READI study group.

Study group	N	Mean MAE	Median MAE
Healthy	78	8.97	8.72
Prediabetes	56	9.42	9.16
T2D oral/non-insulin	64	11.31	11.26
T2D insulin-dependent	23	12.64	12.64

Table 19: Participant-level forecasting performance by baseline HbA1c quartile.

Quartile	HbA1c range (%)	N	Mean MAE	Median MAE
Q1	3.8–5.4	58	8.83	8.41
Q2	5.4–5.8	60	9.06	8.79
Q3	5.8–6.3	45	10.22	10.07
Q4	6.3–12.5	54	12.63	12.23

The difference between the insulin-dependent and healthy groups was 3.67 mg/dL. The difference between the highest and lowest HbA1c quartiles was 3.80 mg/dL.

8.8.2 Insulin-use metadata

Participants with recorded insulin use had a mean participant MAE of 12.83 mg/dL, compared with 9.92 mg/dL among participants without recorded insulin use.

Table 20: Participant-level forecasting performance by insulin-use metadata.

Insulin metadata	N	Mean MAE
No recorded insulin use	204	9.92
Recorded insulin use	17	12.83

The insulin-dependent study-group category and the medication-derived `med_insulin` flag are related but not equivalent definitions. Their sample sizes differ because study-group classification and medication metadata have different availability and inclusion rules.

8.8.3 Clinical site

Site-level mean participant MAE ranged from 9.78 mg/dL at UCSD to 10.55 mg/dL at UAB.

Table 21: Participant-level forecasting performance by clinical site.

Clinical site	N	Mean MAE
UAB	82	10.55
UW	85	9.98
UCSD	54	9.78

Figure 12 presents the four subgroup views together.

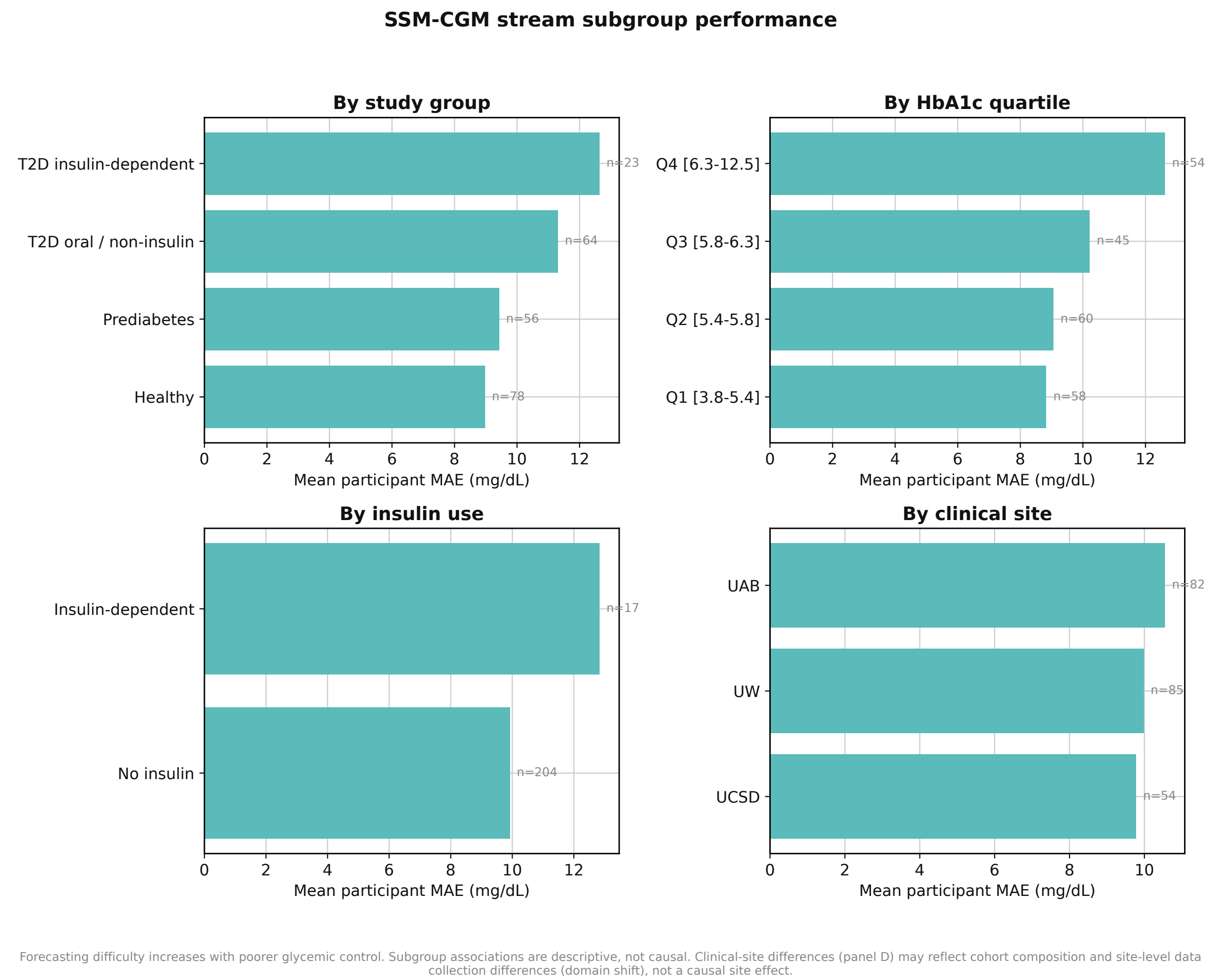


Figure 12: Participant-level SSM-CGM Stream forecasting performance by study group, HbA1c quartile, insulin-use metadata, and clinical site. Forecasting error increases with more complex glycemic phenotypes and higher HbA1c. Clinical-site differences are smaller and may reflect cohort composition, acquisition conditions, missingness, or other domain-shift effects. The associations are descriptive and should not be interpreted causally.

The phenotype and HbA1c gradients were substantially larger than the site gradient. The site range was only:

$$10.55 - 9.78 = 0.77 \text{ mg/dL}, \tag{15}$$

whereas the phenotype and HbA1c ranges were approximately 3.7–3.8 mg/dL.

Possible explanations for the higher error in insulin-dependent and higher-HbA1c participants include greater glucose variability, more frequent rapid excursions, more complex treatment-response dynamics, and unobserved insulin or behavioral events. These interpretations remain hypotheses because treatment status, HbA1c, glycemic variability, and study group are correlated.

8.9 Published SSM-CGM and TFT results as contextual references

The original SSM-CGM paper compared SSM-CGM with a Temporal Fusion Transformer under a common within-participant temporal-holdout protocol [7]. TFT was selected as the principal comparator after outperforming LightGBM, DeepAR, N-HiTS, and a Seq2Seq LSTM in the paper's initial benchmarking.

Under that matched paper protocol, SSM-CGM consistently outperformed TFT at 12-, 24-, and 48-hour context lengths.

Table 22: Published SSM-CGM and TFT performance under the original paper protocol. Values are means with 95% confidence intervals. These values are contextual references and are not directly matched to the current participant-held-out study.

Context	Model	Quantile loss	MAE (mg/dL)	RMSE (mg/dL)
12 h	SSM-CGM	3.87 (3.81–3.93)	5.97 (5.68–6.26)	7.17 (6.83–7.52)
12 h	TFT	3.95 (3.89–4.02)	6.07 (5.76–6.37)	7.32 (6.96–7.69)
24 h	SSM-CGM	3.69 (3.63–3.75)	5.66 (5.39–5.93)	6.86 (6.53–7.19)
24 h	TFT	3.92 (3.86–3.99)	6.06 (5.77–6.36)	7.28 (6.93–7.63)
48 h	SSM-CGM	3.52 (3.47–3.58)	5.40 (5.15–5.65)	6.56 (6.26–6.87)
48 h	TFT	3.70 (3.64–3.76)	5.77 (5.50–6.03)	6.92 (6.61–7.24)

Figure 13 visually separates the current AI-READI participant-held-out experiments from the published within-participant temporal holdout.

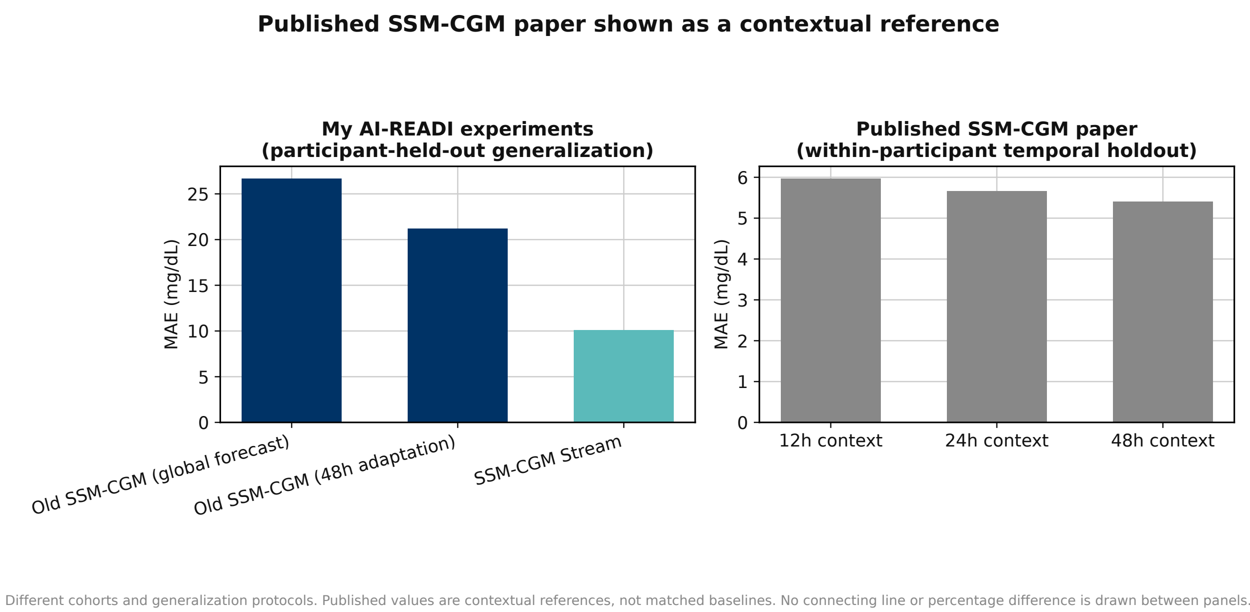


Figure 13: Published SSM-CGM results shown as contextual references. The current AI-READI experiments evaluate generalization to participants who were completely unseen during model training, whereas the published paper used a within-participant temporal holdout. The two panels therefore answer different generalization questions and should not be interpreted as a directly matched numerical comparison.

The lower MAEs reported in the paper do not imply that the published model generalizes better to unseen participants. Conversely, the current results do not demonstrate that SSM-CGM Stream outperforms TFT under the current protocol. A fair architecture comparison would require training TFT on the same current cohort, participant split, covariates, context, horizon, and forecast anchors.

8.10 Discussion

8.10.1 The main gain comes from the redesigned global pipeline

The original participant-held-out Experiment C exposed a major generalization failure. The global model had both high MAE and a large systematic negative bias. A 48-hour adaptation phase partially reduced the error but did not solve the problem.

SSM-CGM Stream substantially improved unseen-participant forecasting without requiring the old adaptation procedure as part of the headline evaluation. The historical progression is consistent with the combined effect of residual-current anchoring, participant-safe streaming, static conditioning, improved segmentation, and train-only normalization. Their individual contributions cannot be isolated from the historical comparison alone.

8.10.2 Reframing personalization

The experiments distinguish several meanings of personalization:

Table 23: Different forms of personalization considered in this work.

Form	Mechanism	Current evidence
Zero-shot participant conditioning	Static clinical covariates initialize and modulate a global model.	Present in SSM-CGM Stream.
Passive state accumulation	The recurrent state carries observed participant history forward.	The current cutoff audit does not directly manipulate state history.
Nominal warm-up scoring	Early anchors are excluded before metrics are calculated.	Apparent gain was explained by anchor selection.
Explicit bias correction	Past residual means are applied to later forecasts.	Reduced bias but worsened MAE.
Convergence-based correction	Adaptation ends when fitted offsets stabilize.	Stability did not imply future forecasting benefit.

The results do not show that the model is non-personalized. Every prediction is conditioned on a participant's static profile, current glucose, and accumulated streaming state. What failed was the narrower assumption that future errors could be corrected using one stable participant-specific offset estimated from the first hours of data.

8.10.3 Why a constant correction is insufficient

A constant offset assumes:

$$e_{i,t,h} \approx b_{i,h} \tag{16}$$

for future timestamps t. The observed degradation indicates that this stationarity assumption is not generally valid. A more realistic residual model would depend on the current state:

$$e_{i,t,h} = f_i\left(G_t, \Delta G_t, \text{time of day}, \text{meal state}, \text{exercise state}, \text{sleep}, \text{treatment context}\right) + \varepsilon_{i,t,h}. \tag{17}$$

A participant may be overpredicted in one physiological regime and underpredicted in another. Averaging these regimes into a single correction can therefore degrade later forecasts.

8.10.4 Implications for subsequent personalization work

The remaining error is concentrated in participants with more complex glycemic profiles and is unlikely to be resolved through a global vertical shift. Subsequent personalization should therefore focus on dynamic response patterns, including:

- exercise and post-exercise glucose responses;
- rapid glucose rise and decline states;
- meal-associated excursions;
- sleep and overnight regulation;
- event-specific residual corrections;
- participant-specific counterfactual response curves.

Together with the meal-state audit, these findings define a consistent dataset boundary: AI-READI supports rigorous participant-held-out forecasting and model-behavior auditing, but fine-grained causal personalization remains limited by the absence of prospectively observed meals, treatment timing, and behavioral actions.

8.11 Limitations

Several limitations qualify the conclusions of this section.

First, the old SSM-CGM and current SSM-CGM Stream results form a historical pipeline comparison. They were not generated as a controlled architecture ablation using identical checkpoints, preprocessing, and common forecast anchors.

Second, the common-anchor warm-up analysis is a scoring audit rather than a controlled state-reset experiment. It establishes that the original apparent gain resulted from changing anchor composition, but it does not directly test the effect of initializing the recurrent state at different times before the same anchor.

Third, the participant correction is intentionally low capacity. Its failure does not exclude nonlinear, event-specific, state-dependent, or counterfactual-response personalization.

Fourth, the relative ordering of the 24-hour, 48-hour, and convergence-based adaptation strategies remains descriptive because paired participant-bootstrap confidence intervals were not used to compare the three mean degradations.

Fifth, subgroup analyses are observational and descriptive. Differences by HbA1c, insulin use, study group, and clinical site may reflect overlapping effects of treatment, glucose variability, cohort composition, missing covariates, and acquisition conditions.

Finally, the published SSM-CGM and TFT results cannot be used as direct benchmarks for the current participant-held-out model because they were obtained using a different cohort construction and a within-participant temporal holdout.

8.12 Conclusion

The original participant-held-out SSM-CGM experiment revealed a major generalization failure, characterized by high MAE and severe systematic underprediction for unseen participants. The redesigned SSM-CGM Stream pipeline reduced MAE to 10.02 mg/dL and reduced bias to -1.446 mg/dL while retaining near-nominal prediction-interval coverage.

The original nominal warm-up curve appeared to improve with longer warm-up, but this pattern was explained by changes in the evaluated anchor set. Common anchor re-scoring yielded identical metrics because the same continuously propagated streaming predictions were scored in every nominal warm-up row. This audit therefore identifies selection bias rather than proving a state warm-up effect.

Explicit participant-level bias corrections removed aggregate bias but increased MAE. Moreover, a correction could converge numerically while still worsening future-anchor forecasts. The remaining error is therefore not well represented by a stable participant-specific offset.

The central conclusion is:

> SSM-CGM Stream substantially improves generalization to unseen participants through the redesigned global forecasting pipeline. The tested nominal warm-up and bias-based correction procedures do not provide evidence of additional forecasting value. Future personalization should target dynamic, event-dependent physiological responses rather than a constant participant offset.

9 Proxy Scenario Evaluation

9.1 AI-READI possible scenario

AI-READI provides rich CGM, wearable, and static clinical data, but it does not contain the action logs needed for fully causal glucose counterfactuals. In particular, there are no timed insulin doses, insulin-on-board values, logged carbohydrate amounts, or prospectively recorded meal times. This is an important limitation because these variables are the actual interventions that would be needed to answer questions such as "what if this participant ate now?" or "what if this participant took insulin?" The meal, activity, and sleep/rest outputs should therefore be described as representative signals rather than true treatment actions. A future scientifically interpretable scenario model would need prospective action logs or stronger supervision of the scenario branch.

The scenario mode comparison confirms this limitation. Forecast-only, factual-future, meal-proxy, activity-proxy, and sleep/rest-proxy evaluation all produce almost identical forecasting metrics. The largest mean absolute prediction change relative to forecast-only mode is only 0.0071 mg/dL, which is far below any clinically meaningful glucose difference. This means that, in the current model checkpoint, the scenario branch is either inactive or only very weakly used.

The prediction-delta audit gives the same conclusion more directly. Even when future proxy information is revealed, the model output barely changes. This is not evidence that meals, activity, or sleep have no effect on glucose. Instead, it shows that the trained decoder has not learned to use these proxy scenario inputs in a meaningful way. The mask and value audit explains part of the issue. The future predmealflag has zero nonzero mask coverage and zero available anchors in this evaluation, while wearable variables such as heart rate and steps are available. Therefore, the meal-proxy pathway is not actually being exercised in the forecast horizon. This makes any meal counterfactual interpretation invalid at this stage.

Table 24: Metrics by evaluation/scenario mode. Proxy scenarios are diagnostics and are not validated causal interventions.

Mode	MAE (mg/dL) ↓	RMSE	Bias	Cov. 80%
activity_proxy	10.02	15.95	-1.443	0.793
factual-future (oracle)	10.02	15.95	-1.445	0.793
forecast-only (deployable)	10.02	15.95	-1.448	0.793
meal_proxy	10.02	15.95	-1.449	0.793
sleep_rest_proxy	10.02	15.95	-1.446	0.793

Table 25: Prediction deltas relative to forecast-only mode. Values near zero indicate weak use of the scenario pathway.

Mode	Mean |delta|	Median |delta|	p95 |delta|	Mean delta
activity_proxy	0.0062	0.0044	0.0186	+0.0044
factual-future	0.0071	0.0054	0.0194	+0.0022
meal_proxy	0.0028	0.0022	0.0073	-0.0010
sleep_rest_proxy	0.0067	0.0049	0.0191	+0.0012

The conclusion from this section is therefore conservative. The current proxy scenario outputs should be described as pathway diagnostics only. They should not be interpreted as treatment recommendations, carbohydrate counterfactuals, insulin counterfactuals, or reliable behavioral simulations. A scientifically interpretable scenario model would require either real prospective action logs or additional training that forces the model to use the scenario channel and validates the resulting direction of effect against observed episodes.

Table 26: Scenario mask and value audit for selected future proxy variables.

Variable	Mask nonzero	Anchors avail.	Value mean	Value SD
predmeal_flag	0.0%	0.0%	0.000	0.000
heart_rate_mean	100.0%	100.0%	0.087	1.007
activity_steps_per_min	100.0%	100.0%	0.015	1.040
stress_level_mean	89.2%	99.3%	0.065	0.998
sleep_stage_awake	34.4%	38.2%	0.025	1.038
sleep_stage_light	34.4%	38.2%	-0.009	0.990
sleep_stage_deep	34.4%	38.2%	0.006	1.011
sleep_stage_rem	34.4%	38.2%	-0.015	0.973

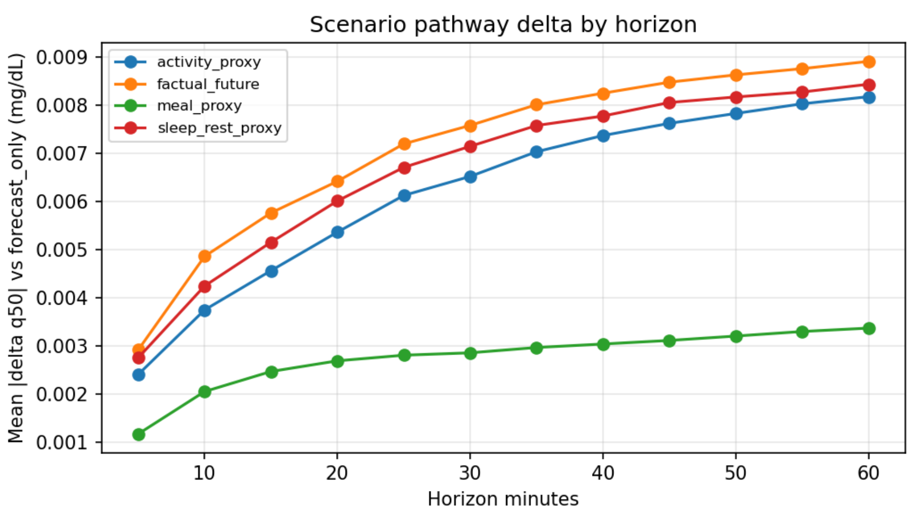


Figure 14: Absolute prediction deltas between proxy scenario modes and forecast-only mode. Deltas are far below clinically meaningful mg/dL changes, indicating that the scenario pathway is currently inactive or weakly used.

10 Passive Meal-State Modeling

10.1 Motivation

AI-READI contains CGM, wearable signals, and static clinical descriptors, but not prospectively logged meals or carbohydrate doses [1]. The existing `predmeal_flag` was therefore only a proxy scenario channel. We rebuilt the meal pathway as a weak-supervision problem and then asked a narrower question, whether any deployable meal feature improves held-out forecasting on the endpoints where a meal actually acts. The pipeline is summarized in Fig. 15 and Table 27.

10.2 Transfer of the CGMacros Meal Detector

The source audit recovered the CGMacros detector as a dilated CNN followed by a two-layer bidirectional LSTM and a per-time-step logit head over 72-sample, six hour windows of 5-minute CGM. Because the LSTM is bidirectional, the teacher probability at an anchor depends on CGM samples after the anchor, up to roughly six hours ahead. That window overlaps the forecast horizon, so the retrospective teacher reads future glucose about the exact quantity being forecast. We therefore treat the bidirectional teacher as a leakage diagnostic, not as a reachable target.

10.3 Correction of the AI-READI Meal-Flag Reconstruction

The original downstream artifact preserved only a thresholded binary flag and discarded the teacher's continuous probability. Reconstructing overlap-averaged probabilities recovered finite values for 99.3% of rows and a corrected teacher flag rate of 0.114, compared with 0.042 for the surviving old binary artifact. The reconstruction is a faithful recovery of the source signal, but the source signal is itself bidirectional, so neither the corrected probability nor the surviving `predmeal_flag` is causal. Figure 16 shows the information loss of the old artifact.

10.4 Weak Target-Domain Supervision

Weak labels were built from train-split quantiles, not hard-coded glucose thresholds. Two changes follow the leakage and pooling critique. First, insulin users are quarantined from meal-label training, because a decoded meal-like response in an insulin user often reflects unobserved insulin dynamics rather than a meal; 305,843 insulin rows are removed from the training labels and reported separately as low-confidence meal-like events. Second, the previously discarded uncertain rows are rescued with a soft, ordinal onset-distance target rather than dropped, because the pre-rise transition is the only region where a meal flag can lead the CGM trend; 389,623 of 2,422,111 uncertain rows now carry a nonzero soft target and stay in training. Response size remains a realized glucose-excursion proxy, not carbohydrate grams.

10.5 Causal Student Model

The target-domain student uses past and current glucose, wearable, clock, and static features only, and is now fit as a regressor on the soft onset-distance target over the quarantined, rescued training set. Teacher outputs and future glucose are used only during label construction. A good classification score does not imply a useful forecasting covariate, so the student is judged by the forecasting gate below, not by its pseudo-label AUC.

10.6 Retrospective and Online Meal-State Decoding

A duration-constrained retrospective decoder converts the student probability into onset, rising, peak, and recovery states, and a strictly forward-only filter produces the deployable online states. The retrospective response-size proxy is physiologically ordered, mean peak rise increasing from 2.1 to 15.9 to 50.8 mg/dL across small, medium, and large events. The retrospective decoder is an event-discovery diagnostic because it uses full-segment summaries, the online decoder is the deployable candidate. Examples appear in Fig. 17.

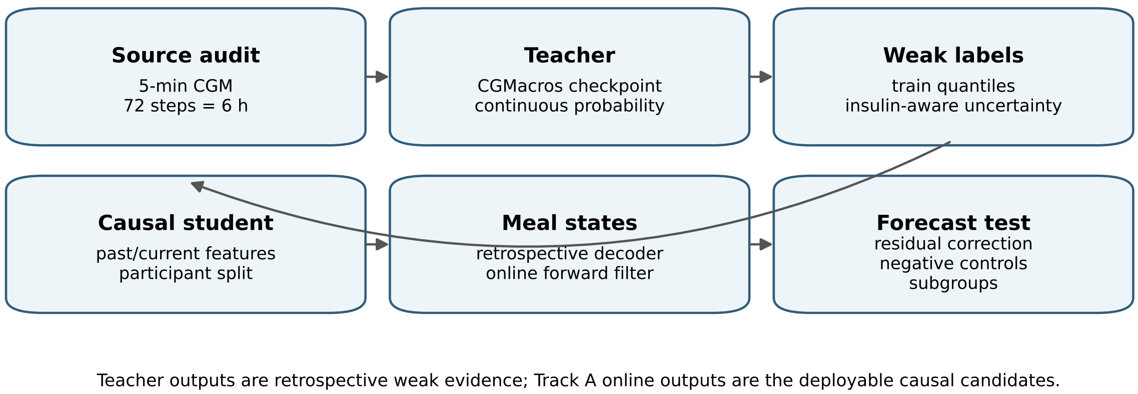


Figure 15: Meal-flag improvement pipeline. The CGMacros teacher and the retrospective state decoder are diagnostic weak-supervision sources, the online Track A decoder and the causal student probability are the deployable candidates.

10.7 Experimental Feature Representations

All setups share a common residual-correction framework. The q50 median from the pretrained SSMCGM-Stream model is used as a base forecast, and each candidate meal representation $M_t^{(m)}$ is evaluated as an additive residual correction:

$$\widehat{y}_{t+h}^{(m)} = \widehat{y}_{t+h}^{q50} + r_m\left(X_t^{\text{base}}, M_t^{(m)}, h\right), \tag{18}$$

where the baseline feature vector is

$$X_t^{\text{base}} = [G_t,\ \Delta G_{15,t},\ \Delta G_{30,t},\ \Delta G_{60,t},\ \sin\theta_t,\ \cos\theta_t], \quad \theta_t = \tfrac{2\pi m_t}{1440}. \tag{19}$$

Here G_t is the current glucose reading, $\Delta G_{\tau,t}$ are τ-minute backward slopes, and $(\sin\theta_t, \cos\theta_t)$ encode time of day continuously.

The retrospective setups build on this with increasingly rich leakage diagnostics:

$$\underbrace{X_t^{\text{base}}}_{A'} \longrightarrow \underbrace{\left[X_t^{\text{base}}, B_t\right]}_{B} \longrightarrow \underbrace{\left[X_t^{\text{base}}, \bar{p}_t^T\right]}_{C}. \tag{20}$$

The leakage-free chain replaces retrospective signals with causal ones:

$$\underbrace{X_t^{\text{base}}}_{A'} \longrightarrow \underbrace{\left[X_t^{\text{base}}, p_t^{T,\text{causal}}\right]}_{C'} \longrightarrow \underbrace{\left[X_t^{\text{base}}, \widehat{p}_t^S\right]}_{D} \longrightarrow \underbrace{\left[X_t^{\text{base}}, \widehat{p}_t^S, \alpha_t, \tau_t, c_t, \mathbf{q}_t\right]}_{H}. \tag{21}$$

The central empirical finding is:

$$C \gg C' \approx D \approx H \approx A'. \tag{22}$$

The large performance advantage of C disappears when future-glucose access is removed; the causal teacher, student, and online state carry little forecasting information beyond what glucose level, slope, and clock already provide.

Table 28 gives the formal definition of each setup's added information, its mathematical representation, and its role in the evaluation.

10.8 Forecasting Evaluation and Selection Gate

Meal features were evaluated as residual corrections on top of the cached 60-minute SSMCGM-Stream q50 forecast,

$$\hat{y}_{i,t+h}^{(m)} = \hat{y}_{i,t+h}^{q50} + r_m(C_{i,t}, M_{i,t}^{(m)}, h), \tag{23}$$

Table 27: Meal-transfer pipeline summary.

Quantity	Value
Participants	1,591
Rows	4,015,355
Teacher probability finite fraction	0.993
Corrected teacher flag rate	0.114
Surviving old binary artifact rate	0.042
Pseudo-label positives	507,830
Pseudo-label negatives	1,085,414
Pseudo-label uncertain	2,422,111
Causal student validation AP	0.863
Causal student validation AUC	0.928
Retrospective decoded events	53,024
Retrospective decoded flag rate	0.290
Online prefix invariance max absolute difference	0.000
Online response-size validation macro-F1	0.767
Independent evaluation events	77,835

fit on the validation cache and evaluated on the participant-disjoint test cache. Alignment passed on 1,660,377 test rows with maximum target residual 0.500 mg/dL and same-segment fraction 1.000.

Whole-horizon MAE60 is not the gate. A 60-minute average washes out the post-prandial excursion, and the negative controls below confirm almost no timing-specific signal exists on that endpoint. The gate is re-anchored on the endpoints where a meal acts, meal-window MAE, one hour peak error, time to peak, hyperglycemia AUPRC, and recall at precision 0.80 for the event $\max(y$ over the next hour$) > 180$ mg/dL. A feature is selected only if, on the non-insulin cohort, it improves at least two of these endpoints over the baseline and beats shuffled, time-shifted, and block-shuffled controls with a participant-bootstrap interval that excludes zero. The decision is computed in code, not asserted in prose.

The baseline is A', a within-harness residual model on CGM slope, level, and clock only. The harness cost of A' relative to untouched `q50` is small on the non-insulin cohort, -0.005 mg/dL on MAE60 and -0.344 mg/dL on peak error, so meal comparisons are made against A' rather than against `q50` to separate the harness penalty from feature value. A' already captures most of the recoverable post-prandial signal, reaching meal-window MAE 13.745 and peak error 10.549 versus 13.856 and 10.893 for `q50`.

The leakage-free causal teacher C' is the reachable ceiling, and it sits essentially at A', meal-window MAE 13.724 and peak error 10.542. The old bidirectional teacher C reaches meal-window MAE 11.286 and peak error 8.150, but that gap is leakage, because the only difference between C and C' is whether the teacher window is allowed to read future glucose. The previously reported gain from C, from 14.846 to 12.460 on MAE60, was therefore an oracle effect rather than reachable headroom. The deployable features cluster at A', the student probability D reaching meal-window MAE 13.711 and peak error 10.541, and the online full state H reaching 13.710 and 10.523.

The orthogonalization test explains the clustering. After regressing each feature on slope, level, and clock, the leakage-free causal teacher C' has partial correlation 0.027 with the `q50` residual, the student probability D has 0.147, while the bidirectional teacher C has 0.519. The large orthogonal signal belongs only to the leaky features, so the deployable features carry little information that the CGM slope does not already provide.

10.9 Leakage Audits and Negative Controls

The provenance table records, for each feature, the maximum CGM input offset relative to the anchor. The student probability and the online states have offset zero, while the bidirectional teacher and the `predmeal_flag` have a positive offset because their value at an anchor is computed from a window that extends past the anchor. A hard assertion fails the run if any teacher probability used as a selectable input has a positive offset, which is why the default teacher mode is causal.

The negative controls are the headline evidence. They compare the real feature timing against shuffled, time-shifted,

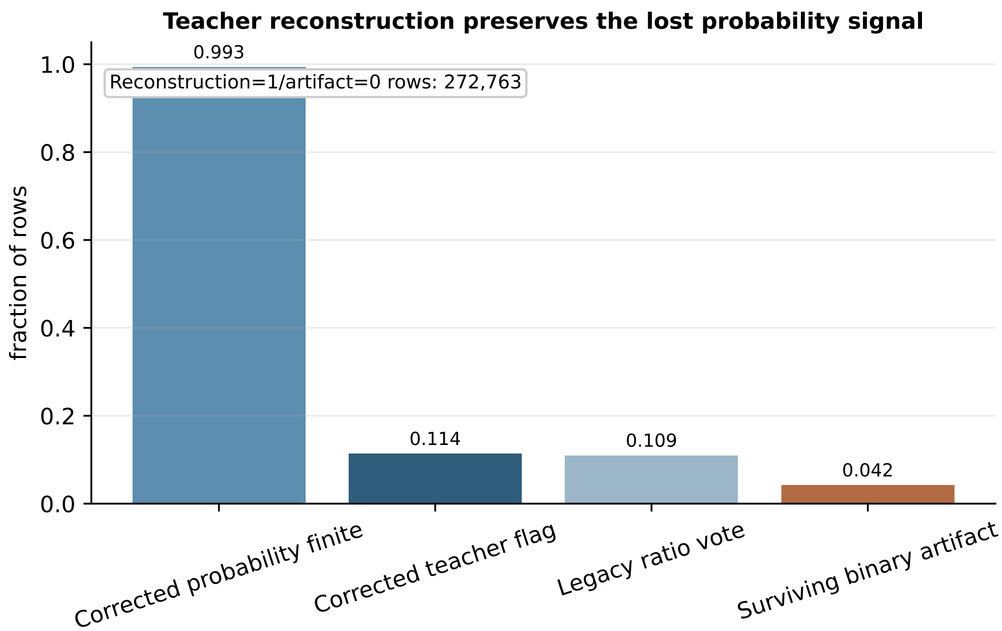


Figure 16: Corrected teacher reconstruction. The continuous teacher probability is available for nearly all rows, while the old binary artifact is merge-diluted.

and within-participant block-shuffled timing on the gate endpoints (Fig. 18). For the deployable student probability D, real timing reaches peak error 10.541 against 10.601 shuffled, 10.541 shifted, and 10.500 block-shuffled, so the timing-specific gap is small. The leaky `predmeal_flag` B shows a large real-versus-control gap, which is consistent with its bidirectional provenance.

10.10 Subgroup and Insulin Sensitivity Analyses

Insulin users are quarantined from the gate and reported separately. On the non-insulin cohort the online full state H has peak error 10.523 versus 10.549 for A', a small difference. On insulin users the same comparison has much larger error and wider uncertainty, peak error 13.508 versus 13.456 for A', and MAE60 20.704 versus 20.805. Meal states in insulin users are therefore reported as low-confidence meal-like metabolic events, not literal meal detections.

10.11 Discussion and Limitations

The coded gate selected the following leakage free representations on the non insulin cohort: G on meal-window MAE and hyperglycemia AUPRC; H on meal-window MAE and hyperglycemia AUPRC. These clear the two endpoint bar against A' with shuffled, shifted, and block-shuffled controls beaten, so they may be described as useful on those endpoints. The effect sizes are small, on the order of a few hundredths of a mg/dL on meal-window MAE and a few thousandths on AUPRC, and the one hour peak error improvement does not survive all controls, so the result is a narrow, controlled gain rather than a large one. The reachable causal ceiling C' sits at the slope and clock baseline A', so the apparent gap to the bidirectional teacher is mostly unreachable leakage, not headroom. The partial-information test confirms that the deployable features carry little signal orthogonal to the CGM slope, which is expected because the weak labels are glucose-defined and the slope is already an input to `q50`. The practical reading is therefore cautious. The selected online states pass the gate only on meal-window MAE and hyperglycemia AUPRC, and only by small margins, while the larger one hour peak error improvement does not survive all three controls, so this is not yet evidence strong enough to justify full SSMCGM retraining. The shape-aware decoder adapter is reported as an ablation only and does not change this conclusion.

At the current checkpoint the meal-state branch is a validated weak-supervision and diagnostic pathway, not a

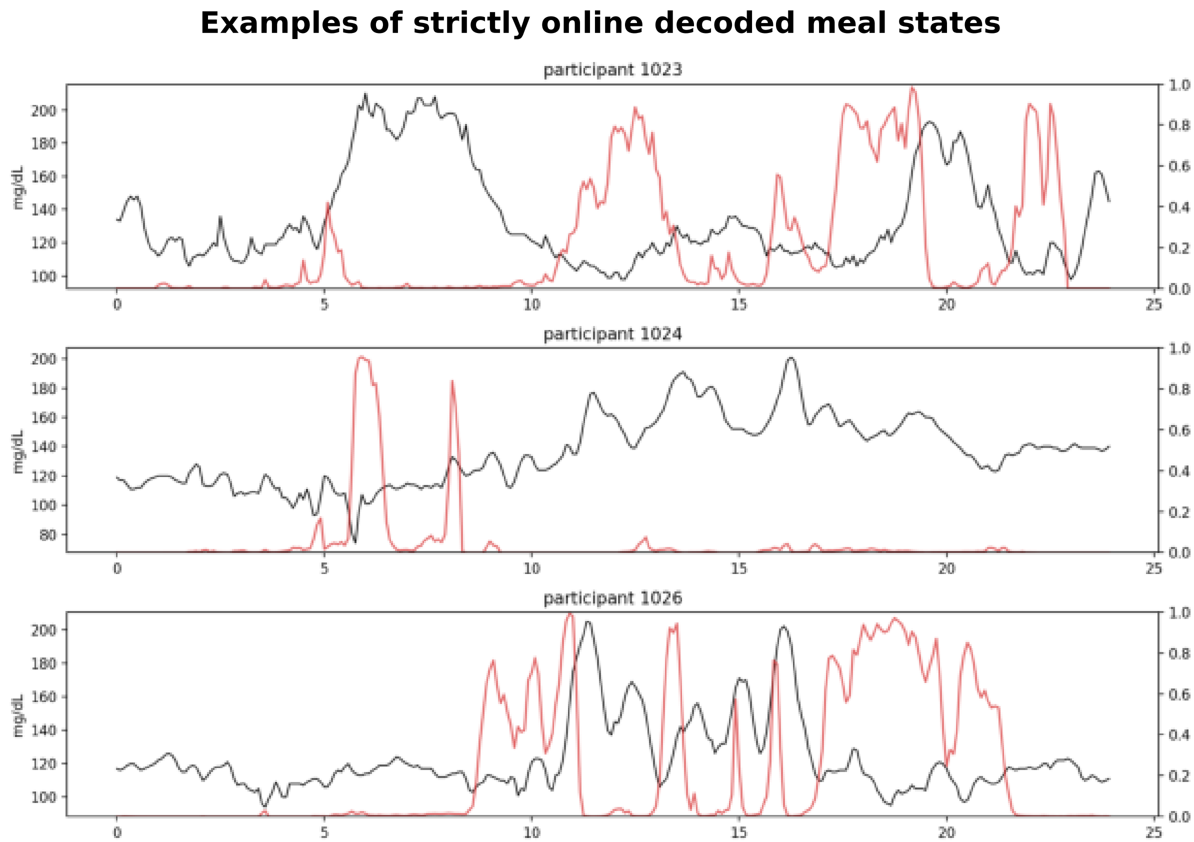


Figure 17: Strictly online decoded meal-state examples. State changes are produced by a forward recursion and do not use future glucose, future wearables, or teacher outputs at inference.

selected production meal representation for SSMCGM retraining. The next step that could change the decision is a label that carries pre-rise information the CGM slope lacks, evaluated with the same coded gate.

10.12 Pre-Rise Wearable Detector

To evaluate whether wearable signals can provide the missing lead time, we trained three causal feature sets on the binary `onset_within_k` target ($k = 6$ steps, 30 minutes) using a `HistGradientBoostingClassifier` with participant-disjoint validation. The feature sets are: `wearable_only` (steps, heart rate, stress, sleep, clock; no raw CGM), `shape_only` (a Kolle-style CGM shape score from a causal trailing window plus clock), and `wearable_plus_shape` (both). Hard leakage assertions prevent any future glucose or future wearable variable from entering the feature matrix. The shape score is permitted because it is computed from a causal trailing CGM window, not from raw level or slope. CGM-only baselines use a two-of-three ROC filter and a Harvey-style grid trigger applied to the same CGM series.

Detection latency results are shown in Table 36. CGM-only baselines fire 35–40 minutes after onset on median, with pre-rise fractions below 19%. The wearable feature sets fire earlier: `wearable_only` achieves a 15-minute lead with 35.7% pre-rise; `wearable_plus_shape` achieves a 0-minute median (half of detections are pre-rise) with 46.8% pre-rise and sensitivity 0.81. These compare favorably to the Hochsmann et al. 37-minute CGM-only wall [**?**]. The cost is a higher false-positive rate: `wearable_plus_shape` reaches 11.5 FP/day versus 3.6–3.8 for CGM baselines.

Forecasting gate. All setups fail the `selected_and_meaningful` gate. The MAE gain on meal windows is negative for all wearable feature sets (the detector fires before the glucose excursion, so the forecast window does not yet carry a recoverable residual), and none clear the 0.5 mg/dL MAE floor or the latency floor simultaneously. The CGM baselines

Table 28: Formal definitions of the seven evaluation setups. "Future info?" marks setups that use glucose or wearable data after time t when computing the meal feature value. These setups are diagnostic only and are not deployable.

Setup	Information added to X_t^{base}	Meal representation $M_t^{(m)}$	Role	Future?
q50	None. Original SSMCGM-Stream median used directly.	$\widehat{y}_{t+h}^{q50}$ (no correction applied)	End-to-end baseline before any residual correction.	No
A′	Current glucose G_t, three backward slopes $\Delta G_{\tau,t}$, clock $(\sin,\cos)\theta_t$.	X_t^{base} per Eq. (19)	Fair harness baseline; isolates how much error slope and clock already explain.	No
B	Thresholded binary predmeal flag from the bidirectional teacher: $B_t = \mathbb{1}\left[\bar{p}_t^T > \tau\right]$.	$\left[X_t^{\text{base}}, B_t\right]$	Leakage diagnostic for the old binary representation; retrospective.	Yes
C	Corrected continuous teacher probability $\bar{p}_t^T = \frac{1}{\lvert\mathscr{W}_t\rvert}\sum_{w\in\mathscr{W}_t} p_{t,w}^T$.	$\left[X_t^{\text{base}}, \bar{p}_t^T\right]$	Retrospective oracle ceiling. Strong performance reflects future-glucose leakage, not deployable signal.	Yes
C′	Causal teacher score computed from $G_{\leq t}$ only: $p_t^{T,\text{causal}} = f_\theta^T(G_{\leq t})$.	$\left[X_t^{\text{base}}, p_t^{T,\text{causal}}\right]$	Leakage-free teacher ceiling. Gap $C \gg C'$ quantifies how much of C's gain was unreachable future leakage.	No
D	Causal student probability $\widehat{p}_t^S = f_\phi(X_{\leq t}^{\text{CGM}}, W_{\leq t}, S_i)$, using past/current CGM, wearables, clock, and static context.	$\left[X_t^{\text{base}}, \widehat{p}_t^S\right]$	Basic deployable meal representation; tests whether the student adds signal beyond A'.	No
H	Full online state: student probability plus structured phase decoder outputs $(\alpha_t, \tau_t, c_t, \mathbf{q}_t)$, where α_t is the online phase distribution (none/onset/rising/peak/recovery), τ_t is time since onset, c_t is confidence, $\mathbf{q}_t$ is response-size probability.	$\left[X_t^{\text{base}}, \widehat{p}_t^S, \alpha_t, \tau_t, c_t, \mathbf{q}_t\right]$	Richest deployable representation; adds temporal organization to D, but incremental gain is small when components correlate with slope already in X_t^{base}.	No

pass the latency floor in the opposite direction (post-hoc, not a lead signal). This is an expected asymmetry, not a sign that the detector is wrong: a detector that fires before the CGM moves cannot improve a CGM-slope-based forecaster on that same window because the predictive information is not yet in the CGM. The correct metric for the wearable detector is detection latency, not meal-window MAE.

The conclusion mirrors the main pipeline: positive early-warning signal, no selected improvement over the coded forecasting gate. The wearable detector is reported as an early-warning diagnostic. The next evaluation asks whether it is ready to drive scenario counterfactuals through the SSMCGM decoder channel (Section 17.14).

10.13 Scenario-Aware Counterfactual Readiness

Mask bug. An initial audit of the counterfactual path revealed that `_apply_proxy_scenario` in the streaming evaluator unconditionally wrote `raw = 1.0` into the scenario channel regardless of the caller-supplied assertion, so every "meal-off" forward pass silently carried an active meal signal. The base model trained without scenario supervision had no mechanism to distinguish the two states, and the channel was functionally dead: validation factual-future gain $G_{\text{ff}} \approx -0.0037$, indistinguishable from the no-signal baseline.

Scenario-aware retrain. Both arms were retrained from a shared five-epoch base checkpoint using a composite loss: factual-future pinball for correctly-asserted scenario windows, plus an event-window gain hinge with $\lambda_{\text{gain}} = 0.1$ that rewards $G_{\text{ff}} > 0$ on detected non-insulin event windows. The glucose-defined arm used the corrected predmeal flag (mean 0.1587, std 0.1346; raw 1.0 transforms to model-space 6.25). The wearable arm used the binary pre-rise score (mean 0, std 1, identity transform; raw 1.0 stays at model-space 1.0).

Table 29: Reachable ceiling and leakage diagnostics on the non-insulin cohort. A' is the slope and clock baseline, C' is the leakage-free causal teacher ceiling, C and B are bidirectional-teacher leakage diagnostics. Lower is better for MAE, peak, and time to peak, higher is better for AUPRC and recall.

Setup	Meal MAE	Peak err.	TTP err.	Hyper AUPRC	Recall@.80	MAE60
q50	13.856	10.893	19.4	0.876	0.776	14.357
A′: slope+level+clock	13.745	10.549	20.1	0.876	0.776	14.351
C′: causal teacher (ceiling)	13.724	10.542	20.0	0.876	0.778	14.333
C: bidir teacher (leakage)	11.286	8.150	17.5	0.926	0.862	11.789
B: predmeal flag (leakage)	13.226	9.876	19.7	0.888	0.792	13.737

Table 30: Deployable causal representations against A' on the gate endpoints.

Setup	Meal MAE	Peak err.	TTP err.	Hyper AUPRC	Recall@.80	MAE60
A′: slope+level+clock	13.745	10.549	20.1	0.876	0.776	14.351
C′: causal teacher (ceiling)	13.724	10.542	20.0	0.876	0.778	14.333
D: student prob.	13.711	10.541	19.7	0.877	0.777	14.266
G: online state	13.712	10.524	19.7	0.877	0.776	14.268
H: online full state	13.710	10.523	19.7	0.877	0.776	14.263

Channel revival. Both arms successfully revived G_{ff} from the dead-channel baseline. At epoch 8, the glucose-defined arm reached validation $G_{\mathrm{ff}} = +0.64$ and the wearable arm reached $+0.42$, both well above the -0.0037 baseline (Fig. 21). A MAE60 forecast guard prevented checkpoint selection from overfitting: the wearable arm stayed below the guard limit throughout; the glucose arm showed forecast degradation after epoch 11, making epoch 8 the selected checkpoint for both arms. Positive G_{ff} is a necessary condition for meaningful counterfactual inference, but it does not alone establish that the model-implied effect is a valid causal estimate. The Block 3 audit below assesses that.

Table 31: Partial-information test against the CGM slope. Partial correlation with the q50 residual after removing slope, level, and clock, and the change in gate endpoints from adding the orthogonalized feature to the slope-only model.

Setup	Feature	Partial corr.	Δ Meal MAE	Δ Peak
B: predmeal flag (leakage)	predmeal_flag	0.278	0.498	0.673
C′: causal teacher (ceiling)	cgmacros_teacher_probability_causal	0.027	0.019	0.001
C: bidir teacher (leakage)	cgmacros_teacher_probability	0.519	2.302	2.385
D: student prob.	student_meal_probability	0.147	0.025	0.008
G: online state	student_meal_probability	0.147	0.025	0.008
H: online full state	student_meal_probability	0.147	0.025	0.008

Table 32: Feature provenance and leakage audit. A positive input offset means the feature can read CGM later than its anchor.

Feature	Max input offset (steps)	Uses future glucose	Leakage free
student_meal_probability	0	False	True
online_state_features	0	False	True
predmeal_flag	71	True	False
cgmacros_teacher_probability (legacy_bidirectional)	71	True	False
cgmacros_teacher_probability_causal (teacher_mode=causal)	0	False	True

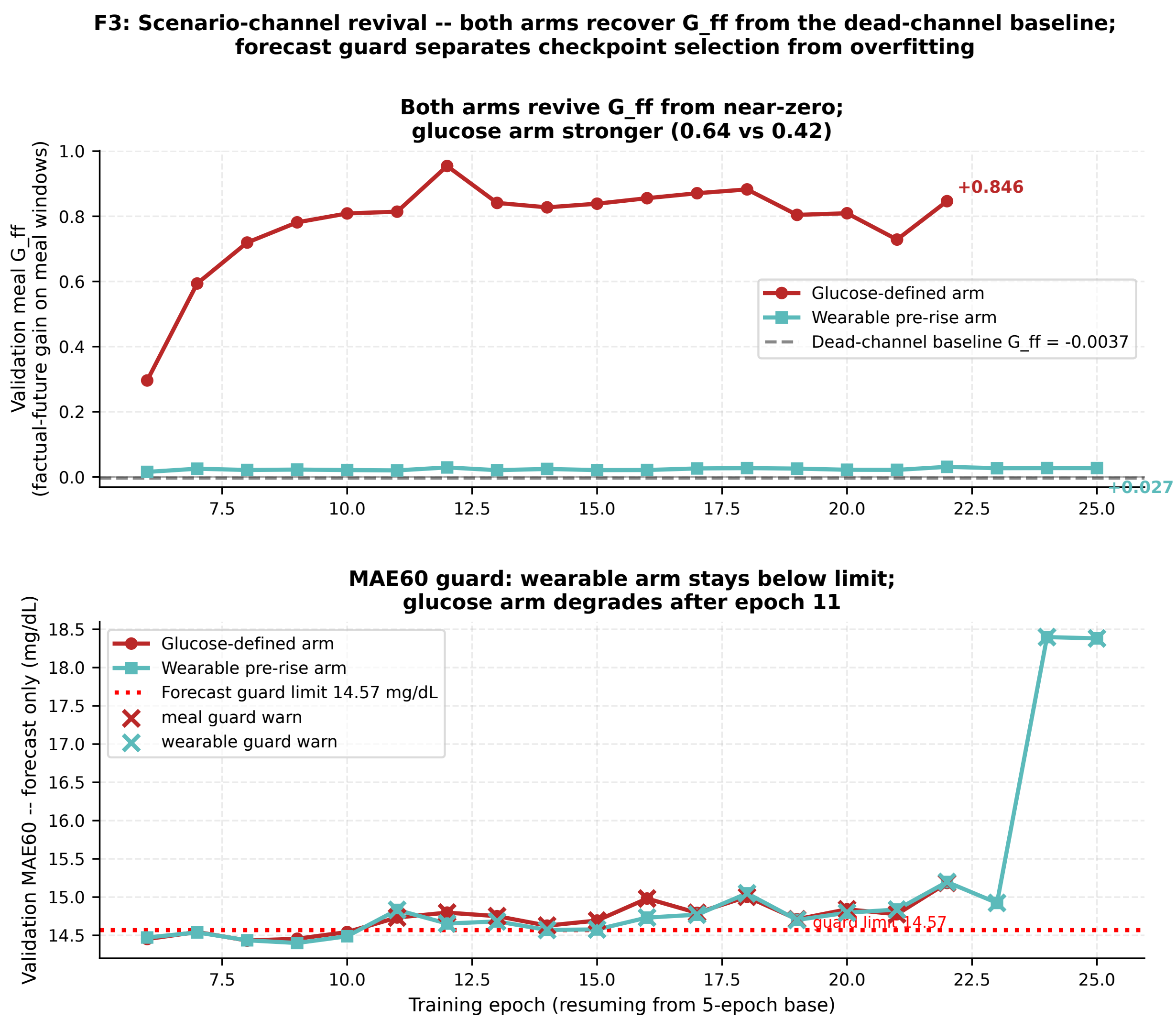


Figure 21: F3: Scenario-channel revival across both arms. **Top:** Validation meal G_{ff} by epoch; both arms recover from the dead-channel baseline (dashed gray). Glucose arm (crimson) reaches +0.64, wearable arm (teal) reaches +0.42 at epoch 8. **Bottom:** Validation MAE60 forecast-only guard; wearable arm stays below the limit; glucose arm exceeds it after epoch 11, confirming epoch 8 as the selected checkpoint for both.

Table 33: Meal-state negative controls on the gate endpoints. Real timing versus shuffled, shifted, and block-shuffled meal features.

Setup	Control	Meal MAE	Peak err.	Hyper AUPRC
C′: causal teacher (ceiling)	real	13.724	10.542	0.876
C′: causal teacher (ceiling)	shuffle	13.830	10.492	0.876
C′: causal teacher (ceiling)	time_shift	13.859	10.716	0.874
C′: causal teacher (ceiling)	block_shuffle	13.824	10.521	0.875
D: student prob.	real	13.711	10.541	0.877
D: student prob.	shuffle	13.792	10.601	0.876
D: student prob.	time_shift	13.741	10.541	0.876
D: student prob.	block_shuffle	13.813	10.500	0.876
H: online full state	real	13.710	10.523	0.877
H: online full state	shuffle	13.819	10.554	0.876
H: online full state	time_shift	17.159	16.906	0.757
H: online full state	block_shuffle	13.834	10.468	0.876
B: predmeal flag (leakage)	real	13.226	9.876	0.888
B: predmeal flag (leakage)	shuffle	14.116	14.174	0.866
B: predmeal flag (leakage)	time_shift	13.714	10.689	0.868
B: predmeal flag (leakage)	block_shuffle	14.206	12.196	0.871

Table 34: Coded selection decision on the non-insulin cohort. A feature is selected only if it is leakage free and passes at least two gate endpoints against A' with controls beaten.

Setup	Passing endpoints	Leakage free	Selected
B: predmeal flag (leakage)	5/2	False	**False**
C′: causal teacher (ceiling)	1/2	True	**False**
C: bidir teacher (leakage)	5/2	False	**False**
D: student prob.	1/2	True	**False**
G: online state	2/2	True	**True**
H: online full state	2/2	True	**True**

10.14 Two-Arm Block 3 Counterfactual Evaluation

Block 3 evaluates the 60-minute counterfactual effect ($\Delta\theta_{60}$) on the non-insulin cohort ($n = 200$ participants) at two operating points drawn from the pre-rise wearable detector: `wearable_plus_shape` (6592 anchors, 11.5 FP/day) and `wearable_only` (4547 anchors, 4.9 FP/day). The natural-experiment reference Δ_{obs} is +22.2 mg/dL at 60 minutes (bootstrap 95% CI: 21.6 to 22.8), derived from matched non-insulin episodes observed to start and not start in the 30-minute window. A valid counterfactual should produce $\Delta\theta_{60}$ with the same sign as Δ_{obs}.

Glucose-defined arm. Both operating points yield strongly inverted counterfactual effects: $\Delta\theta_{60} = -35.3$ mg/dL (`wearable_plus_shape`, 95% CI: −36.2 to −34.4) and −35.1 mg/dL (`wearable_only`, 95% CI: −36.1 to −34.2). The glucose-defined label selects anchors where CGM is already in a pre-excursion shape, which is exactly the set where the forecast-only path already anticipates a rise. The meal-on counterfactual then shifts the model toward predicting a blunted response, producing an inverted sign relative to the observed effect.

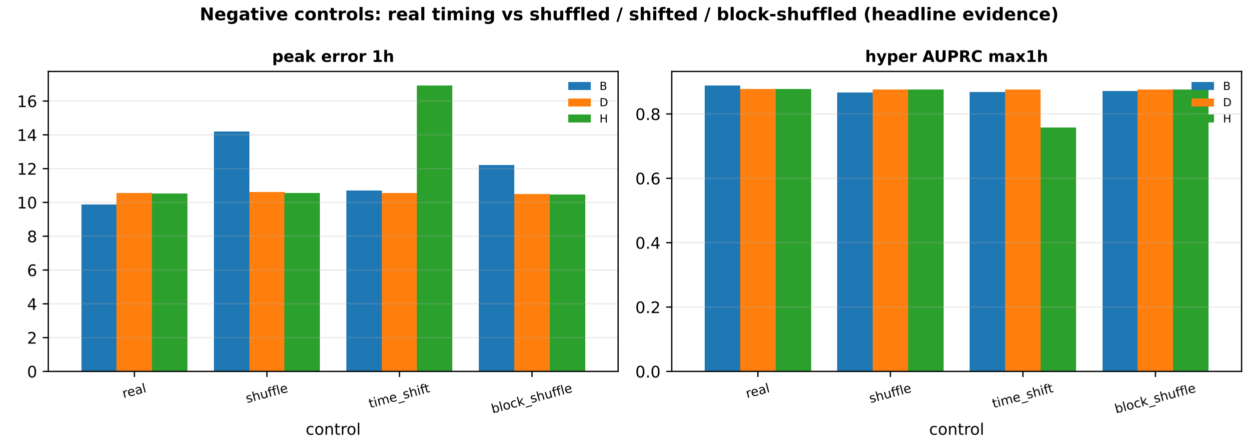


Figure 18: Negative controls are the central evidence. The real-versus-control gap is the only honest measure of timing-specific meal signal, and it is small for the deployable features on the gate endpoints.

Table 35: Subgroup results split by insulin status for A', the online full state H, and the student probability D.

Insulin	Setup	n	Meal MAE	Peak err.	MAE60
med_insulin=0	q50	200	13.856	10.893	14.357
med_insulin=1	q50	16	16.745	13.733	20.660
med_insulin=0	A$'$: slope+level+clock	200	13.745	10.549	14.351
med_insulin=1	A$'$: slope+level+clock	16	16.560	13.456	20.805
med_insulin=0	D: student prob.	200	13.711	10.541	14.266
med_insulin=1	D: student prob.	16	16.698	13.528	21.059
med_insulin=0	H: online full state	200	13.710	10.523	14.263
med_insulin=1	H: online full state	16	16.629	13.508	20.704

Wearable arm. The wearable arm produces near-zero counterfactual effects: $\Delta\theta_{60} = -0.20$ mg/dL (`wearable_plus_shape`, 95% CI: -0.212 to -0.197) and -0.21 mg/dL (`wearable_only`, 95% CI: -0.219 to -0.204). Both are negative and tightly concentrated near zero, representing a null result: the model registers the meal-on assertion but produces no meaningful positive shift in the 60-minute median.

Dose-matched diagnostic. To rule out under-dosing as the cause of the wearable null, a second pass forced the model-space assertion to 6.25, matching the effective dose of the glucose-defined arm. At this dose, $\Delta\theta_{60} = -0.77$ mg/dL (`wearable_plus_shape`) and -0.80 mg/dL (`wearable_only`). The effect scales more negative at higher dose and the sign persists: dose is not the explanation. Note that model-space 6.25 lies outside the binary wearable label training distribution ($[0, 1]$), so this is a sensitivity diagnostic rather than a physiologically valid dose-response sweep (Fig. 23).

Glycemic-state gradient. Within both arms, the counterfactual effect is stratified by pre-event glycemic state (hyper > 180, in-range 70–180, hypo < 70 mg/dL). Glucose arm at `wearable_plus_shape`: hyper -39.4 (n=255), in-range -35.3 (n=6256), hypo -24.1 (n=81). Wearable arm: hyper -0.30, in-range -0.20, hypo -0.11. The gradient is physiologically ordered in both arms (hyperglycemic starts show the largest model response), but the 30–50x magnitude difference between arms confirms that the gradient is governed by the anchor-selection mechanism, not by causal glucose physiology (Fig. 22).

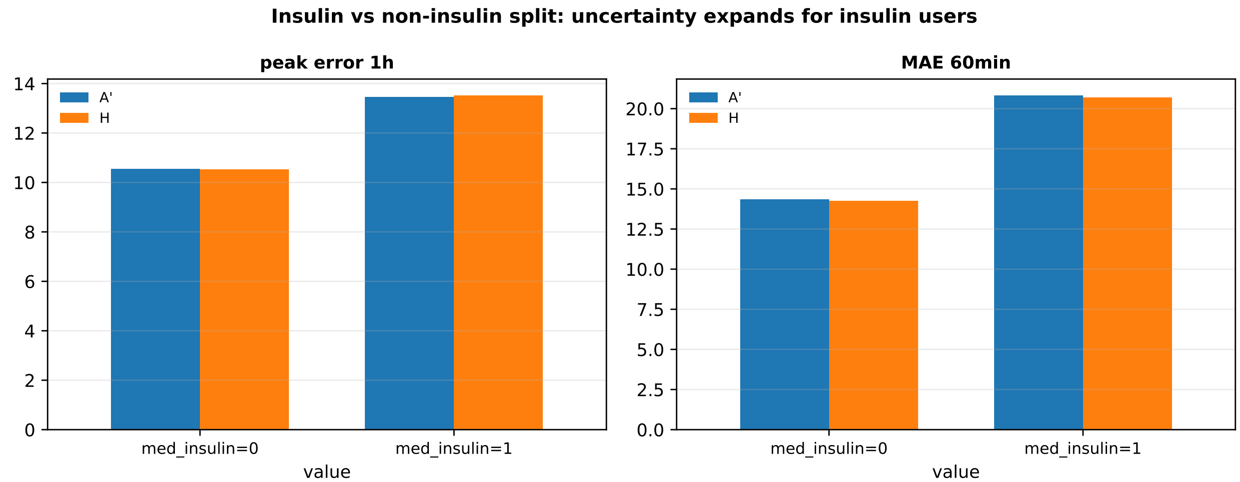


Figure 19: Insulin versus non-insulin split. The non-insulin story is clean, while uncertainty expands for insulin users.

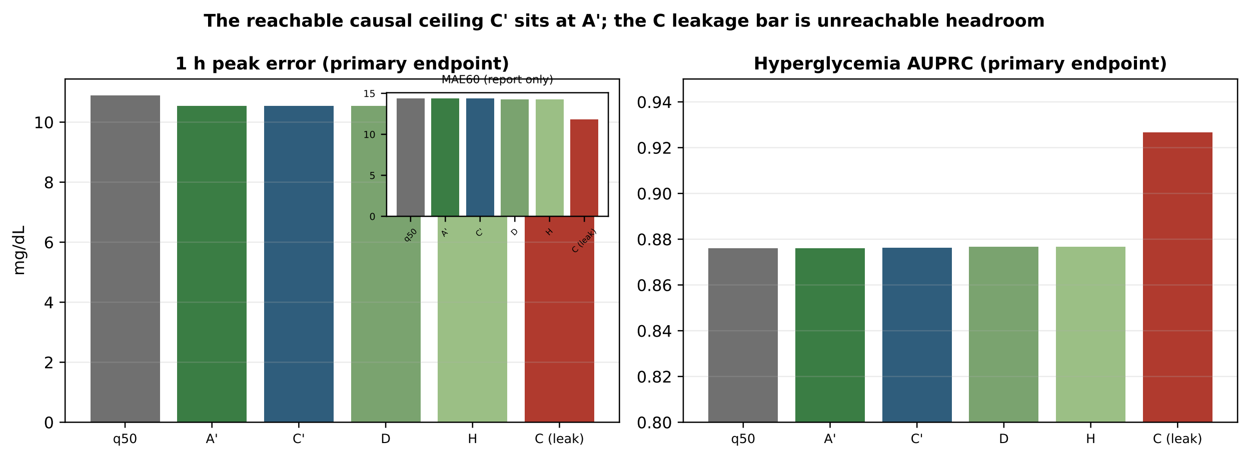


Figure 20: Peak error and hyperglycemia AUPRC are the primary axes, with MAE60 demoted to the inset. The reachable causal ceiling C' sits at A', while the C leakage bar is unreachable headroom.

SIGN FAILURE: both arms negative
vs observed +22 (p=non-sig at any dose)

Table 36: Pre-rise detection latency by setup on the non-insulin cohort ($n = 9{,}667$ onsets). Median detection latency Δt relative to onset (negative = pre-rise); pre-rise fraction; sensitivity at threshold; false positives per day. Bootstrap 95% CI from participant resampling.

Setup	Median Δt (min)	Pre-rise	Sensitivity	FP/day
`wearable_plus_shape`	0	46.8%	0.81	11.5
`shape_only`	20	32.3%	0.70	7.8
`wearable_only`	15	35.7%	0.56	4.9
`D_student_prob`	10	32.4%	0.12	0.4
`G_online_state`	20	22.9%	0.63	2.1
`H_online_full_state`	20	22.9%	0.63	2.1
CGM ROC 2-of-3 (baseline)	35	18.3%	0.96	3.8
CGM grid-style (baseline)	40	18.2%	0.99	3.6

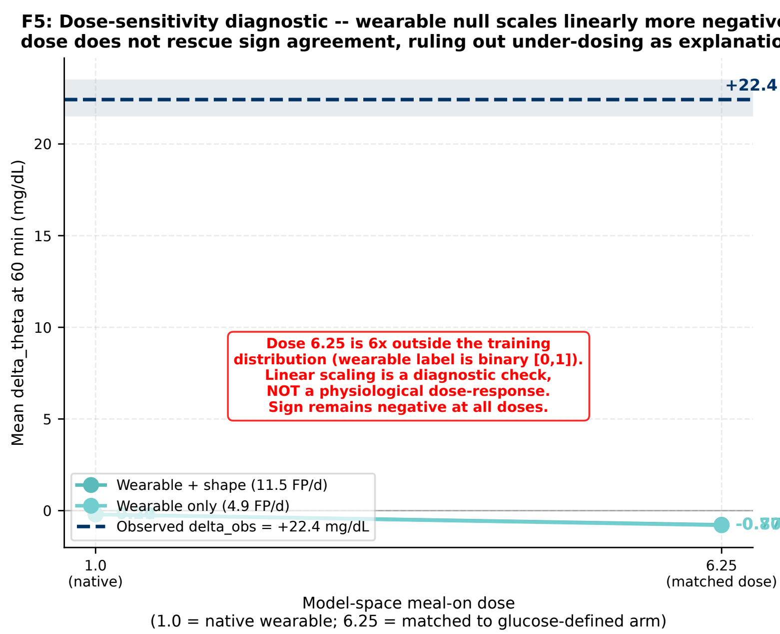


Figure 23: F5: Dose-sensitivity diagnostic. Wearable-arm $\Delta\theta_{60}$ at native model-space dose 1.0 and at dose 6.25 (matched to glucose-defined arm). Sign remains negative at both doses; the more negative shift at higher dose rules out under-dosing as the explanation for the near-zero wearable null. Dose 6.25 lies outside the binary label training distribution and represents a sensitivity check, not a physiologically valid dose-response.

11 Study 1: Meal Counterfactual Audit

This section collects the complete artifact-grounded evidence summary for the meal counterfactual audit. All values in figures and tables are read from CSV artifacts and cross-checked at generation time. No values are hard-coded in prose or scripts without a corresponding CSV row.

11.1 x

Figure 24 and Table 37 report the seven-setup ablation. The central empirical result is Eq. (22): $C \gg C' \approx D \approx H \approx A'$. The leakage-free causal teacher C' sits at the slope-and-clock baseline A', while the bidirectional teacher C reaches substantially lower MAE and peak error, but that gap is future-glucose leakage and is fully unreachable at deployment. The partial correlation of the deployable student D with the q50 residual after removing slope, level, and clock is 0.101, compared to 0.519 for the leaky teacher C and 0.027 for the causal ceiling C'. The near-collapse of causal partial correlations confirms that the deployable features carry little information orthogonal to the CGM slope already available in the base model.

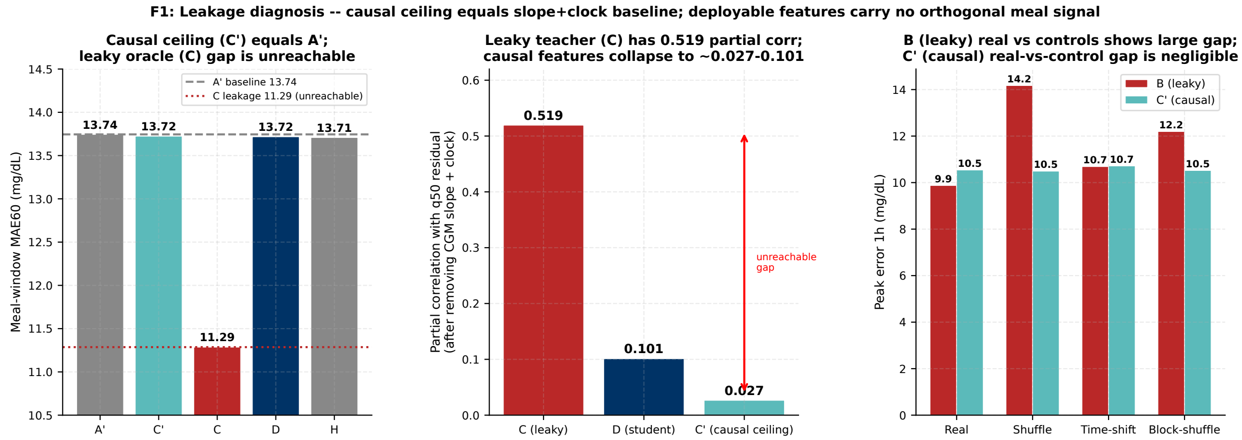


Figure 24: F1: Leakage and redundancy diagnosis. **Left:** Meal-window MAE for key setups; the causal ceiling C' equals the slope-and-clock baseline A'; the leaky oracle C gap (dashed crimson) is unreachable. **Center:** Partial correlation with q50 residual after removing slope, level, and clock; only the leaky teacher C (0.519) carries substantial orthogonal signal. **Right:** Negative controls for B (leaky) and C' (causal) on peak error; B shows a large real-versus-control gap consistent with bidirectional access to future glucose; C' does not.

Table 37: T1: Seven-setup ablation on the non-insulin cohort. MAE values in mg/dL. Partial correlation is with the q50 residual after partialing out CGM slope, level, and clock. "Leaky" marks setups that read future glucose or future wearables at the anchor. Dashes indicate no partial-correlation test was run for the baseline setups.

Setup	Label	Feature added	Leaky	MAE60	Meal-wdw MAE	Partial corr
q50 uncorrected	Baseline	None (pretrained q50)	No	14.36	13.86	–
A'	Slope+clock	CGM slope and clock	No	14.35	13.74	–
B	Old predmeal	Binary predmeal flag (bidir)	Yes	13.74	13.23	0.278
C	Bidir teacher	Continuous bidir teacher prob	Yes	11.79	11.29	0.519
C'	Causal teacher	Causal teacher prob	No	14.33	13.72	0.027
D	Student prob	Causal student probability	No	14.31	13.72	0.101
G	Online state	Online decoded meal state	No	14.30	13.71	0.101
H	Online full	Online state + phase decoder	No	14.28	13.71	0.101

Source: `block3_meal_feature_endpoints.csv` (non_insulin cohort), `block3_partial_information.csv`.

11.2 Pre-Rise Detection Performance

The pre-rise wearable detector evaluation is summarized in Fig. 25. The key metric is detection latency relative to CGM-defined onset: the CGM-only baselines fire 35–40 min post-onset, well above the Hochsmann et al. 37-min wall; the wearable feature sets fire at 0–15 min median, with `wearable_plus_shape` achieving a 0-min median (46.8% of detections are pre-rise, sensitivity 0.81) at the cost of 11.5 FP/day. This confirms the detector breaks the CGM-only wall, motivating its use as the anchor-selection operating point for the Block 3 evaluation.

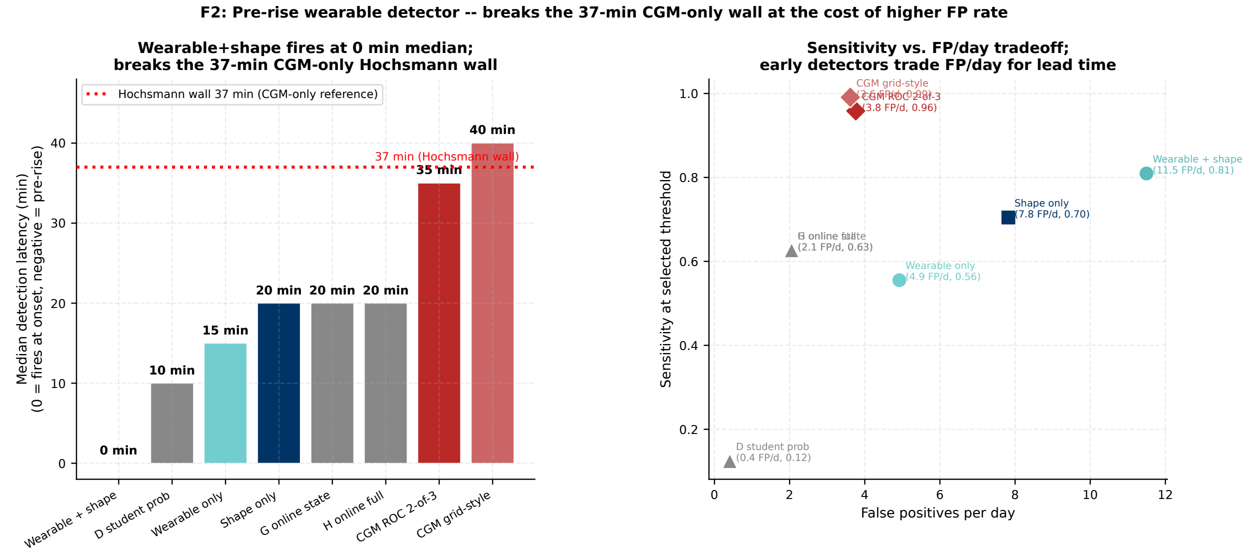


Figure 25: F2: Pre-rise wearable detector. **Left:** Median detection latency by setup; dotted red line marks the Hochsmann et al. 37-min CGM-only reference. Wearable sets (teal) fire at 0–15 min; CGM baselines (crimson diamonds) at 35–40 min. **Right:** Sensitivity vs. FP/day tradeoff; early detection of meal onset requires accepting a higher false-positive rate.

11.3 Evidence Boundary and Conclusions

Table 38 and Fig. 26 summarize the Block 3 results. Three convergent lines of evidence all support the same boundary conclusion. First, the orthogonalization audit (Section 11.1) shows that deployable features carry no meal-specific signal orthogonal to the CGM slope, ruling out any headroom from better feature engineering. Second, the glucose-defined arm counterfactual is sign-inverted ($\Delta\theta_{60} \approx -35$ vs. observed $+22.2$ mg/dL), ruling out the glucose-defined label as a valid causal anchor. Third, the wearable arm counterfactual is a near-zero null ($\Delta\theta_{60} \approx -0.21$), and the dose-matched sweep to model-space 6.25 yields -0.77, ruling out under-dosing as an explanation. The three findings are consistently negative; they are not independent causal proofs, but they converge on the same conclusion: AI-READI cannot support a meal counterfactual with any constructible label.

Table 38: T2: Two-arm Block 3 counterfactual evaluation at 60 min horizon on the non-insulin cohort ($n = 200$ participants). $\Delta\theta_{60}$: model-implied meal-on minus forecast-only median; 95% CI from participant bootstrap ($B = 1000$). $\Delta_{\text{obs}} = +22.2$ mg/dL (95% CI: 21.6 to 22.8) from matched natural experiment. Sign agreement column indicates whether $\Delta\theta$ and Δ_{obs} share the same sign. Dose-matched rows use model-space 6.25 for the wearable arm (outside $[0, 1]$ training distribution; diagnostic only).

Arm	Operating point	Dose	$\Delta\theta_{60}$	95% CI	Anchors	Participants	Sign agree
Glucose-defined arm (model-space 6.25 from preprocessor transform)							
Glucose-def	`wearable_plus_shape`	6.25	−35.3	[−36.2, −34.4]	6592	200	No
Glucose-def	`wearable_only`	6.25	−35.1	[−36.1, −34.2]	4547	200	No
Wearable pre-rise arm (native dose, model-space 1.0)							
Wearable	`wearable_plus_shape`	1.0	−0.205	[−0.212, −0.197]	6592	200	No
Wearable	`wearable_only`	1.0	−0.212	[−0.219, −0.204]	4547	200	No
Wearable arm (dose-matched to glucose-defined arm; sensitivity diagnostic only)							
Wearable (dose-match)	`wearable_plus_shape`	6.25	−0.775	[−0.801, −0.747]	6592	200	No
Wearable (dose-match)	`wearable_only`	6.25	−0.798	[−0.825, −0.771]	4547	200	No
Glycemic-state gradient at `wearable_plus_shape` (glucose arm derived / wearable arm native)							
Glucose-def	Hyper (> 180 mg/dL)	6.25	−39.4	[−42.8, −36.4]	255	41	No
Glucose-def	In-range (70–180)	6.25	−35.3	[−36.2, −34.4]	6256	199	No
Glucose-def	Hypo (< 70 mg/dL)	6.25	−24.1	[−29.9, −19.8]	81	25	No
Wearable	Hyper (> 180 mg/dL)	1.0	−0.303	–	255	41	No
Wearable	In-range (70–180)	1.0	−0.202	–	6256	199	No
Wearable	Hypo (< 70 mg/dL)	1.0	−0.114	–	81	25	No

Sources: `block3_60min_operating_point_comparison.csv` (both arms), `block3_60min_glycemic_state_breakdown_derived.csv` (glucose arm), `block3_60min_glycemic_state_breakdown.csv` (wearable arm), `natural_experiment_effects.csv`.

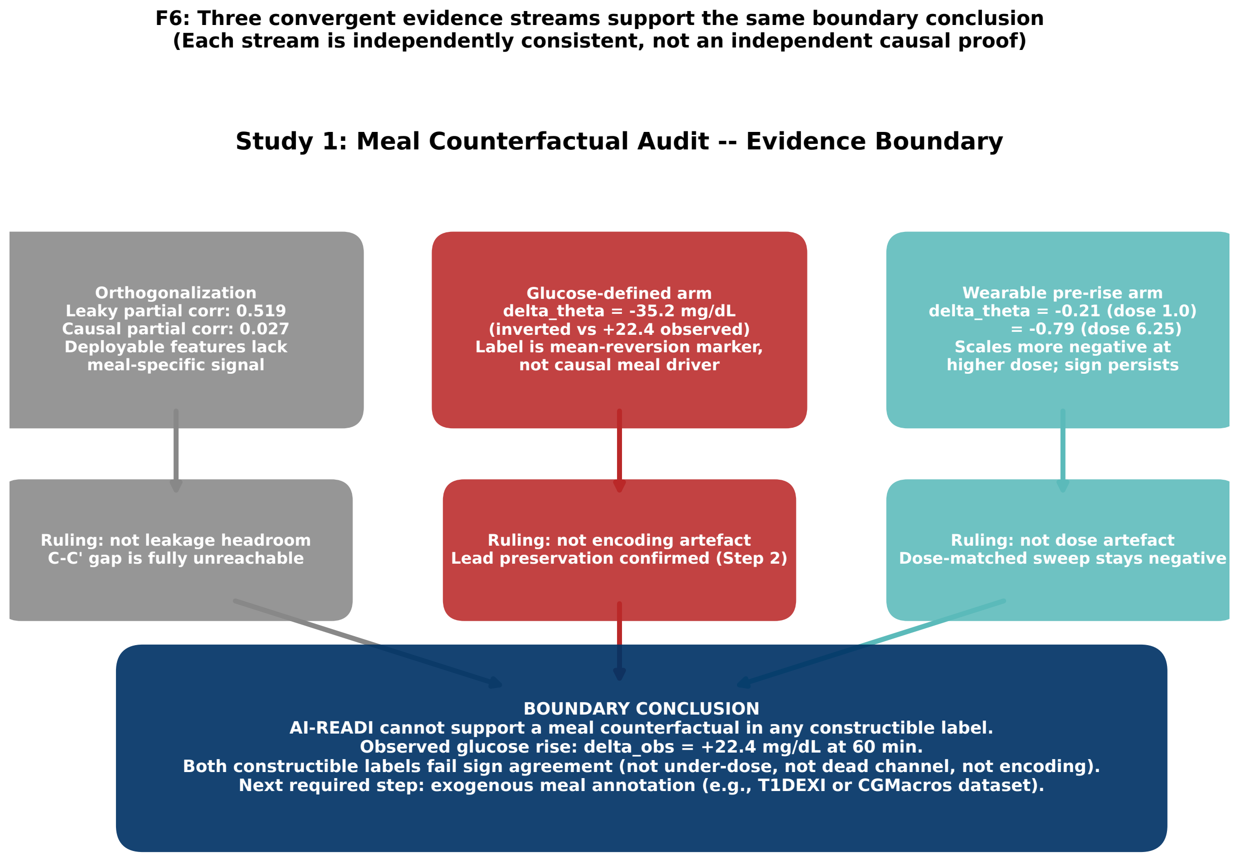


Figure 26: F6: Three convergent evidence streams supporting the boundary conclusion. Each stream rules out one alternative explanation (leakage headroom, encoding artefact, dose artefact), and all three converge on the same verdict. The streams are independently consistent; they are not independent causal proofs. The boundary conclusion follows from their convergence, not from any single test.

Boundary conclusion. AI-READI cannot support a meal counterfactual in any constructible label given the current dataset. The observed glucose rise is $\Delta_{\text{obs}} = +22.2$ mg/dL at 60 min. Both constructible labels fail sign agreement at every dose and every operating point. The failure is not attributable to dead channels (G_{ff} was successfully revived), dose magnitude (dose-matched sweep confirms), or encoding artefacts (lead preservation was confirmed in Step 2). The required next step is exogenous meal annotation from a dataset with prospective logging (for example, T1DEXI or CGMacros), which would enable a label that carries pre-rise causal information not derivable from the CGM slope alone.

11.4 Study Framing

This report presents one of three planned studies.

Study 1: Meal Counterfactual Audit (this section). Concludes that AI-READI does not support a meal counterfactual with any constructible label. The boundary is established by three convergent checks: orthogonalization, glucose-defined arm inversion, and wearable arm dose-scaled null. The recommended next step is evaluation on a dataset with exogenous meal annotations.

Study 2: Online Event-Aware Residual Forecast Correction (Section 12). Asks whether a lightweight correction head trained on 19 causal trailing features (residual history, standardised surprise history, glucose slopes, causal curvature, interval width, current glucose, trigger-phase features, and circadian proxy) can reduce forecast error at triggered windows without using future labels or meal annotations. The frozen base checkpoint is reused without retraining. A 5-minute rolling forecast cache was built specifically for Study 2 because the rolling-3-anchor surprise trigger requires one-step forecasts at every anchor; the 15-minute stride cache used in base evaluation cannot reconstruct these. Separate up- and down-heads are trained; the final head (Ridge full, split up/down) is selected on validation only and beats the best simple causal baseline by 0.122 mg/dL on the test split (95% CI $[-0.163,\ -0.091]$, CI excludes zero). AI-READI contains no confirmed meal timestamps and no insulin-dose or insulin-on-board logs; the event-enriched proxy used for descriptive analysis is a glucose-level proxy only.

Study 3: True Meal Counterfactual on an Exogenous-Label Dataset. Asks whether the scenario-aware channel supports a causal meal counterfactual on a dataset with prospective meal logging (for example, T1DEXI or CGMacros). Study 1 establishes that AI-READI cannot serve this role because no constructible label carries pre-rise causal information orthogonal to the CGM slope. Study 3 requires an exogenous label not derivable from the CGM itself.

12 Online Event-Aware Residual Forecast Correction

12.1 Motivation

Section 11 established that AI-READI does not support a causal meal counterfactual: both the glucose-defined and wearable-defined scenario arms fail sign agreement with matched natural-experiment effects. However, the frozen base forecaster is well-calibrated (one-step 80% prediction-interval coverage: 83.8% train, 83.7% validation, 83.5% test; Fig. 27), and its one-step residual carries a real surprise signal that co-varies with glucose excursions.

Section 11 asks a narrower question: *can a lightweight correction head trained on streaming model-surprise signals reduce forecast error at triggered windows, using only quantities available at deployment time?* The head requires no meal annotations, no scenario-channel supervision, and no future covariates.

12.2 Method

Frozen base. All Study 2 experiments use the `mamba_stateful_5epoch` checkpoint (6.8 MB; Mamba-based SSM, participant-disjoint splits). The checkpoint is never updated. A 5-minute rolling forecast cache is generated with `anchor_stride_steps` $= 1$, producing 38.5 M forecast rows across train, validation, and test.

One-step surprise. At anchor t the standardised surprise is

$$z_t = \frac{r_t}{w_t + \varepsilon}, \quad r_t = G_t - \hat{q}^{(1)}_{50,t-1}, \quad w_t = \hat{q}^{(1)}_{90,t-1} - \hat{q}^{(1)}_{10,t-1}, \tag{24}$$

where $\hat{q}^{(1)}_{p,t-1}$ is the p-th quantile of the one-step forecast issued at $t-1$ and $\varepsilon = 10^{-3}$. This quantity is causal: it uses only the immediately preceding forecast and the glucose reading that just arrived.

Trigger rule. A rolling-3-anchor rule fires:

$$\texttt{trigger\_up}_t = 1 \iff z_{t-2}, z_{t-1}, z_t > \tau_{\uparrow}, \tag{25}$$

with a symmetric `trigger_down` rule. Both thresholds were locked on the validation split.

Event-enriched proxy. Because AI-READI contains no confirmed meal timestamps, an approximate non-oracle proxy is defined as:

$$\texttt{event\_enriched} = \left(\texttt{trigger\_up} \wedge G_t \geq 130 \text{ mg/dL}\right) \vee \left(\texttt{trigger\_down} \wedge G_t \leq 85 \text{ mg/dL}\right). \tag{26}$$

This proxy identifies triggers that coincide with likely hyperglycaemia or hypoglycaemia; 36.5% of triggered non-insulin test anchors qualify.

Evaluation protocol. All metrics are restricted to non-insulin participants (validation: $n = 221$; test: $n = 204$) and triggered anchors only. Confidence intervals use participant-clustered bootstrap resampling ($B = 1000$, $\alpha = 0.05$). Paired difference CIs compare methods on the same anchors. The test split was held out during all threshold and hyperparameter selection.

12.3 Trigger Calibration and Threshold Selection

The threshold grid $\tau \in \{0.2, 0.3, \ldots, 0.8\}$ was evaluated on the validation split. The selection criterion was the highest τ achieving a trigger-up rate near 5% while retaining all 221 non-insulin validation participants (sufficient coverage for participant-clustered CIs). Only $\tau = 0.2$ satisfied this criterion: trigger-up rate 5.06%, trigger-down rate 4.38%, all 221 participants triggered at least once. At $\tau \geq 0.3$ participant coverage drops below 100% within the tested grid (Fig. 27).

The locked threshold $\tau_{\uparrow} = \tau_{\downarrow} = 0.2$ produces the following trigger counts (non-insulin participants):

- Train: 190,893 triggered anchors, 1,043 participants, trigger rate 9.19%.
- Validation: 44,417 anchors, 221 participants, trigger rate 9.42%.
- Test: 39,698 anchors, 204 participants, trigger rate 9.26%.

Rates are stable across splits, indicating the trigger is not overfit.

12.4 Rich Causal Residual-Correction Head

Feature set. Each triggered anchor t contributes the following causal trailing features to the correction head. All features are computed from quantities available strictly before or at time t; no future glucose, future label, or future covariate is used.

- **Residual history**: $r_t, r_{t-1}, r_{t-2}, r_{t-3}$ (lagged one-step residuals, mg/dL).
- **Standardised surprise history**: z_t, z_{t-1}, z_{t-2}.
- **Glucose slopes**: $\text{slope}_{15} = (G_t - G_{t-3})/15$, $\text{slope}_{30} = (G_t - G_{t-6})/30$, $\text{slope}_{60} = (G_t - G_{t-12})/60$ (mg/dL/min).
- **Causal curvature**: $(\text{slope}_{15} - \text{slope}_{30})/15$ (mg/dL/min^2).
- **Interval width**: $w_t = \hat{q}_{90} - \hat{q}_{10}$ (model uncertainty, mg/dL).
- **Current glucose**: G_t (mg/dL).
- **Trigger-phase features**: time-since-trigger onset (anchors), cumulative residual since trigger onset (mg/dL).
- **Circadian proxy**: $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$, where $h = \texttt{hours\_since\_start} \bmod 24$.
- **Direction flag**: $\texttt{trigger\_up} \in \{0, 1\}$.

Two features were excluded as zero-variance after inspection: the column `steps_since_start` is a sequential anchor counter (not a pedometer), so its 15- and 30-minute differences are constant. Heart-rate change features are absent from the cache parquet. The final feature set contains **19 features**.

Architecture and training. Separate correction heads are trained for upward and downward triggers. Each head consists of 12 independent Ridge regression models ($\alpha = 1.0$, with intercept, features standardised), one per forecast horizon $h \in \{1, \ldots, 12\}$. The target for horizon h is $G_{t+h} - \hat{q}^{(h)}_{50,t}$, the correction the head must add to the frozen-base median. Both up- and down-heads are trained on the non-insulin train split (190,893 triggered anchors); all 1,043 train participants contribute triggers in both directions.

Head selection. Candidate heads and baselines were evaluated on the validation split only. The two top-ranked methods – Ridge full (split up/down) and HGBT full (combined) – were statistically tied (validation paired difference CI: -0.0009 mg/dL, 95% CI $[-0.040, +0.029]$, CI includes zero). Following a pre-specified tie-break rule of preferring the simpler model, **Ridge full (split up/down)** was selected as the final head. Test metrics were examined only after this selection was fixed.

12.5 Baseline Ladder

Table 40 and Fig. 28 present MAE with participant-clustered 95% CIs for all methods on non-insulin triggered anchors. Anchor counts are reported as distinct $(participant\_id,\ segment\_id,\ anchor\_ds)$ tuples; the frozen-base anchor is never counted more than once per participant and anchor index.

Key findings.

- **Current shift is worse than the frozen base** ($+0.217$ mg/dL on test). Applying the full residual r_t as a constant offset overcorrects: the base model already mean-reverts toward the historical glucose distribution over longer horizons, and a flat shift counteracts this.
- **Decaying shift matches the frozen base**: test MAE 11.381 vs 11.383 mg/dL (difference negligible).
- **Linear z_t beats decaying shift** (-0.025 mg/dL). Standardising by the model's own uncertainty width w_t provides a better correction scale than a fixed decay.
- **The rich head (Ridge split) beats the best simple baseline** (linear z_t + slope + curvature) by 0.122 mg/dL (test), 95% CI $[-0.163,\ -0.091]$ (CI excludes zero).
- **Ridge and HGBT are statistically tied**: HGBT combined minus Ridge split $= -0.005$ mg/dL, 95% CI $[-0.029,\ +0.023]$ (CI includes zero).

Ablation. Fig. 31 shows validation MAE for Ridge trained on six feature groups. Residual history (r_t–r_{t-3}) is the strongest single group (val MAE 11.340, test 11.302), outperforming z_t alone (val 11.403). The full 19-feature set achieves val MAE 11.290, lower than any single group in isolation, confirming complementary information across feature families.

12.6 Event-Enriched Proxy versus Unmatched Control

This comparison is descriptive, not a matched causal contrast. Event-enriched labels are glucose-level proxies because AI-READI does not contain confirmed meal timestamps. The control group is not distribution-matched on glucose level, slope, z_t, or interval width. The structural confound arises from the proxy definition itself: Equation (26) selects on glucose level, so event and non-event anchors differ by construction.

Table 41 and Fig. 30 compare correction gain (base MAE minus final-head MAE) for the event-enriched and unmatched-control groups on the test split.

- **Event-enriched** ($n = 14{,}505$ anchors): base MAE 14.355, head MAE 13.941, gain $+0.415$ mg/dL, 95% CI $[+0.216,\ +0.603]$ (CI excludes zero).
- **Unmatched control** ($n = 25{,}193$ anchors): base MAE 9.671, head MAE 9.635, gain $+0.036$ mg/dL, 95% CI $[+0.003,\ +0.070]$ (CI excludes zero).

The event group shows larger absolute gain, but this is expected: the event definition selects anchors with high glucose, which are inherently harder to forecast (higher base MAE) and where large one-step residuals are more common. The control group gain, while small, also excludes zero, indicating that the correction head is not harmful outside event windows. Neither comparison should be interpreted as evidence that the head detects or responds specifically to physiological events.

12.7 Deployment Behavior and Abstention

Fig. 33 and Table 42 summarise deployment characteristics across splits.

- **Trigger rate**: 9.19% (train), 9.42% (validation), 9.26% (test) of non-insulin anchors. Stability across splits confirms the trigger is not overfit.

- **Episode duration**: median 5 minutes, P90 15 minutes in all splits. With $\tau = 0.2$ the rolling-3 rule rarely sustains for more than one or two consecutive anchors.
- **Abstention rate**: $< 0.09\%$ in all splits, arising exclusively from stream boundaries (first anchor of each segment) and rare missing prediction-interval rows ($< 0.001\%$).

12.8 Interpretation and Limitations

What the head achieves. The rich correction head reliably reduces MAE at triggered windows. The improvement over the frozen base (0.175 mg/dL test, 95% CI $[-0.259, -0.094]$) and over the best non-trivial no-learning baseline (decaying shift: 0.172 mg/dL test, 95% CI $[-0.238, -0.109]$) are both statistically clear. The improvement over the best simple linear baseline (0.122 mg/dL, CI $[-0.163, -0.091]$) is also statistically clear, showing that the richer causal feature set adds genuine value beyond a single z_t term. The ablation confirms that residual history (r_t–r_{t-3}) is the most informative feature group, followed by trigger-phase features and the combined set of slopes and curvature.

What the head does not achieve. The gap between Ridge and HGBT is not statistically distinguishable (-0.005 mg/dL, CI $[-0.029, +0.023]$), indicating that the linear form of Ridge already captures the dominant structure in this feature space. The event-vs-control comparison is descriptive and structurally confounded; it should not be read as evidence that the head specifically targets physiological events. AI-READI lacks confirmed meal or exercise timestamps, so no oracle event evaluation is possible.

Per-horizon profile. Fig. 32 shows per-horizon MAE for the final head and selected baselines. The correction provides the largest absolute benefit at short horizons (0–30 min): the rich head reduces 0–30 min MAE from 8.112 (base) to 7.844 mg/dL. The benefit narrows at long horizons: 30–60 min MAE is 14.573 mg/dL (head) versus 14.654 mg/dL (base). This pattern is expected because the one-step residual is most informative about the immediate future, and its signal decays with horizon.

Scope. The correction head is appropriate only when triggered ($\approx 9.3\%$ of deployment anchors). At non-triggered anchors the frozen-base median should be used. The head should be retrained if the base checkpoint is updated or if the target glucose population changes substantially.

12.9 Link to Interpretability

The correction head's 19-feature linear Ridge structure remains interpretable by design: each coefficient $b_{h,j}$ quantifies how much feature j shifts the horizon-h prediction. The dedicated interpretability section now focuses on the frozen base model's two reliance views: output-level LOMO attribution and rigorous hidden-state construction through h_t. This separates base-model reliance from the Study 2 correction-head audit.

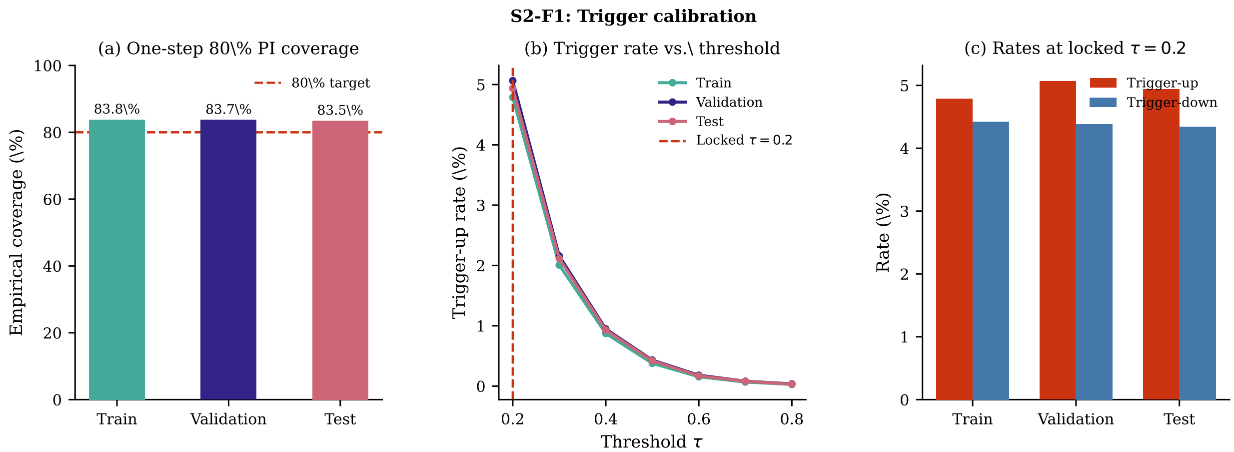


Figure 27: S2-F1: Trigger calibration. (a) Empirical one-step 80% PI coverage by split; the dashed line marks the nominal 80% target. (b) Trigger-up rate as a function of threshold τ; the locked threshold $\tau = 0.2$ is marked. (c) Trigger-up and trigger-down rates at $\tau = 0.2$ by split. Coverage exceeds 83% in all splits; $\tau = 0.2$ is the only candidate achieving near-5% trigger rate with 100% participant coverage.

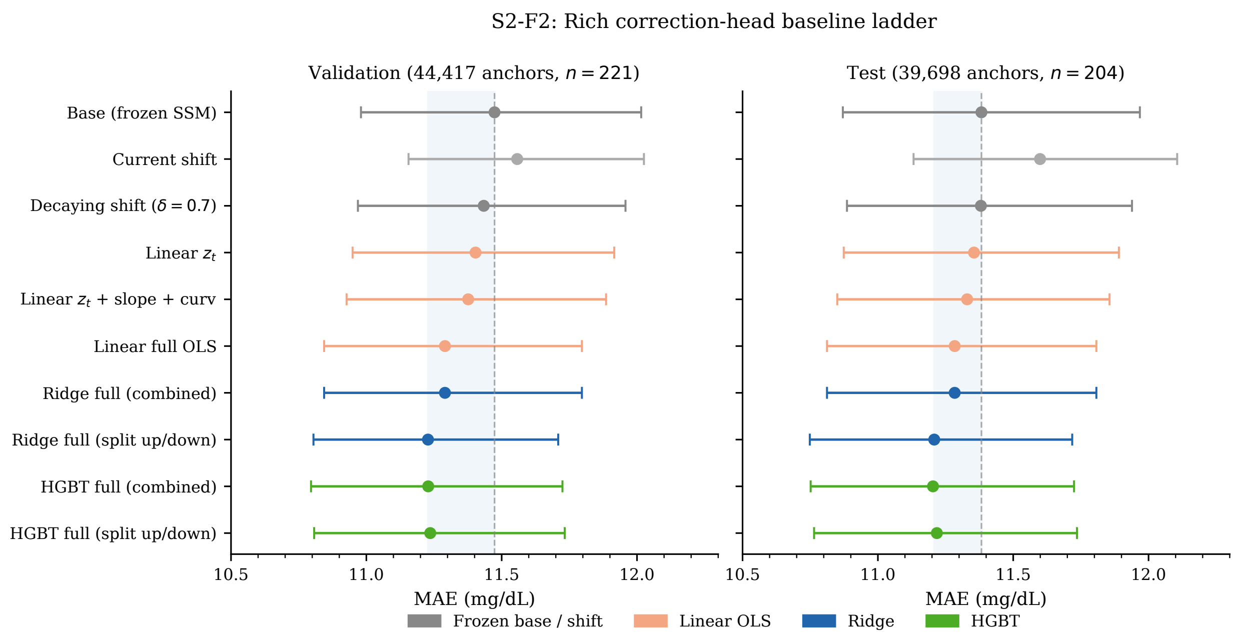


Figure 28: S2-F2: Full correction-head baseline ladder. MAE (mg/dL) with participant-clustered 95% CIs for all ten correction methods on non-insulin triggered windows. Left: validation (44,417 anchors, $n = 221$); right: test (39,698 anchors, $n = 204$). Dashed vertical line: frozen-base MAE. Shaded band: gain region of the selected final head (Ridge split up/down). Anchor counts are distinct $(participant, segment, anchor)$ tuples.

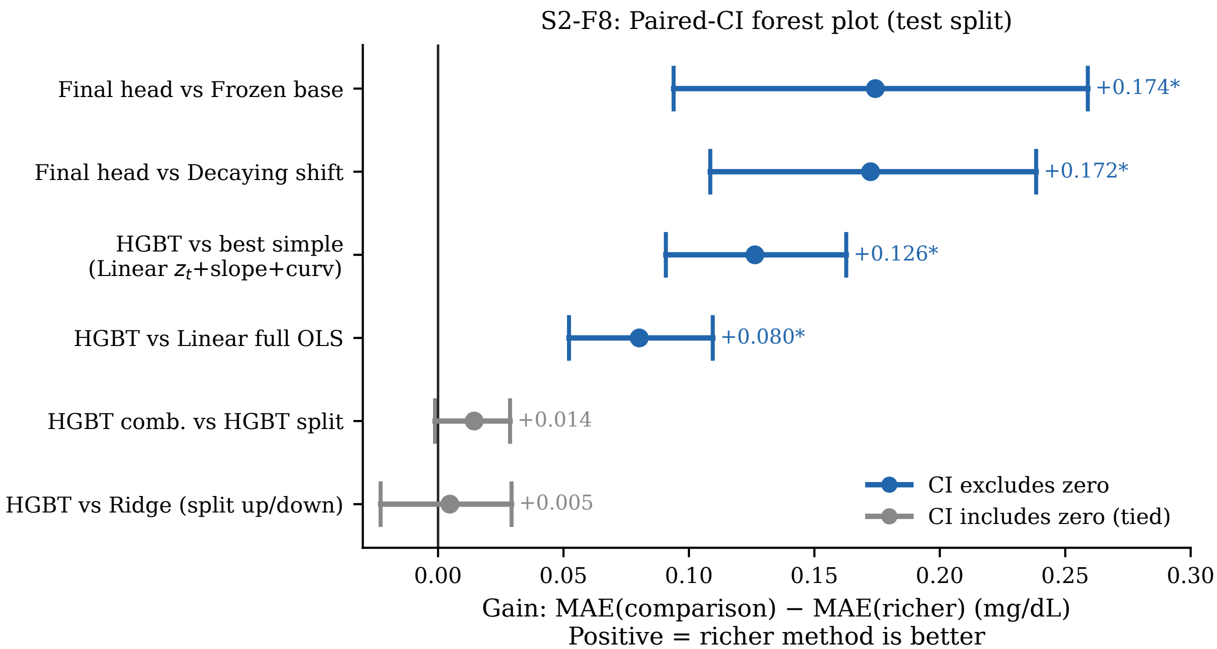


Figure 29: S2-F8: Paired-CI forest plot (test split). Each bar shows the gain of the richer method over the comparison method, with participant-clustered 95% CI. Blue: CI excludes zero (statistically distinguishable). Grey: CI includes zero (statistically tied). Asterisk (*): CI excludes zero. All comparisons against simpler baselines favour the rich head; HGBT and Ridge are indistinguishable.

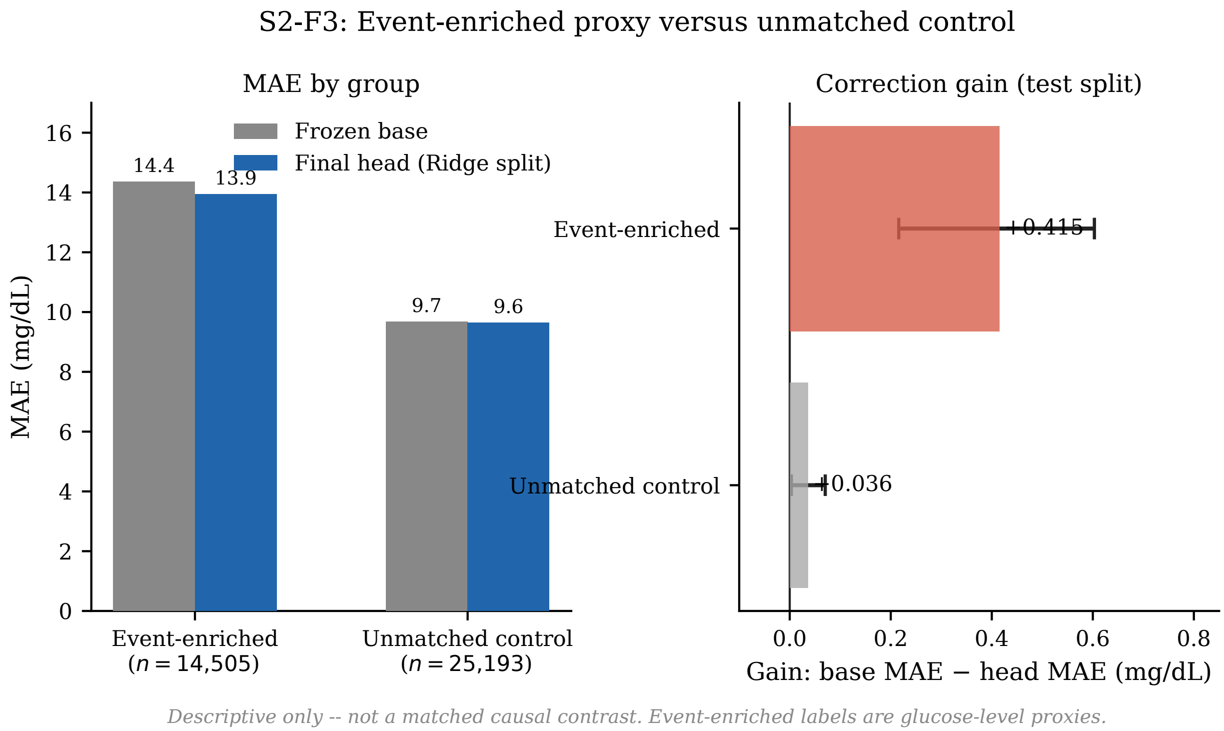


Figure 30: S2-F3: Event-enriched proxy versus unmatched control (test split). Left: MAE of the frozen base and final head for each group. Right: correction gain (base MAE − head MAE) with 95% CI. *Descriptive only – not a matched causal contrast. Event-enriched labels are glucose-level proxies because AI-READI does not contain confirmed meal timestamps. The control group is not distribution-matched on glucose level, slope, z_t, or interval width.*

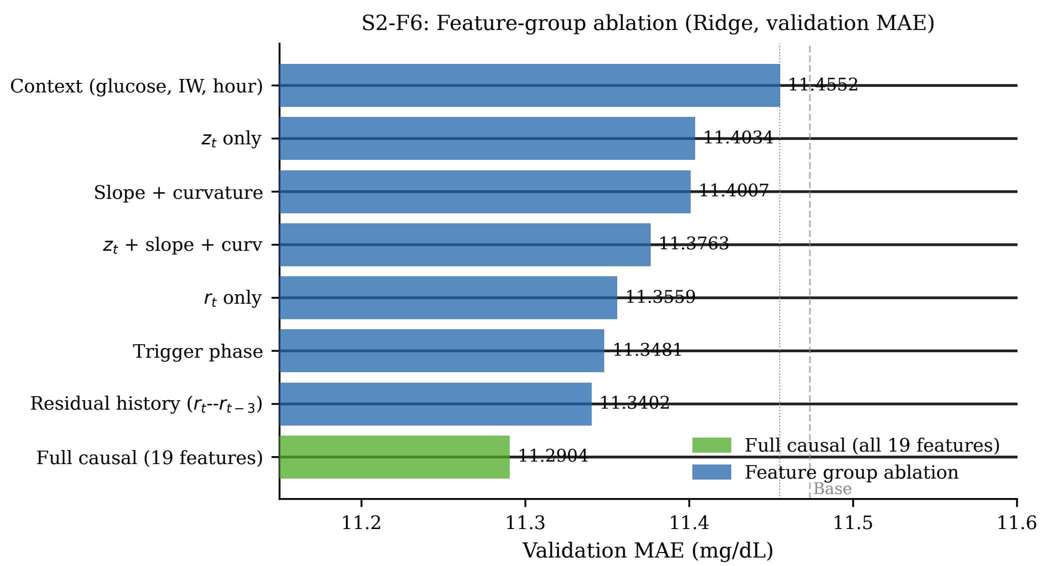


Figure 31: S2-F6: Feature-group ablation (Ridge, validation MAE). Each bar shows validation MAE for a Ridge head trained on the named feature subset only (19-feature full-causal set in green). Error bars: participant-clustered 95% CI. Dashed vertical line: frozen-base validation MAE. The full causal set achieves the lowest validation MAE; residual history (r_t–r_{t-3}) is the strongest single group.

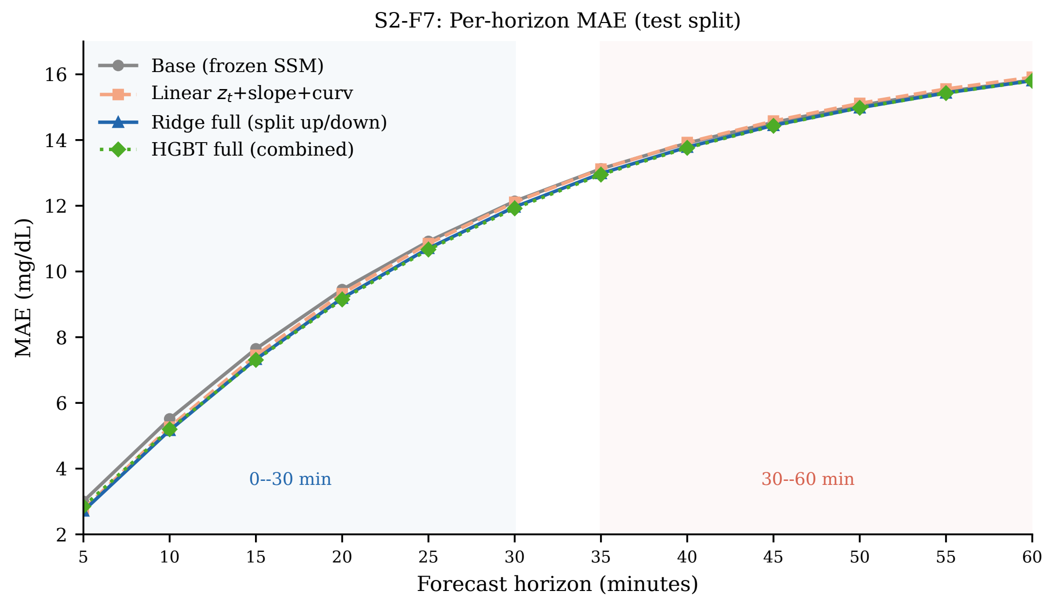


Figure 32: S2-F7: Per-horizon MAE (test split). Lines show mean absolute error at each 5-minute forecast horizon for the frozen base, best simple baseline, and the two best rich heads. The correction provides the largest benefit at short horizons (0–30 min) and narrows at longer horizons.

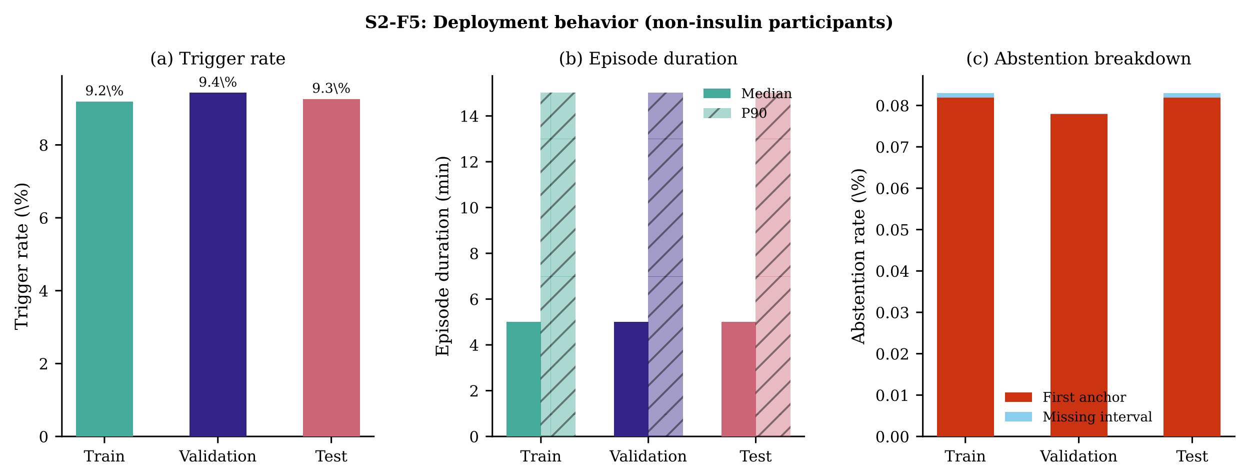


Figure 33: S2-F5: Deployment behavior (non-insulin participants). (a) Trigger rate by split. (b) Episode duration: median and P90. (c) Abstention rate. Trigger rates and episode durations are stable across splits; abstention is negligible ($< 0.09\%$ in all splits).

Table 39: S2-T1: Final model selection (validation-based). All candidate models ranked by validation MAE (lower is better). Anchor counts are distinct $(participant, segment, anchor)$ tuples. The paired difference CI (best vs. second-best on validation) did not exclude zero, so the simpler model (Ridge split up/down) was selected.

Rank	Method	n_{anchors}	Val MAE	Val 95% CI	Selected
1	Ridge full (split up/down)	44,417	11.228	[10.805, 11.709]	✓
2	HGBT full (combined)	44,417	11.229	[10.796, 11.725]	
3	HGBT full (split up/down)	44,417	11.236	[10.807, 11.733]	
4	Ridge full (combined)	44,417	11.290	[10.844, 11.796]	
5	Linear full causal OLS	44,417	11.291	[10.844, 11.796]	
6	Linear z_t + slope + curv	44,417	11.376	[10.927, 11.886]	

Tie-break: paired CI (Ridge split − HGBT combined) $= -0.0009$*,*
95% CI $[-0.040, +0.029]$ *(includes zero)* $\Rightarrow$ *prefer simpler.*

Table 40: S2-T2: Full baseline ladder. MAE (mg/dL) with participant-clustered 95% CIs for all ten correction methods on non-insulin triggered windows. "n" is distinct $(participant, segment, anchor)$ tuples. Starred paired differences in the lower panel are CIs excluding zero.

Split	Method	n	$n_{\text{part.}}$	MAE	95% CI
Val.	Base (frozen SSM)	44,417	221	11.474	[10.980, 12.016]
	Current shift	44,417	221	11.557	[11.156, 12.026]
	Decaying shift ($\delta = 0.7$)	44,417	221	11.434	[10.969, 11.958]
	Linear z_t	44,417	221	11.403	[10.949, 11.916]
	Linear z_t + slope + curv	44,417	221	11.376	[10.927, 11.886]
	Linear full causal OLS	44,417	221	11.291	[10.844, 11.796]
	Ridge full (combined)	44,417	221	11.290	[10.844, 11.796]
	Ridge full (split up/down)	44,417	221	11.228	[10.805, 11.709]
	HGBT full (combined)	44,417	221	11.229	[10.796, 11.725]
	HGBT full (split up/down)	44,417	221	11.236	[10.807, 11.733]
Test	Base (frozen SSM)	39,698	204	11.383	[10.871, 11.968]
	Current shift	39,698	204	11.599	[11.132, 12.106]
	Decaying shift ($\delta = 0.7$)	39,698	204	11.381	[10.886, 11.939]
	Linear z_t	39,698	204	11.355	[10.874, 11.891]
	Linear z_t + slope + curv	39,698	204	11.330	[10.850, 11.856]
	Linear full causal OLS	39,698	204	11.284	[10.812, 11.807]
	Ridge full (combined)	39,698	204	11.284	[10.812, 11.807]
	Ridge full (split up/down)	39,698	204	11.208	[10.748, 11.718]
	HGBT full (combined)	39,698	204	11.204	[10.752, 11.725]
	HGBT full (split up/down)	39,698	204	11.218	[10.764, 11.736]

Selected paired differences (test, participant-clustered bootstrap, $B = 1000$*):*

Final head vs base	-0.174 mg/dL, CI $[-0.259, -0.094]$*
Final head vs decaying shift	-0.172 mg/dL, CI $[-0.238, -0.109]$*
HGBT combined vs best simple	-0.126 mg/dL, CI $[-0.163, -0.091]$*
HGBT combined vs linear OLS	-0.080 mg/dL, CI $[-0.110, -0.052]$*
HGBT combined vs Ridge split	-0.005 mg/dL, CI $[-0.029, +0.023]$ (tied)

* CI excludes zero.

Table 41: S2-T3: Event-enriched proxy versus unmatched control (test split). Correction gain (base MAE − final-head MAE, mg/dL) for event-enriched and unmatched-control groups. CIs are participant-clustered bootstrap ($B = 1000$). *This comparison is descriptive, not a matched causal contrast. The control group is not distribution-matched on glucose level, slope, z_t, or interval width.*

Group	n	Base MAE	Head MAE	Gain	95% CI
Event-enriched	14,505	14.355	13.941	+0.415	[+0.216, +0.603]
Unmatched control	25,193	9.671	9.635	+0.036	[+0.003, +0.070]

Table 42: S2-T4: Deployment characteristics (non-insulin participants). Trigger rates and episode statistics are stable across splits.

Split	$n_{\text{part.}}$	Trigger rate	Episodes	Dur. med.	Dur. P90	Abstention
Train	1,043	9.19%	111,288	5 min	15 min	0.083%
Validation	221	9.42%	25,244	5 min	15 min	0.078%
Test	204	9.26%	23,035	5 min	15 min	0.083%

Table 43: S2-T5: Feature-group ablation (Ridge, validation MAE). Each row shows validation and test MAE for a Ridge head trained on the named feature subset only. The full 19-feature set achieves the lowest validation MAE. All n_{anchors} are the same 44,417 validation and 39,698 test triggered tuples.

Feature group	$n_{\text{feat.}}$	Val MAE	Val 95% CI	Test MAE
z_t only	1	11.403	[10.949, 11.916]	11.355
z_t + slope + curv	3	11.376	[10.927, 11.886]	11.330
r_t only	1	11.356	[10.908, 11.859]	11.310
Slope + curvature	4	11.401	[10.942, 11.909]	11.349
Trigger phase	4	11.348	[10.902, 11.853]	11.314
Residual history (r_t–r_{t-3})	4	11.340	[10.896, 11.844]	11.302
Context (glucose, IW, hour)	4	11.455	[10.988, 11.977]	11.383
Full causal (19 features)	19	**11.290**	[10.844, 11.796]	**11.284**

13 Exercise detection

13.1 Motivation

The meal counterfactual is not identifiable on AI-READI (Section 11), and the event aware residual correction (Section 12) a statistically significant but modest improvement of 0.175 mg/dL on triggered windows. AI-READI participants wore Garmin wrist devices that record heart rate (HR), step counts, and respiratory rate (RR) continuously, making ambulatory exercise a second direction for counterfactuals. Unlike meal events, physical activity produces large, time series physiological signals in HR and steps that are detectable without any glucose information.

The goals: construct a detector of episodes that are ambulatory exercise using only wearable signals; characterize the descriptive glucose response at detected onsets; and analyse the frozen base model's residuals at episodes that are exrcices to determine whether systematic signed bias justifies a correction or advisory.

No causal claims about exercise physiology are made, and all findings are scoped to non-insulin participants unless noted.

13.2 Available wearable wignals and design constraints

The Garmin wrist stream provides HR (bpm), step counts (steps per 5-minute epoch), RR (breaths/min), a sleep-stage label, and device-derived fields including estimated calories per minute and a Garmin activity intensity score.

Four design constraints govern signal use in the detector. First, glucose is never used to detect or gate episodes; glucose appears only as an outcome variable. Second, calories per minute and the Garmin activity intensity score are excluded from all gates because both are partly derived from HR; using them would circularly reinforce the HR criterion. Third, a rolling local HR delta was considered as an onset refinement feature but was reduced to visualization and onset diagnosis only because a backward looking delta over exercise windows self-contaminates the anchor test. Fourth, RR is used descriptively but not as a gate: its noise level on wrist PPG is too high for a reliable onset criterion at 5-minute resolution.

The working signals for detection are therefore HR (excess above a stable resting floor), step count rate, and the sleep label (for daytime restriction).

Stable resting HR floor: Each participant's resting HR floor is defined as the 5th percentile of all HR observations recorded during sedentary, awake windows (step rate below threshold, sleep label absent). This is a person-specific baseline computed once per participant; it is never a rolling window, which would absorb exercise induced HR elevation and suppress the excess signal.

HR excess: At each 5-minute anchor t, the HR excess above the stable floor is

$$\Delta\mathrm{HR}_t = \mathrm{HR}_t - \mathrm{HR}_{\mathrm{floor}}, \tag{27}$$

where $\mathrm{HR}_{\mathrm{floor}}$ is the 5th-percentile stable resting HR for that participant.

Anchor gate: A window is flagged as an anchor (potential exercise start) when all three of the following hold simultaneously:

$$\overline{\mathrm{steps}}_{t,15\mathrm{min}} \geq 40 \text{ steps/min} \quad (\text{STEP\_RATE\_FLOOR}), \tag{28}$$
$$\overline{\Delta\mathrm{HR}}_{t,15\mathrm{min}} \geq 8 \text{ bpm} \quad (\text{HR\_EXCESS\_FLOOR}), \tag{29}$$
$$\mathrm{sleep}_t = 0. \tag{30}$$

Steps lead the gate; HR excess confirms physiological arousal.

Strain grading: Strain is graded against a person specific 20th-percentile HR excess floor (not the detection gate 5th-percentile floor), creating three bands: light ($\Delta\mathrm{HR} \in [8,15)$ bpm), moderate ($[15,30)$ bpm), and vigorous (≥ 30 bpm). A sub threshold class captures anchors that clear the 5th percentile step criterion but fall below the 8-bpm excess floor; these are retained in cadence analysis only.

Episode construction. Consecutive anchor windows are merged into sessions (merge gap ≤ 5 min, minimum episode duration $= 10$ min, maximum $= 120$ min). Exercice episodes are excluded if they overlap with more than 5% of sleep stage annotations or form part of a long active envelope (≥ 3 hr of continuous activity), leaving the clean set (primary definition). A stricter sensitivity subset additionally requires the post episode HR-recovery fraction to exceed 30% within 30 minutes (strict recovery definition).

The clean set contains **2,171 episodes** across **788 participants**. The strict recovery sensitivity subset contains **1,015 episodes** across **574 participants**.

Confidence score and refined start of exercise: A confidence score weights five signals: activity quality ($w = 0.40$), HR excess ($w = 0.30$), RR elevation ($w = 0.10$, descriptive), duration quality ($w = 0.10$), and recovery ($w = 0.10$); a sleep overlap penalty overrides all positive weights ($w = -1.0$). A high confidence episode exceeds a score of 0.65; medium confidence falls between 0.40 and 0.65. The confidence window start (when the episode first exceeds the anchor gate) is not used as the onset: the detector backtracks from the confidence start to the earliest sustained step rise within 45 minutes, producing a **refined start** that precedes the confidence gate.

Across all 3,131 onset-refined bouts, the **median refined-onset advance is 30 minutes** (mean 41.6 min) relative to the confidence-window start. Onset and offset are symmetric: if the start is pulled back, the end is extended by the same logic, subject to a sleep-label clamp and the 120-min duration cap.

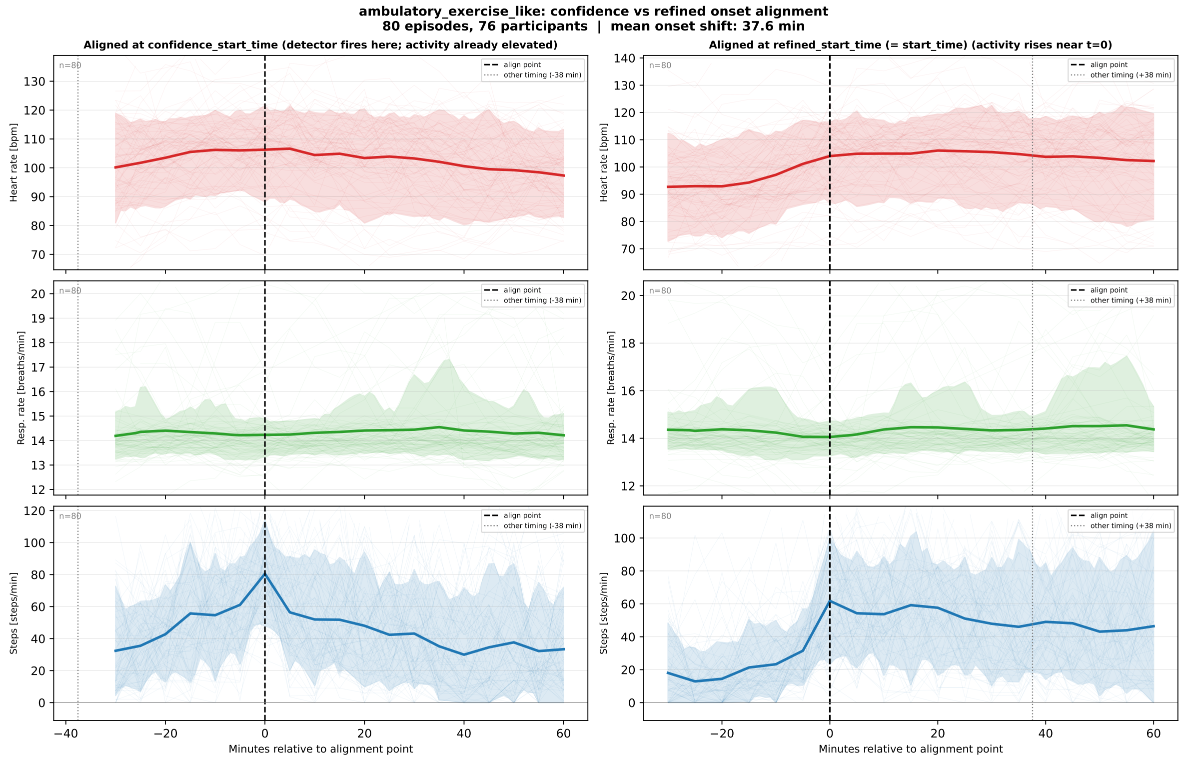


Figure 34: Confidence-window onset versus refined onset. Each episode is aligned to its confidence-window start (vertical dashed line at $t = 0$); traces show mean step rate (top) and HR excess (bottom) for 90 min before and after that anchor. The median refined onset (30 min before the confidence start) captures the sustained step rise that precedes the confidence gate, confirming that the confidence-window start systematically lags the earlier wearable-defined activity signal.

13.3 Detector Validation

Episode timelines: The first alignment below shows the per-episode wearable signal alignment at the refined onset, pooled across all clean episodes. Steps rise sharply at $t = 0$ and HR excess follows within 5–10 minutes, consistent with an activity led gate in which step elevation precedes the cardiovascular response. The paired timelines below illustrate clean detection across an active day for participants 1029 and 1030; participant 4022 is shown as a diagnostic example with more complex episode structure and is discussed in supplementary context only.

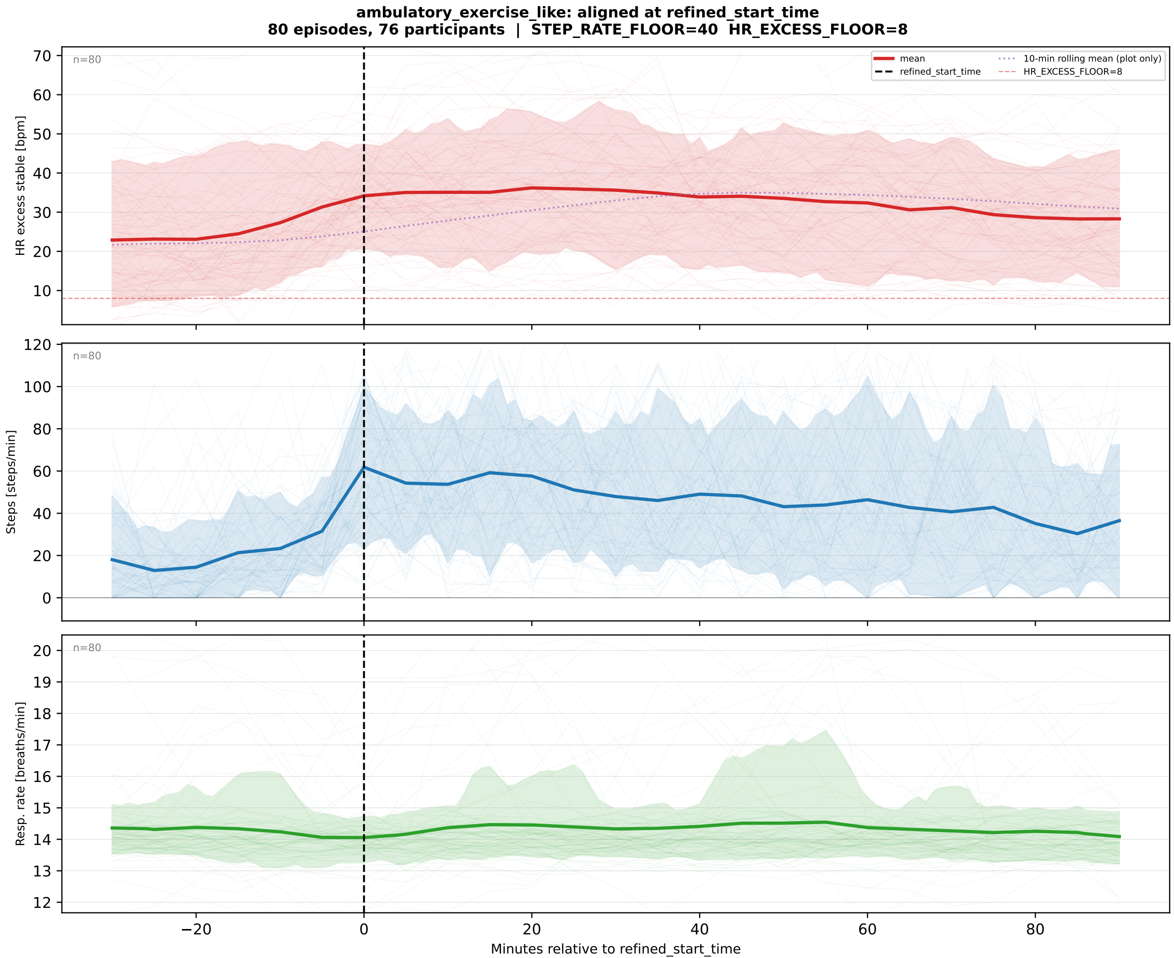


Figure 35: Wearable signals aligned at the refined episode onset. Mean and 95% bootstrap CI for step rate (top) and HR excess above the stable resting floor (bottom), aligned at $t = 0$ (refined onset) across all 2,171 clean ambulatory exercise-like episodes. Steps rise sharply at onset; HR follows within 5–10 minutes, consistent with the activity-led gate design in which steps lead and HR confirms. No glucose signal is used in detection; the glucose trace visible in the full notebook is shown post-hoc only.

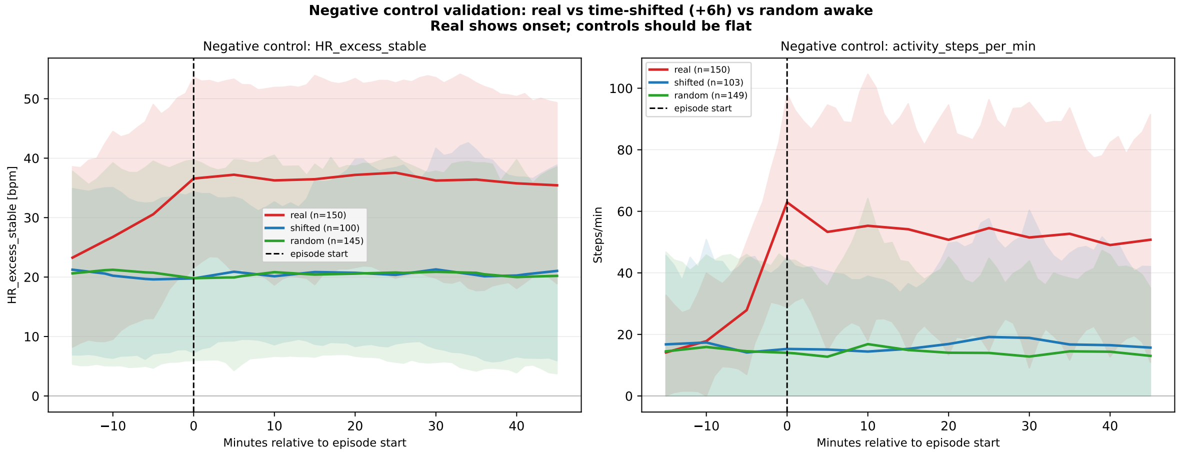
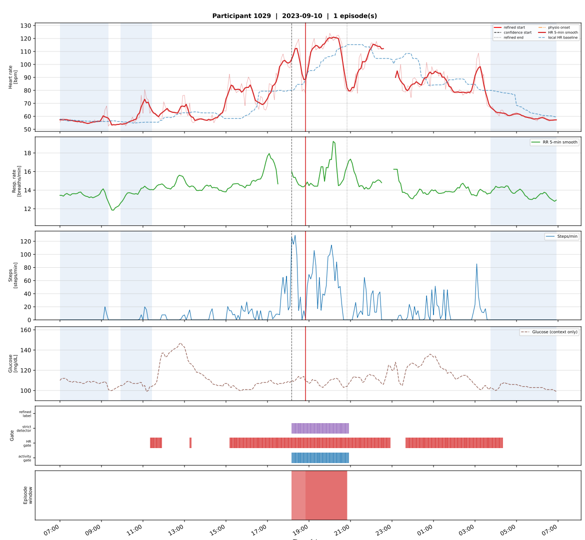
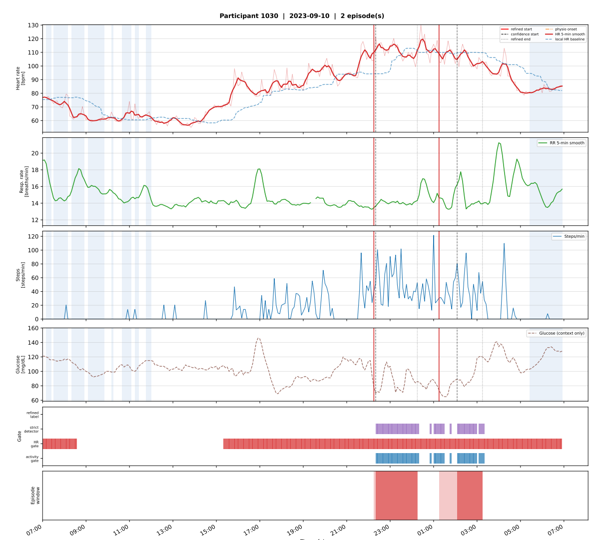


Figure 36: Negative control validation. Three conditions aligned at a common reference: (i) real refined exercise like onsets; (ii) the same onsets time-shifted by $+6$ hours (temporal negative control); and (iii) random awake anchors matched on time-of-day (distributional negative control). Real onsets show a sharp, coordinated HR and step rise. Both negative controls are flat across the alignment window, validating that the detector captures a real physiological signal rather than circadian variation or device artefact. Exact mean HR-excess and step-rate values at $t = 0$ for each condition are readable from the figure.

Using the pre registered rise contrast, defined as the mean over $[0, 15]$ minutes minus the mean over $[-15, 0]$ minutes, real onsets showed an HR excess rise of 8.01 bpm and a step-rate rise of 29.44 steps/min. The $+6$ hour control showed 0.27 bpm and -0.93 steps/min, while the random-awake control showed -0.46 bpm and -0.23 steps/min. The resulting separation from the strongest control was 7.74 bpm for HR excess and 29.67 steps/min for step rate.



(a) Participant 1029, 2023-09-10



(b) Participant 1030, 2023-09-10

Figure 37: Representative episode timelines (participants 1029 and 1030). Each panel shows a full day of HR, step rate, sleep label, and detected episode boundaries (refined onset/offset, shaded region). Both participants show clear HR-step coupling at episode boundaries. Participant 4022 (available as a supplementary figure) illustrates a participant with more complex episode structure and is used for diagnostic illustration only.

13.4 Exercise type classification

The episode type is characterized along two independent axes. **Cadence** is the mean step rate over active core minutes (minutes with ≥ 20 steps/min within the episode) and is person-invariant: it reflects movement pattern independent of the participant's cardiovascular fitness. Four cadence classes are defined: slow walk (40–80 steps/min), walk (80–110), brisk walk (110–130), and run-like (≥ 130). "Run-like" denotes high-cadence ambulation and does not confirm a specific activity mode.

Strain is the HR excess above the person-specific 20th-percentile sedentary-awake floor and is person-specific: it reflects cardiovascular load relative to that participant's own baseline. Three strain bands are defined: light (8–15 bpm excess), moderate (15–30 bpm), and vigorous (≥ 30 bpm).

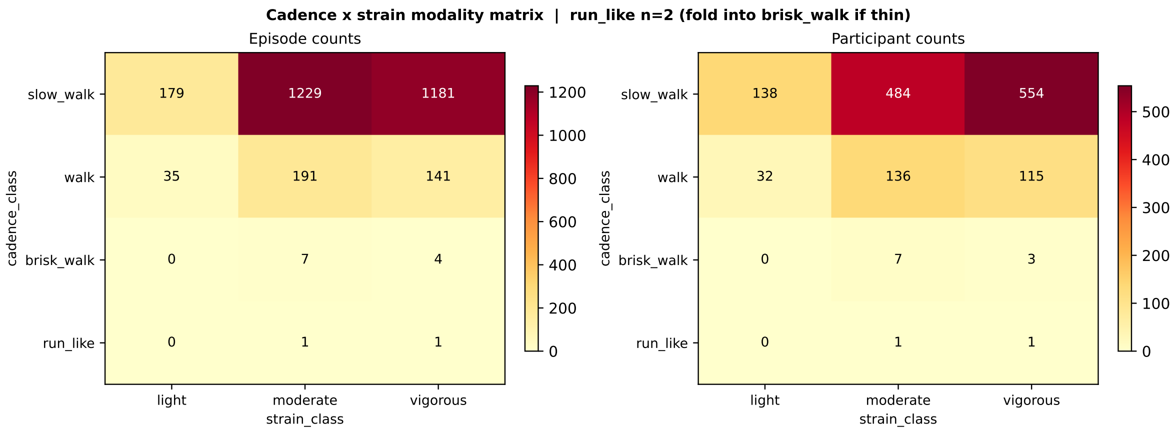


Figure 38: Cadence–strain episode heatmap. Each cell shows episode count (or a derived metric) for the cadence-class (columns) by strain class (rows) joint distribution across the 2,171 clean episodes. Slow walk and walk cadence classes dominate numerically. A notable feature is that many slow-walk episodes carry moderate or vigorous strain, reflecting inter-individual variability: a slow gait can be cardiovascularly demanding for participants with low fitness or high resting HR variability. This heterogeneity motivates stratified analysis later (Section 13.7) but does not support inference about insulin resistance or cardiovascular fitness.

13.5 Recovery and Quality Control

HR recovery is assessed as the fraction of the HR excess that dissipates within 30 minutes post episode (recovery_frac_thresh = 0.30). Recovery is used as a *sensitivity marker*, not a hard detection criterion: episodes failing recovery are retained in the primary set but excluded from the strict-recovery sensitivity subset. This design avoids circularity in which a recovery criterion would preferentially select short, high-intensity bouts that happen to have large post-episode HR drops.

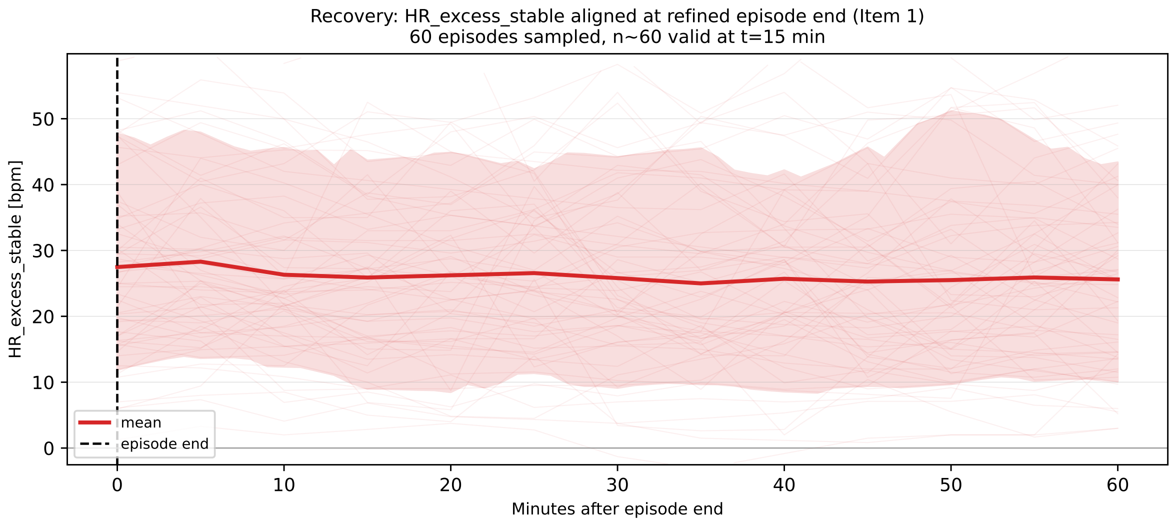


Figure 39: HR recovery aligned at episode end. Mean HR excess (with 95% CI) in the 60 minutes following the episode end, stratified by strict-recovery pass/fail. Episodes that pass the strict-recovery criterion (fraction ≥ 0.30 within 30 min post-end) show a clear HR return toward baseline; episodes that fail show a slower or partial return, consistent with extended moderate-activity transitions rather than detector error. Recovery rate is interpreted as a sensitivity marker for episode quality, not a physiological claim.

13.6 Preliminary glucose-response kernel

The aligned kernel below shows mean glucose at the refined episode onset, pooled across all clean episodes with available CGM. The descriptive profile shows a **16.66 mg/dL drop** (the moderate-strain, non-insulin anchor used in the planning response in Section 15.3), with a minimum reached at **28.31 minutes** post onset, followed by a partial rebound within the observation window.

This is a **purely descriptive, associational pattern**. It is not a causal estimate. The observed trajectory is potentially confounded by meals occurring during or near exercise starts, pre exercise glucose level (regression to the mean), circadian time-of-day, and selection of episode type. Insulin treated participants are excluded from causal interpretation throughout; the kernel here is descriptive across all participants with CGM coverage. The kernel is later used as a magnitude anchor for the planning response (Section 15.3), where this descriptive-associational role is stated explicitly.

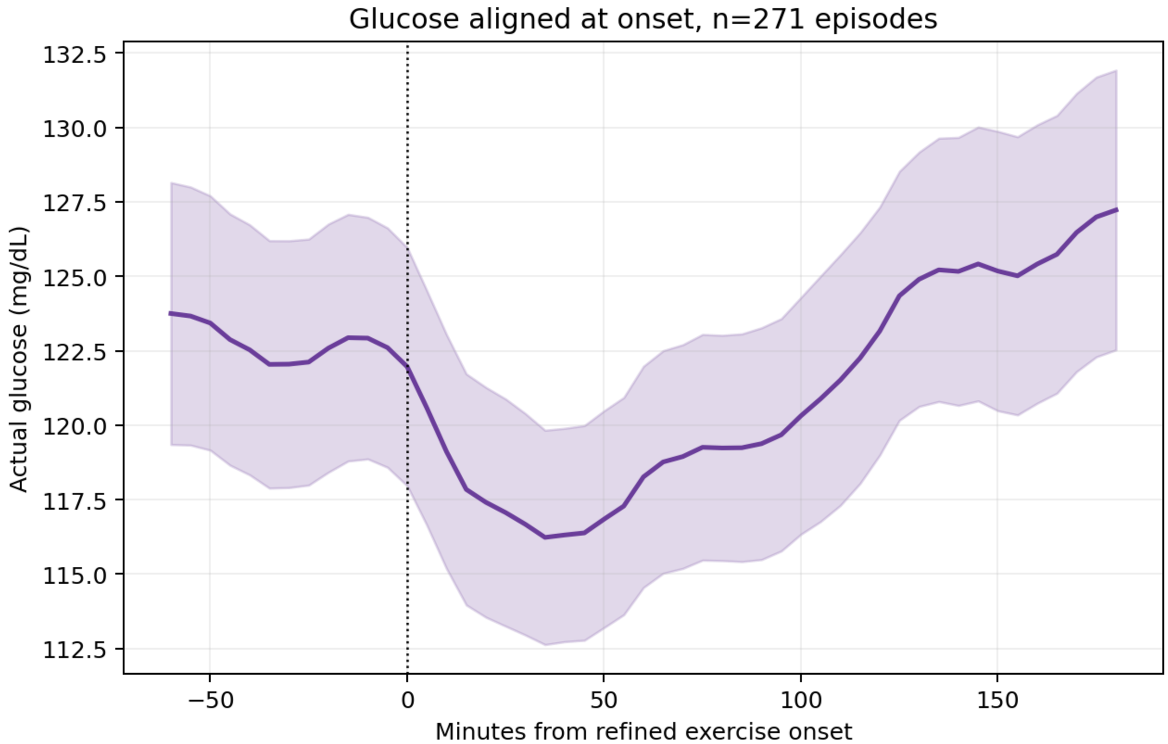


Figure 40: Descriptive glucose kernel aligned at refined episode onset. Mean glucose (mg/dL) with 95% bootstrap CI, aligned at $t = 0$ (refined onset) across clean episodes with CGM coverage. The profile shows an approximate 16.7 mg/dL drop and a partial rebound. This is a descriptive, associational pattern: confounding by meals, baseline glucose, and time-of-day is possible and not controlled. Glucose is never used to detect episodes; this is a post-hoc summary.

13.7 Comparison to the frozen base model

The parts focuses on whether the frozen `mamba_stateful_5epoch` base model (the same checkpoint as Study 2) makes systematically larger errors inside exercise like windows than outside. For each clean episode with forecast cache coverage, a matched non-exercise control window is drawn from the same participant, matched within ± 60 min time-of-day and ± 20 mg/dL baseline glucose. The signed residual follows the notebook convention:

$$b_h = \hat{q}^{(h)}_{50,t} - G_{t+h}, \tag{31}$$

where $\hat{q}^{(h)}_{50,t}$ is the frozen base median forecast at horizon h issued from anchor t. Positive b_h indicates that the base model over predicts actual glucose, so the observed glucose is lower than forecast. The horizon-growing positive bias therefore indicates that the frozen model under represents the exercise associated drop.

Matching yielded **212 episode–control pairs** (316 episodes passed Step 0, 271 had cache coverage, and 212 found a within-participant matched control at the selected tolerance).

MAE comparison. Episode and control windows show no statistically distinguishable MAE gap: paired MAE gap (all strains): $+0.29$ mg/dL, 95% CI $[-0.68, +1.28]$ (CI includes zero). The null MAE finding holds within each strain class (vigorous: $+0.67$ $[-0.51, +1.83]$; moderate: -0.17 $[-1.72, +1.56]$). The frozen model is not less accurate during exercise like windows on the absolute error metric.

Signed bias The signed bias reveals a different picture. The model's forecast progressively over predicts the actual glucose trajectory during exercise-episode horizons compared to matched controls, growing with forecast horizon. At the matched-pairs level:

- **Vigorous** ($n = 104$ pairs): episode signed bias $+3.82$ mg/dL (95% CI $[+2.35, +5.21]$); control signed bias -0.37 mg/dL ($[-1.51, +0.68]$); **paired bias gap** $+4.19$ **mg/dL** (CI $[+2.33, +5.97]$, CI excludes zero).
- **Moderate** ($n = 102$ pairs): episode signed bias $+2.20$ mg/dL (95% CI $[+0.09, +4.09]$); control signed bias -1.85 mg/dL ($[-3.40, -0.33]$); **paired bias gap** $+4.05$ **mg/dL** (CI $[+1.83, +6.29]$, CI excludes zero).
- **Light** ($n = 6$ pairs): paired bias gap $+4.43$ mg/dL (CI $[-4.47, +15.11]$, CI includes zero). *Do not interpret: very low sample size.*

The bias gap for vigorous and moderate strain both exclude zero, indicating provisional, recoverable headroom above matched controls. The verdict from the diagnostic pipeline is **JUSTIFIED** (vigorous headroom exceeds the 2 mg/dL material-gap threshold; moderate also exceeds it but with a wider CI). The headroom is termed *provisional* here because: the analysis is scoped to non-insulin T2D participants (descriptive for insulin-treated; the insulin-treated subgroup has $n = 32$ pairs and wide CIs); episode count is modest; matching tolerance is coarse.

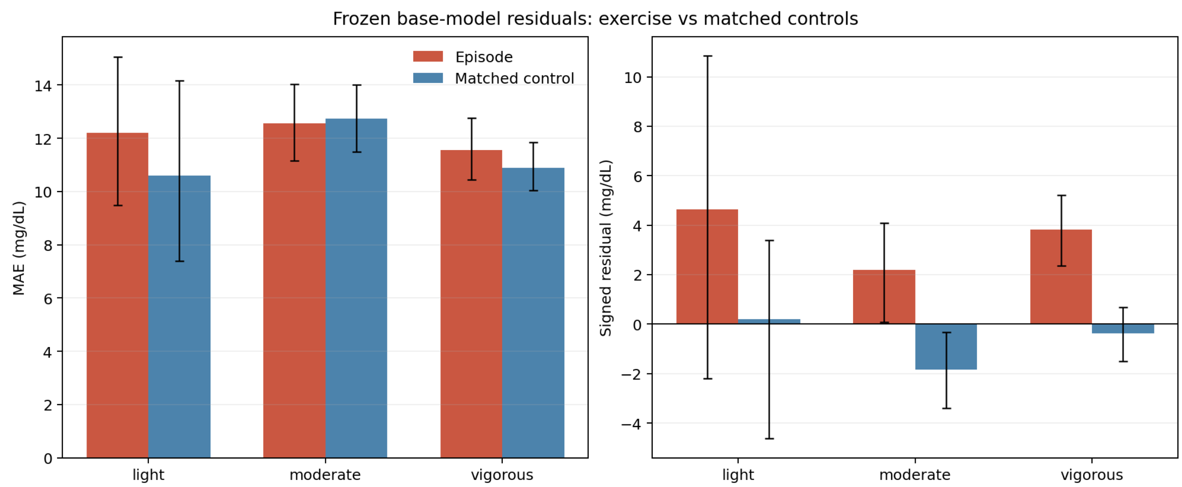


Figure 41: Episode versus matched-control signed residual by strain class. Each panel shows the mean signed residual (mg/dL) for exercise-like episodes (red) versus matched non-exercise controls (blue) at the median horizon, stratified by strain class. Error bars are 95% bootstrap CIs. Vigorous and moderate episodes show a positive bias gap relative to matched controls (CIs excluding zero), while light episodes ($n = 6$ pairs) show a wide CI. No causal claim is made: the bias reflects the frozen model's inability to anticipate the exercise-associated glucose recovery, not a confirmed exercise effect.

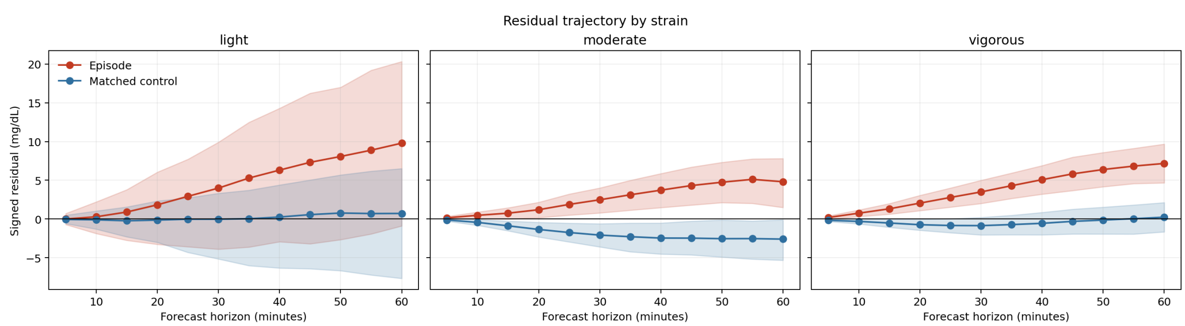


Figure 42: Signed residual trajectory by forecast horizon and strain class. Mean signed residual ($b_h = \hat{q}_{50}^{(h)} - G_{t+h}$, mg/dL) for episodes (red) and matched controls (blue) as a function of forecast horizon (5–60 min), stratified by strain class (light, moderate, vigorous). The episode residual grows monotonically with horizon for moderate and vigorous strain, while the control residual remains near zero. Light episodes ($n = 6$) show a wide CI that precludes interpretation. This horizon-growing bias is the provisional headroom that motivates both the direction of planning response(counterfactual) and advisory correction in Section 15.

13.8 Limitations bridging to the counterfactual

The detector is wearable defined, not ground truth validated against prospective exercise logs. Detection captures only ambulatory, step based activity; low step activities such as resistance training, isometric exercise, stationary cycling are not captured. RR is too noisy for reliable onset detection at wrist-PPG resolution and is used descriptively only. Meal confounding is possible within the glucose kernel: episode windows of 10–120 min can overlap with eating, particularly at mealtimes.

The residual analysis is scoped to non insulin participants for causal interpretation; insulin treated results are reported descriptively because unobserved insulin dosing is a direct confounder of the glucose trajectory. Subgroup analyses by cadence class, time-of-day, and participant phenotype are deferred until the detector and kernel are confirmed stable on a holdout population.

The insulin resistance hypothesis was tested and closed as a negative result: baseline glucose was the only predictor of the subsequent drop, and no recoverable participant exercise sensitivity term was found. The relationship between cadence/strain heterogeneity and participant metabolic profiles is therefore framed as a fairness and heterogeneity issue to characterize in future work, not as evidence for a biological mechanism.

Table 44: EX-T1: Detector and residual audit summary. All values are sourced from notebook or pipeline output artifacts.

Item	Result	Interpretation
Clean episodes (primary)	2,171	Primary detection set
Participants (clean)	788	Unique participants with ≥ 1 clean episode
Strict-recovery subset	1,015 episodes	574 participants; sensitivity check
Median onset shift	30 min	Refined onset precedes confidence-window start
Negative-control HR rise	8.01 vs 0.27 / -0.46 bpm	Real vs shifted / random-awake
Negative-control steps rise	29.44 vs -0.93 / -0.23 steps/min	Real vs shifted / random-awake
Cadence distribution	1,859 slow; 223 walk; 81 sub-threshold; 6 brisk; 2 run-like	Slow walk dominates
Glucose kernel drop	16.66 mg/dL	Descriptive-associational; anchor for Path B
Glucose kernel nadir	28.31 min	Moderate-strain, non-insulin anchor
MAE gap (vigorous)	$+0.67$ [-0.51, $+1.83$] mg/dL	CI includes zero; no error increase
MAE gap (moderate)	-0.17 [-1.72, $+1.56$] mg/dL	CI includes zero; no error increase
Signed bias gap (vigorous)	$+4.19$ [$+2.33$, $+5.97$] mg/dL	CI excludes zero; provisional headroom
Signed bias gap (moderate)	$+4.05$ [$+1.83$, $+6.29$] mg/dL	CI excludes zero; provisional headroom
Residual audit verdict	JUSTIFIED	Vigorous headroom $>$ 2 mg/dL threshold

14 Descriptive Heterogeneity of the Exercise Glucose Response

This section characterizes how glucose behaves after wearable-defined ambulatory exercise like episodes and to understand whether the observed glucose response is homogeneous or whether it differs across episode intensity, baseline glucose, glycemic phenotype, clinical variables, and demographic subgroups.

The detector is wearable-defined rather than ground-truth exercise-labeled, and glucose is used only as an outcome after episode detection. Groups with fewer than 20 episodes or fewer than 10 unique participants are suppressed. Legend entries report both the number of episodes (n) and the number of unique participants (p).

The main purpose of this section is to answer three questions:

Is there a reproducible glucose pattern after detected exercise-like episodes? Which sources of heterogeneity are largest? Does the frozen forecasting model miss the post-exercise glucose trajectory in a systematic way?

The overall interpretation is that the post exercise glucose response is real but heterogeneous. The largest visible driver is baseline glucose at exercise onset. Several participant characteristics show apparent differences, but most of these differences are partly explained by starting glucose, diabetes phenotype, medication status, and cohort composition. Therefore, these subgroup figures are best used to guide future stratified validation, not to make biological claims.

14.1 Overall Glucose-Response Kernel

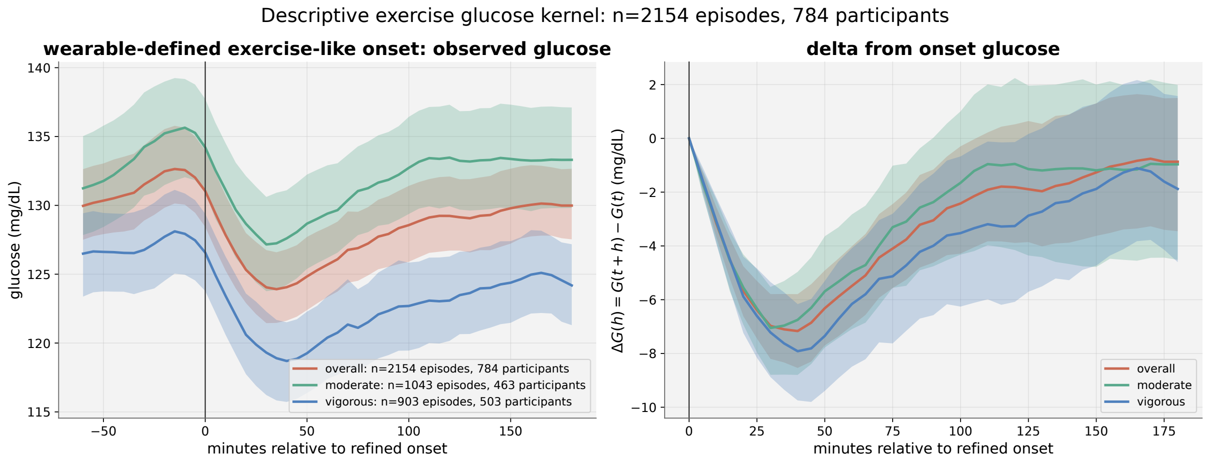


Figure 43: Overall wearable-defined exercise glucose kernel. Descriptive glucose kernel around wearable-defined ambulatory exercise-like episodes, with participant-clustered bootstrap 95 % intervals. Curves are additionally stratified by moderate and vigorous strain where eligible. This is not a causal exercise estimate.

Glucose does not fluctuate randomly around the exercise onset. Instead, there is a coordinated drop after the refined onset, with a lowest point occurring after the start of the wearable defined episode and then a rebound. This supports the detector as identifying physiologically meaningful activity periods rather than arbitrary wearable noise. However the aligned curve is vulnerable to confounding by baseline glucose, time of day, meals close to exercise, and medication status. The decrease may partly reflect regression to the mean, especially because people with higher glucose at onset have more room to decrease. Therefore, Figure 34 is best described as a descriptive response kernel: it gives the observed shape and timing of glucose around detected exercise-like episodes, but it does not identify what would have happened under a formal do-exercise intervention.

14.2 Cadence and Strain Heterogeneity

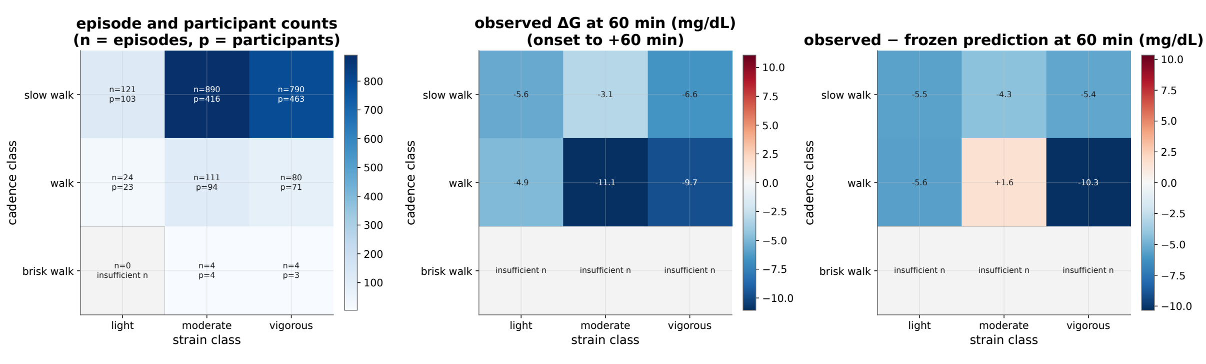


Figure 44: Exercise glucose response by cadence and strain class. Left: episode (n) and participant (p) counts. Centre: observed glucose change from refined onset to +60 min (ΔG_{60}, mg/dL). Right: observed minus frozen-model prediction at +60 min (signed residual, mg/dL). Cells below the eligibility threshold are marked “insufficient n”. Run-like cadence is folded into brisk walk when sparse. Text colour adapts to cell luminance for readability.

The dataset is numerically dominated by slow-walk and walk episodes. Brisk-walk and run-like categories are sparse, and some cells are suppressed because they do not meet the minimum sample size rule. This already shapes the interpretation: the exercise analysis is primarily an ambulatory, low to moderate activity analysis. It is not a robust analysis of running or high cadence exercise.

The centre panel shows the observed glucose change at 60 minutes. The largest decrease appears in the walk/moderate and walk/vigorous cells, rather than in the sparse brisk walk cells. This suggests that the post exercise glucose drop is not controlled by cadence alone. A moderate walking bout can be more glucose relevant than a higher cadence episode if it is longer, more sustained, or occurs at a higher starting glucose level.

The right panel compares observed glucose with the frozen model prediction at 60 minutes. Negative values mean that observed glucose is below what the frozen model predicted, so the model overestimates glucose after the episode. The walk/vigorous cell shows the strongest negative residual, indicating that the frozen model under-represents the glucose-lowering pattern in this subgroup. The most informative cells are those where both movement and strain are substantial, but the sparse high-cadence cells mean that conclusions about running-like activity remain unsupported.

14.3 Glycemic Stratum

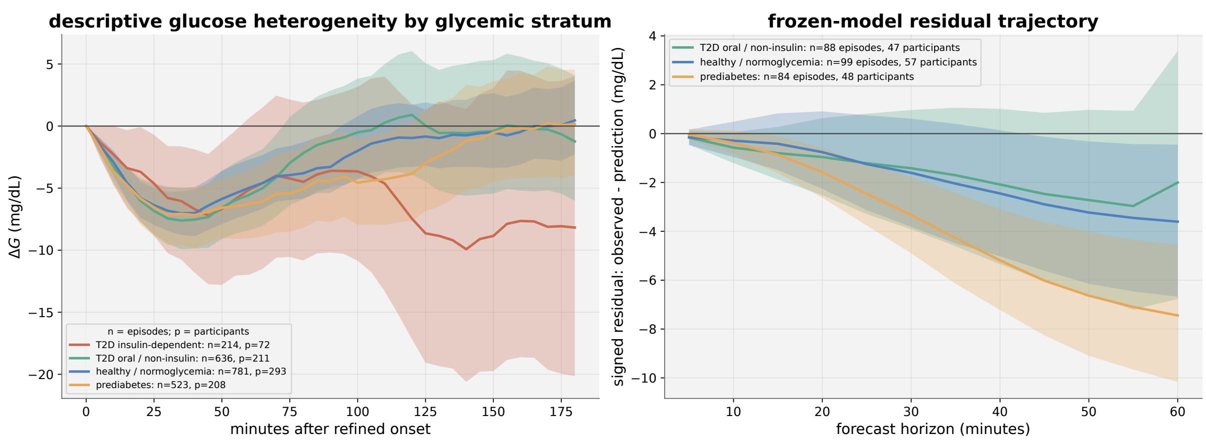


Figure 45: Exercise glucose response by glycemic stratum. Descriptive heterogeneity across normoglycaemia, prediabetes, T2D oral/non-insulin, and T2D insulin-dependent strata. Right panel: frozen-model residual trajectory (observed − prediction) from the matched residual diagnostic; available only for non-insulin strata. Differences are not causal effects of diabetes status.

The insulin-dependent group has wider uncertainty and should be interpreted with the most caution. This is expected because insulin timing and insulin-on-board are not available in AI-READI. The healthy and prediabetes groups show smoother average trajectories, while the T2D oral/non-insulin group appears more variable. This variability may reflect higher baseline glucose, greater glycemic instability, or medication-related heterogeneity.

14.4 Baseline Glucose Dependence

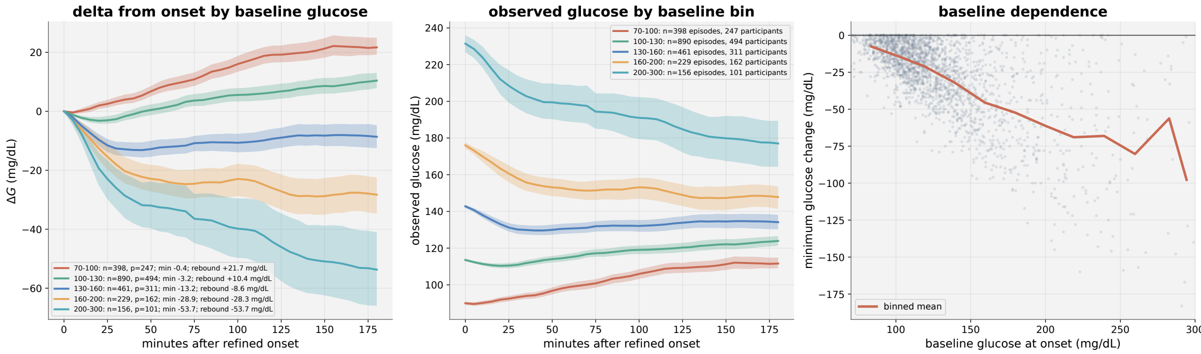


Figure 46: Exercise glucose response by baseline glucose at episode onset. Left: ΔG trajectory by 30 mg/dL baseline bin. Centre: observed glucose level after onset. Right: LOWESS trend of within-episode glucose nadir vs. baseline glucose. Baseline dependence is descriptive and may reflect regression to the mean, safety gating, and selection, not a causal dose-response.

The figure shows that baseline glucose at exercise onset strongly structures the observed response. The left panel plots (ΔG) by baseline glucose bin. Low baseline glucose bins show little drop and may even rebound upward later in the window. Higher baseline bins show much larger decreases. The 160–200 mg/dL and 200–300 mg/dL groups show the strongest downward trajectories. This is clinically intuitive because high starting glucose leaves more room for glucose to fall, while low starting glucose is constrained by physiological and behavioral safety limits.

The centre panel confirms that even after the drop, high-baseline groups often remain at higher absolute glucose than lower-baseline groups. This means that a large negative (ΔG) does not necessarily imply hypoglycemia. A participant starting above 200 mg/dL can drop substantially and still remain in or above the target range.

This baseline dependence may reflect true physiology, but it can also reflect regression to the mean and selection. If episodes are selected during periods of high glucose, later values may decrease even without exercise. This is why baseline glucose should be treated as a key matching variable in any natural-experiment analysis and as a safety gate in any planning response. It also explains why a single average exercise response is misleading: the expected observed trajectory depends strongly on where glucose starts.

14.5 Anthropometric and Metabolic Quartile Subgroups

Each figure below uses a three-panel layout: left panel shows the ΔG response kernel over time (0–180 min after refined onset) with participant-clustered bootstrap 95 % bands; centre panel shows ΔG_{60} (filled circles) and minimum ΔG (open squares) with 95 % bootstrap error bars; right panel shows the observed minus frozen prediction at +60 min. Legend labels include the quartile name, effective participant-level value range (with unit), episode count (n), and participant count (p).

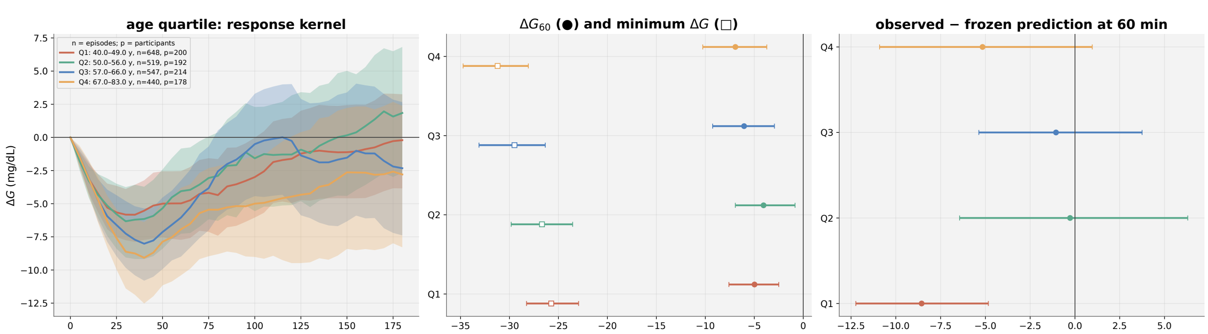


Figure 47: Age quartile: exercise glucose response kernel. Descriptive subgroup comparison across age quartiles (unit: years). Only quartiles with ≥ 20 episodes and ≥ 10 participants are shown.

The response curves suggest that age groups differ in both drop magnitude and recovery shape, but the uncertainty bands overlap substantially. The main analytical point is that age alone does not define a clean exercise-response phenotype. Older participants may have different baseline glucose, medication exposure, cardiovascular response, and activity type. Therefore, age should be treated as a stratification variable for robustness and fairness, not as a direct explanatory mechanism. Of older or younger quartiles show larger observed-minus-predicted deviations, this would suggest that the frozen model is less calibrated for those age groups after exercise-like activity. The figure should therefore be interpreted as a model-audit tool more than as evidence for an age-specific exercise physiology.

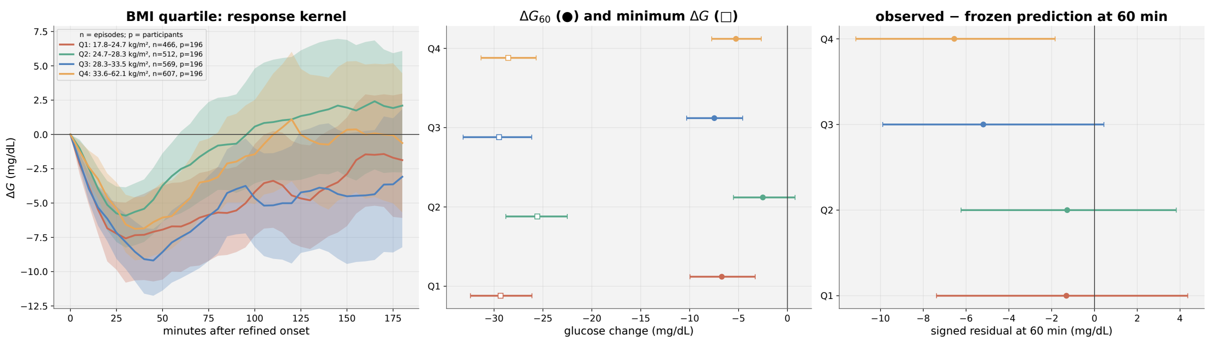


Figure 48: BMI quartile: exercise glucose response kernel. Descriptive subgroup comparison across BMI quartiles (unit: kg/m^2).

Higher BMI does not automatically imply a larger or smaller glucose decrease in this descriptive setting. This is expected because BMI is a broad anthropometric marker and does not directly encode insulin sensitivity, fitness, exercise intensity, meal timing, or medication exposure.

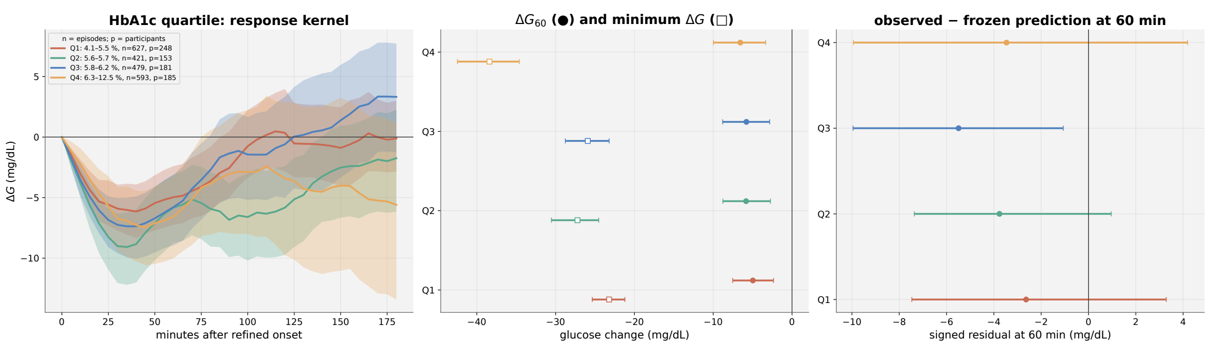


Figure 49: HbA1c quartile: exercise glucose response kernel. Descriptive subgroup comparison across HbA1c quartiles (unit: %).

This is one of the more clinically relevant subgroup plots because HbA1c reflects longer-term glycemic exposure. Higher HbA1c groups tend to have higher baseline glucose and more variable glucose trajectories, so a larger observed drop can partly reflect starting higher rather than having a stronger physiological exercise response. The key analytical point is that HbA1c may influence the apparent exercise kernel indirectly through baseline glucose. Therefore, HbA1c subgroup differences should be interpreted together with Figure 37. A high-HbA1c quartile showing a larger drop does not necessarily mean exercise is more effective in that group. It may mean that the group starts higher and has more downward room.

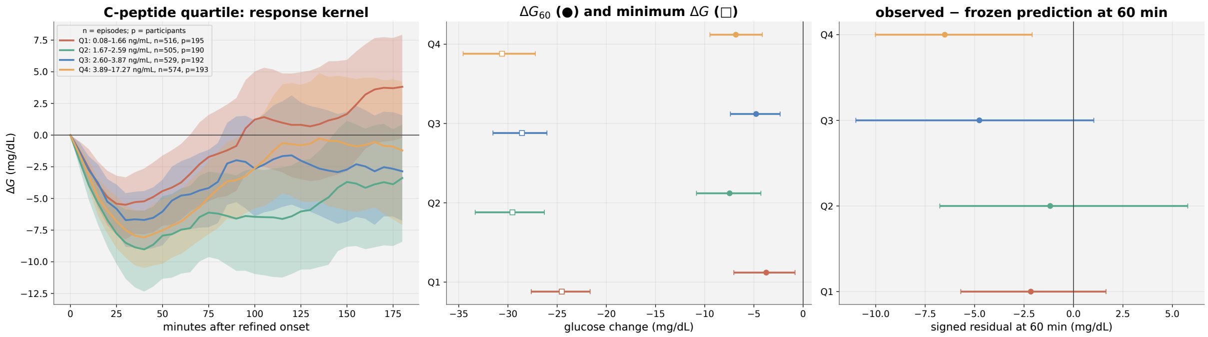


Figure 50: C-peptide quartile: exercise glucose response kernel. Descriptive subgroup comparison across C-peptide quartiles (unit: ng/mL).

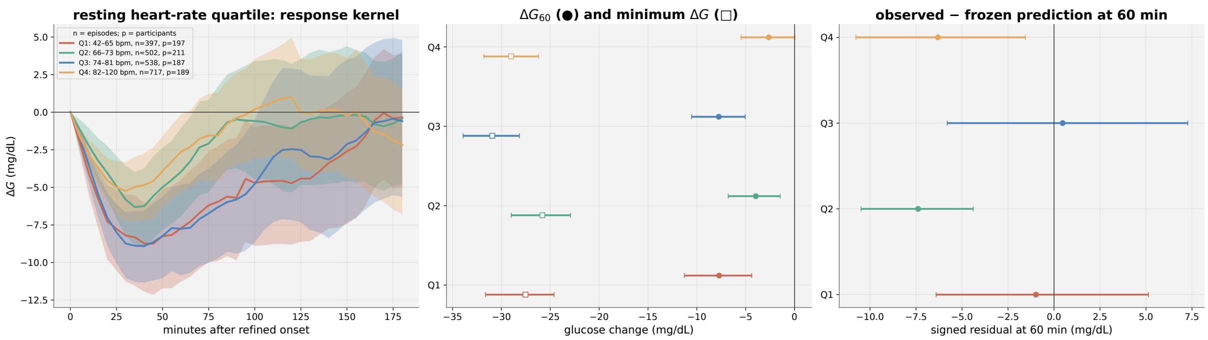


Figure 51: Resting heart-rate quartile: exercise glucose response kernel. Descriptive subgroup comparison across resting HR quartiles (unit: bpm).

Figure 42 stratifies by resting heart-rate quartile. Resting HR is relevant because the detector defines strain relative to a participant-specific HR floor. Participants with high resting HR may reach high absolute HR values with different physiological meaning than participants with low resting HR.

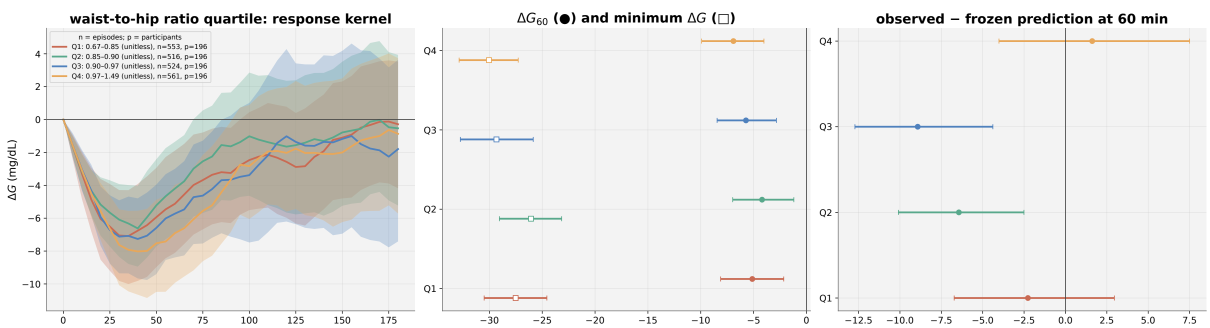


Figure 52: Waist-to-hip ratio quartile: exercise glucose response kernel. Descriptive subgroup comparison across waist-to-hip ratio quartiles (dimensionless).

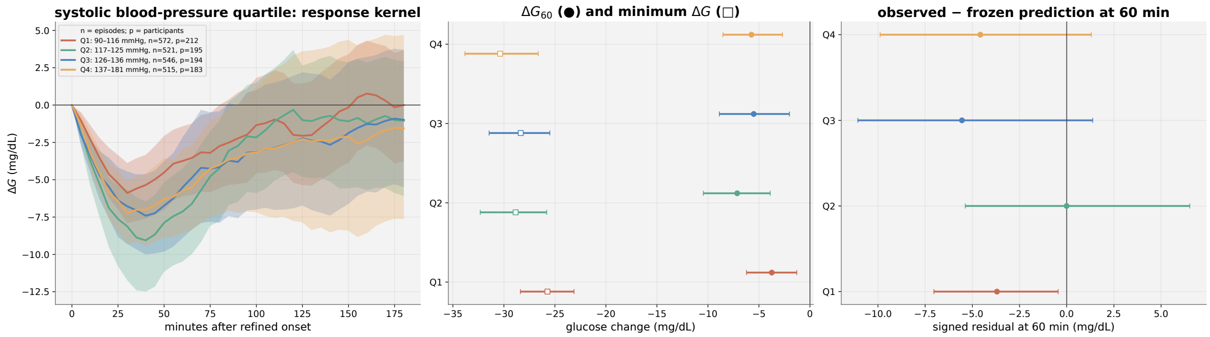


Figure 53: Systolic blood-pressure quartile: exercise glucose response kernel. Descriptive subgroup comparison across systolic BP quartiles (unit: mmHg).

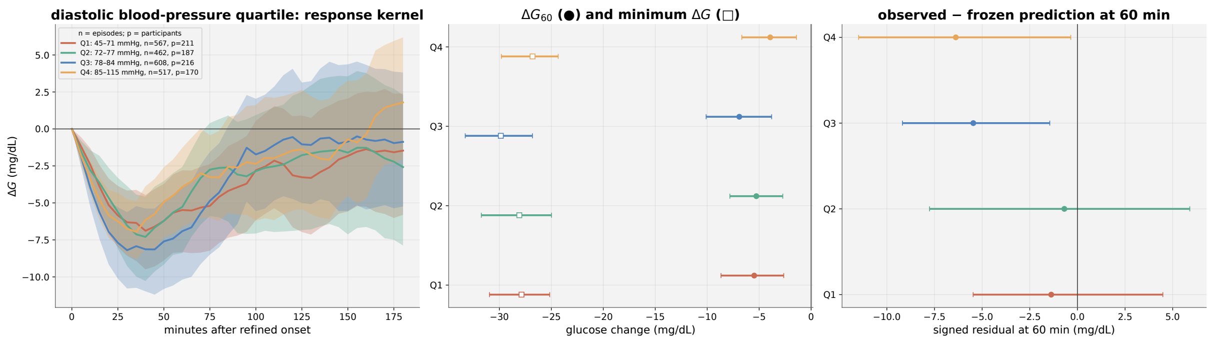


Figure 54: Diastolic blood-pressure quartile: exercise glucose response kernel. Descriptive subgroup comparison across diastolic BP quartiles (unit: mmHg).

14.6 Categorical Subgroups

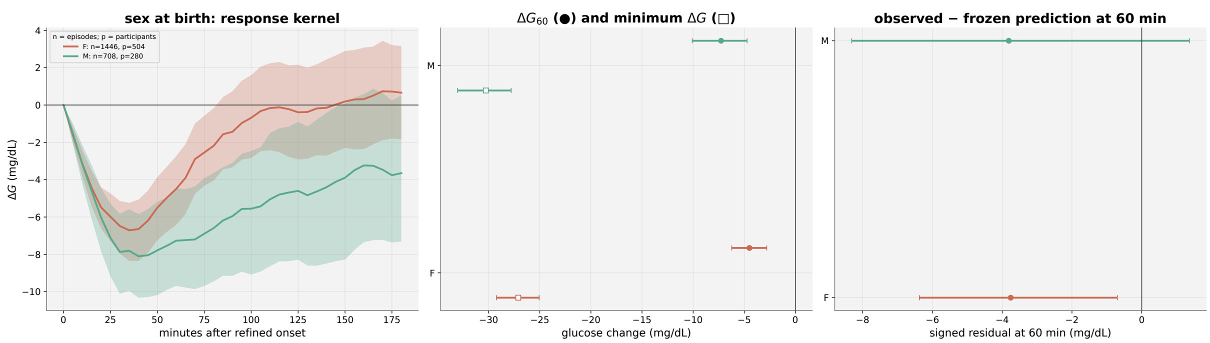


Figure 55: Sex at birth: exercise glucose response kernel.

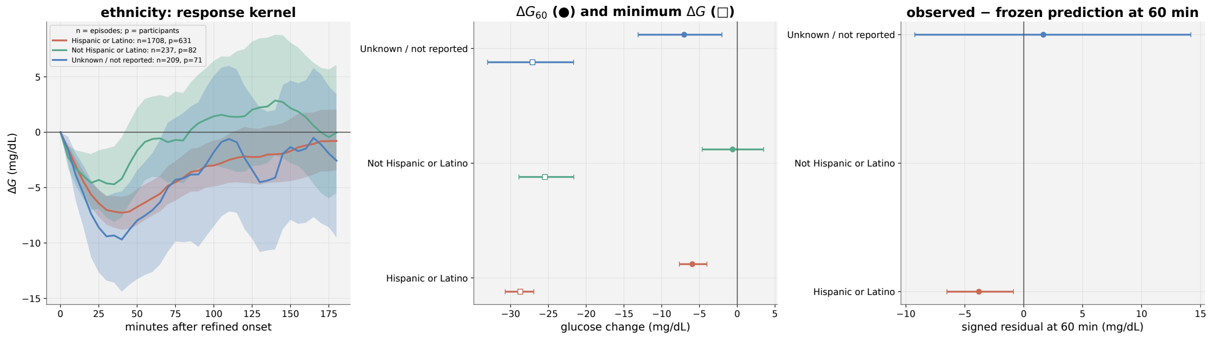


Figure 56: Ethnicity: exercise glucose response kernel. Categories derived from the four AI-READI ethnicity indicator columns (`hispanic_or_latino`, `not_hispanic_or_latino`, `dont_know`, `prefer_not_to_answer`). Groups with insufficient episodes are suppressed.

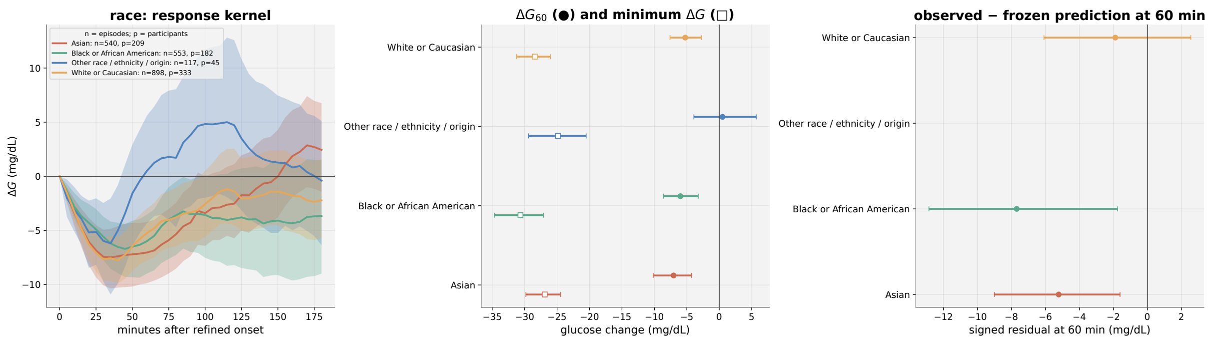


Figure 57: Race: exercise glucose response kernel. Derived from the primary race field; “Prefer not to say” and “Not answered” are unified as “Unknown / not reported”.

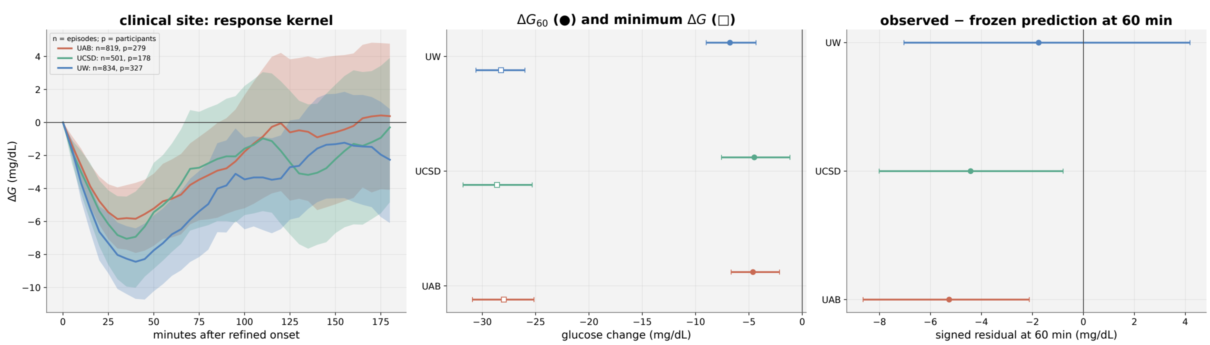


Figure 58: Clinical site: exercise glucose response kernel.

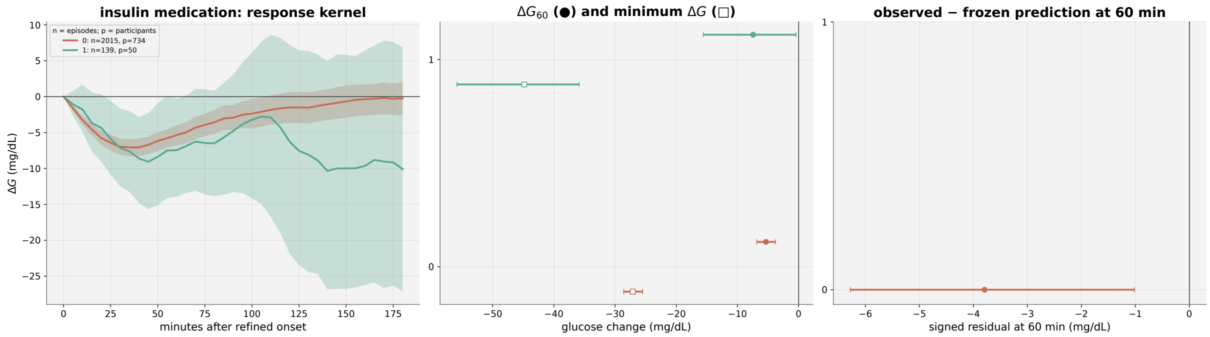


Figure 59: Insulin medication: exercise glucose response kernel. Episodes stratified by whether the participant was on insulin therapy at the time of the study.

This is one of the most important cautionary plots. Insulin users are fewer and have wider uncertainty. More importantly, insulin timing and dose are not observed in AI-READI, so the post-exercise glucose trajectory in insulin users is heavily confounded.

A large drop in insulin users cannot be attributed to exercise alone because unobserved insulin-on-board could be the direct driver. Therefore, insulin-user results should remain descriptive only. The model residual panel is still useful: if the frozen model misses insulin-user trajectories more strongly, this supports the earlier decision to avoid causal interpretation in this subgroup

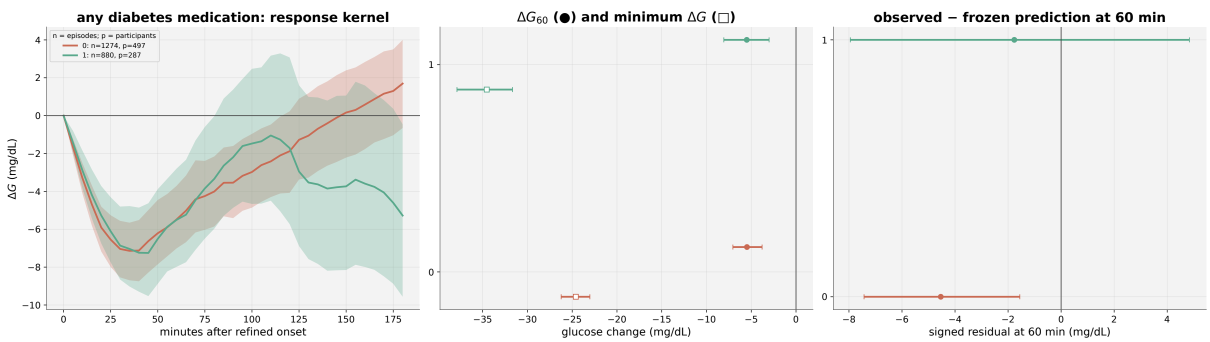


Figure 60: Any diabetes medication: exercise glucose response kernel. Episodes stratified by whether the participant was on any glucose-lowering medication.

The right residual panel is the most relevant for forecasting. If medication users have more negative observed-minus-predicted residuals, the frozen model may overestimate post-exercise glucose in this group. This would support including medication status or medication interactions in an exercise-aware correction model. However, the finding remains descriptive because medication status is confounded with diabetes severity and baseline glucose.

14.7 Frozen-Model Residual Summary

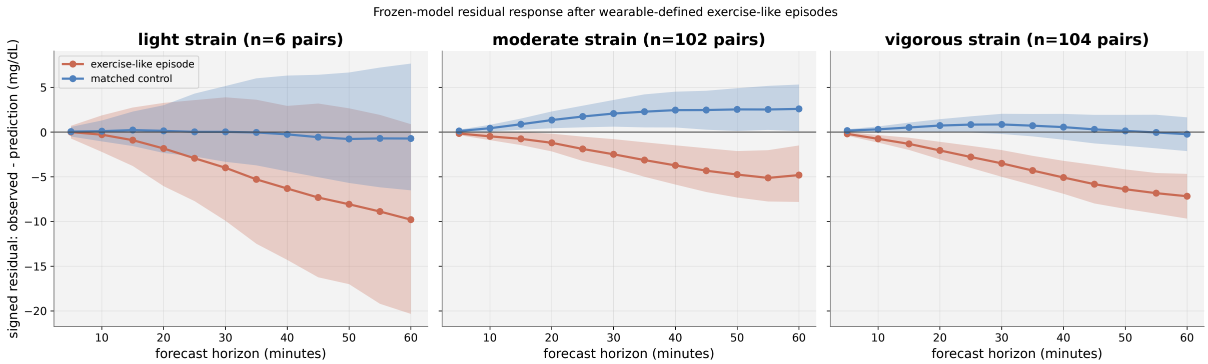


Figure 61: Frozen-model residual trajectory: episodes vs. matched controls. Signed residual is observed glucose minus frozen-model prediction. Positive values indicate the frozen model underpredicts observed glucose. Confidence intervals are participant-clustered bootstrap 95 % bands.

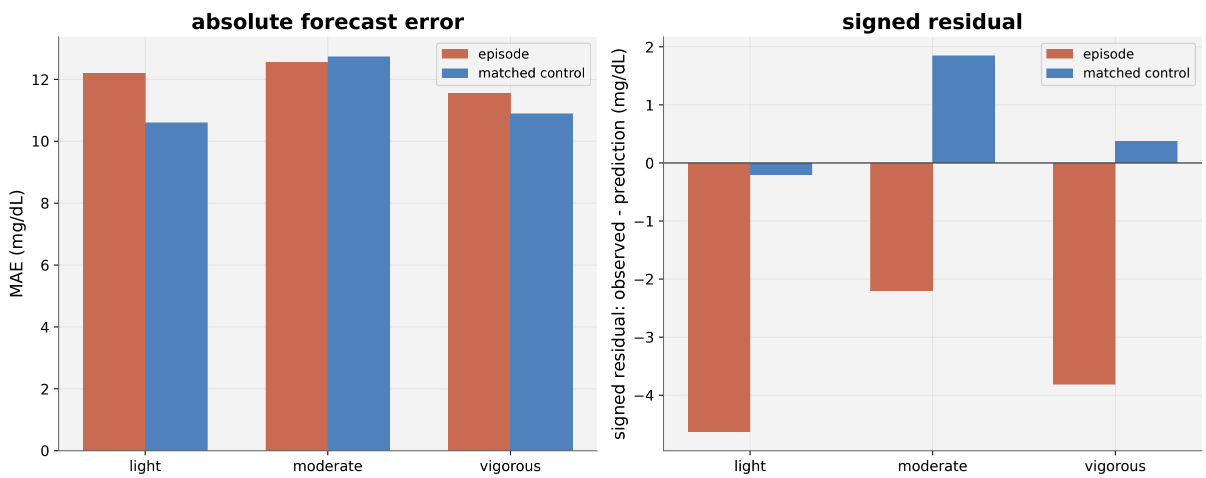


Figure 62: Forecast error summary by strain class. Left: mean absolute error (MAE, mg/dL) for exercise-like episodes and matched controls. Right: signed residual (observed − prediction, mg/dL). Light-strain estimates are sparse and are not interpreted.

14.8 Exercise Planning Head: Response Surface and Horizon Shapes

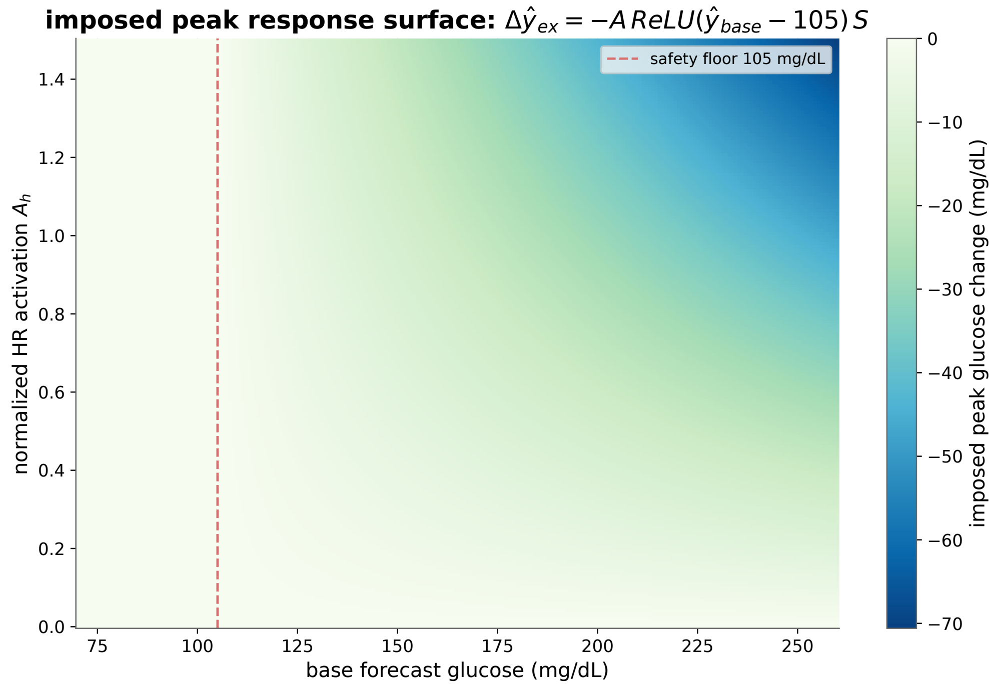


Figure 63: Imposed planning-head response surface. The surface shows $\Delta\hat{y}_{\text{ex}} = -A\,\text{ReLU}(\hat{y}_{\text{base}} - 105)\,S$ as a function of base-forecast glucose and HR activation A. The safety floor at 105 mg/dL suppresses the overlay at low glucose. Magnitude is imposed, not identified from AI-READI data.

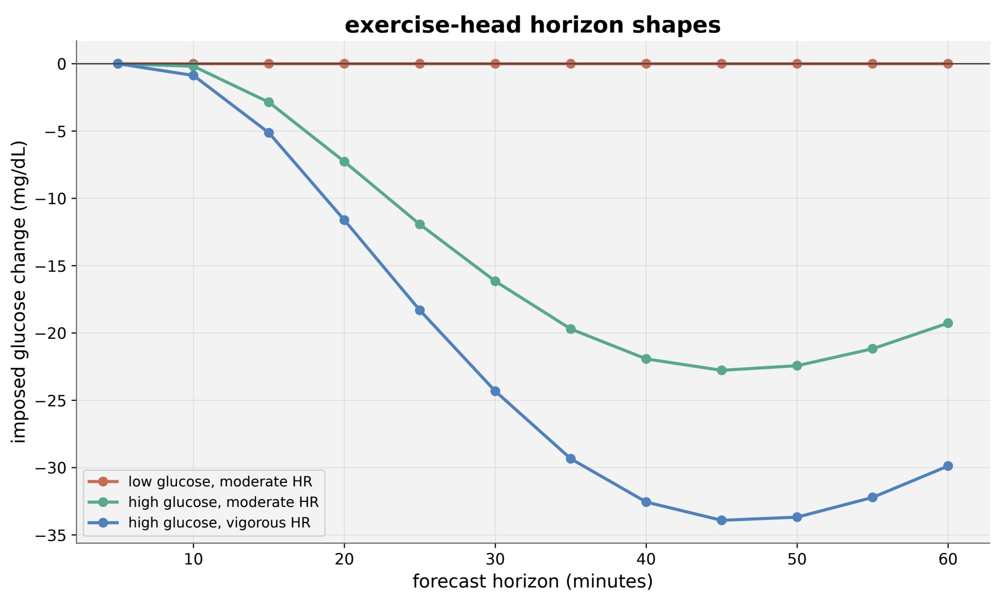


Figure 64: Imposed planning-head horizon shapes. Three canonical scenarios: low glucose with moderate HR (safety gate active), high glucose with moderate HR, and high glucose with vigorous HR. No causal claim; the shape is prescribed by the planning head formula.

15 Exercise Counterfactual: Imposed planning response, and a deployable advisory

15.1 Motivation for counterfactual

The frozen model residual analysis found no exercise specific MAE penalty, but it did find a horizon growing positive signed residual in moderate and vigorous wearable defined episodes. Under the notebook convention $b_h = \hat{q}^{(h)}_{50,t} - G_{t+h}$, this means that the model over-predicts the later glucose trajectory and under represents the exercise-associated drop. This motivates a natural question: what would glucose do if this person exercised?

That question is not answered by detector validation. Detection is a signal processing problem: steps and HR can locate an ambulatory exercise-like state. Identification is a confounding and overlap problem: comparable non exercise outcomes must exist for the same glucose state, time, and participant context. A detector can be accurate while the magnitude of the associated counterfactual response remains non identifiable.

15.2 Why the exercise effect size cannot be reliably estimated on AI-READI

Here we tested whether the observed glucose change after detected exercise like episodes could be converted into a deconfounded estimate of the exercise response. The idea was to compare each exercise episodes with matched non exercise windows that had a similar baseline glucose level and occurred at a similar time of day. However, this matching strategy exposed a limitation of the AI-READI data. Cross patient matching produced many controls with similar baseline glucose and time of day, but most controls came from different participants. With the ± 10 mg/dL glucose caliper, 92.23% to 97.24% of controls were drawn from another participant across the five g_0 bins. In the 70–100 mg/dL bin, the apparent effect was significantly positive, $+2.50$ mg/dL with 95% CI $[+0.73, +4.31]$. Because the contrast is largely between people, this is a signature of residual between person confounding, not evidence of a physiological rise.

Within participant matching removed that low-g_0 result. The corresponding estimates were $+0.33$ mg/dL $[-5.40, +6.19]$ in the 70–100 bin and -0.10 mg/dL $[-3.53, +3.23]$ in the 100–130 bin. The cross-patient low-g_0 result therefore does not survive within-patient matching. This is a selection aware statement, not evidence of a zero effect. Comparability came at the cost of positivity: only 48 of 510 low-g_0 bouts were retained (9.4%), and the two highest-g_0 bins retained only 17 and 7 bouts. Retention in the 200–300 mg/dL bin was 7.1%.

Table 45: Stage 0 matching audit. Cross-patient matching supplies overlap but relies almost entirely on other participants. Within-patient matching supplies comparability but loses positivity. The low-g_0 positive contrast does not survive the within-patient restriction.

Scheme and g_0 bin	Caliper	Controls	Cross-patient	Effect at 60 min, mg/dL
Cross, 70–100	± 5	15,141	97.60%	$+3.78$ $[+2.09, +5.62]$
Cross, 70–100	± 10	15,300	96.12%	$+2.50$ $[+0.73, +4.31]$
Within, 70–100	± 5	364	0%	$+0.33$ $[-5.40, +6.19]$
Within, 100–130	± 5	1,272	0%	-0.10 $[-3.53, +3.23]$
Within, 160–200	± 5	82	0%	$+9.91$ $[-16.95, +37.49]$
Within, 200–300	± 5	44	0%	-31.45 $[-63.24, -2.26]$

The within-patient curve is non-monotonic and changes sign across baseline bins. A true exercise counterfactual is not identifiable from AI-READI because the dataset is observational and does not contain randomized or externally assigned exercise interventions. The detected exercise episodes are inferred from wearable signals, mainly steps and heart rate, rather than recorded as ground-truth exercise sessions with known type, intensity, and intention. More importantly, the post exercise glucose response is affected by several unobserved or only partially observed factors, including recent meals, carbohydrate intake, insulin timing, insulin on-board, medication effects, stress, sleep, and baseline glucose trend. Therefore, comparing exercise windows with non exercise windows does not isolate the effect of exercise alone. Cross participant matching provides enough controls but compares different people with different physiology, while within participant matching improves comparability but leaves too few matched controls to estimate a stable effect. For this reason, AIREADI can support wearable defined exercise detection and predictive residual analysis, but it cannot provide a reliable causal estimate of the exact glucose change that would have occurred under a do exercise intervention.

15.3 Counterfactual: Exercise head with imposed magnitude

The magnitude is imposed from the descriptive, associational moderate strain kernel of 16.66 mg/dL, while AI READI supplies timing, shape, and live glucose state gating. The output is an imposed planning response, not a causal estimate. At forecast horizon h, the head is

$$\Delta\hat{y}_{\mathrm{ex},h} = -A_h R_h S, \tag{32}$$

with

$$A_h = \frac{[k * \mathrm{ReLU}(\Delta\mathrm{HR} - 15)]_h}{A_{\mathrm{ref}}}, \tag{33}$$

$$R_h = \mathrm{ReLU}\left(\hat{y}_{\mathrm{base},h} - 105\right), \tag{34}$$

$$S = \frac{16.7}{160 - 105} = 0.303636. \tag{35}$$

Here A_h is an HR-intensity activation with a 15-bpm dead zone and a causal short-lag kernel k; R_h is a one-sided glucose-state safety ramp; and S is one cohort-wide scalar. There is no per-participant sensitivity term. Participant exercise sensitivity was not recoverable from the state, and the insulin-resistance hypothesis closed as a negative result: baseline glucose was the only predictor of the subsequent drop. So since AI-READI could not identify the true exercise magnitude, I imposed a reasonable magnitude from the descriptive kernel, then used AI-READI wearable signals to control timing, shape, and safety gating.

The attempted gain optimization is itself informative. Training against the scenario loss moved S from 0.303636 to 0.123238. The scenario pinball loss was 31,213 times the magnitude-prior loss initially and 496 times the prior loss at the final epoch. This reproduces the known magnitude-muting dynamic. It corroborates that the imposed magnitude cannot be learned through this objective, so the drifted checkpoint was rejected and S was frozen at 0.303636.

With frozen gain, the median effect in the 70–100 mg/dL bin is 0.0 mg/dL because of safety gating. The response grows monotonically in magnitude with g_0, with $\mathrm{corr}(g_0, \Delta\hat{y}_{\mathrm{ex}}) = -0.9402$ at the 45-minute peak. In the 160–200 mg/dL bin, the median peak is -17.50 mg/dL at 45 minutes and has begun to recover by 60 minutes (-14.05 mg/dL). Forecast MAE is unchanged at 10.0665 mg/dL. The realized 17.50 mg/dL peak can exceed the 16.66 mg/dL static anchor because the live R_h trajectory can exceed the static reference $R = 55$; it was reported as observed and not retuned.

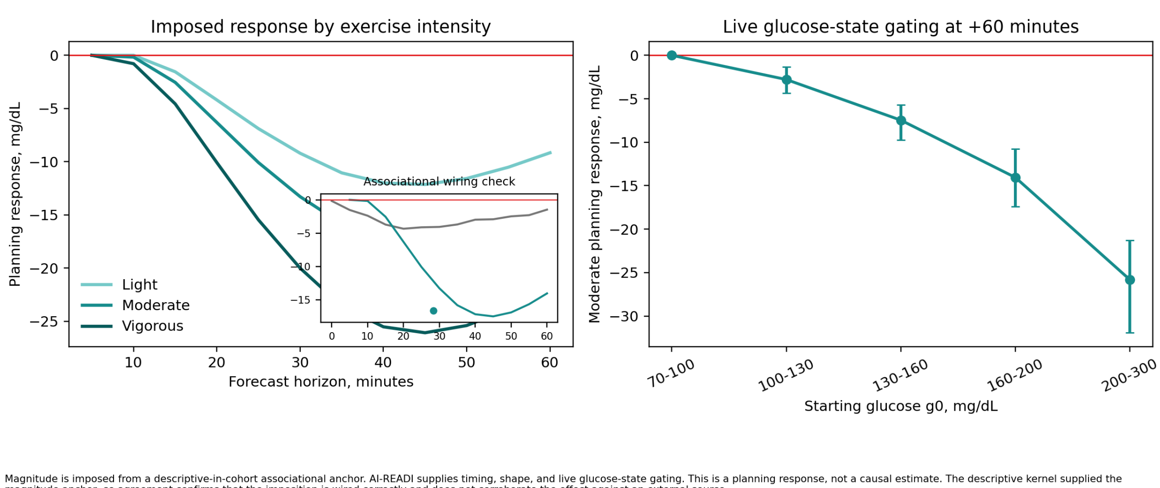


Figure 65: Imposed planning response Dose response across wearablevdefined HR intensity and glucose-state gating at 60 minutes. Magnitude is imposed; AIREADI supplies timing, shape, and live glucose state gating. This is a planning response, not a causal estimate. Comparison with the descriptive kernel is a wiring check, not independent validation, because that kernel supplied the magnitude anchor.

Figure [66](#) is an answer to the question: given the current forecast and exercise state defined on wearable signals, what would the imposed "do exercise" look like?

This figure shows three example forecast cards where the frozen SSM-CGM forecast is overlaid with an imposed "do exercise" response: the overlay lowers glucose when baseline glucose is high, is blocked when hypoglycemia risk is present, and remains descriptive rather than causal because the exercise magnitude was manually imposed rather than identified from AI-READI.

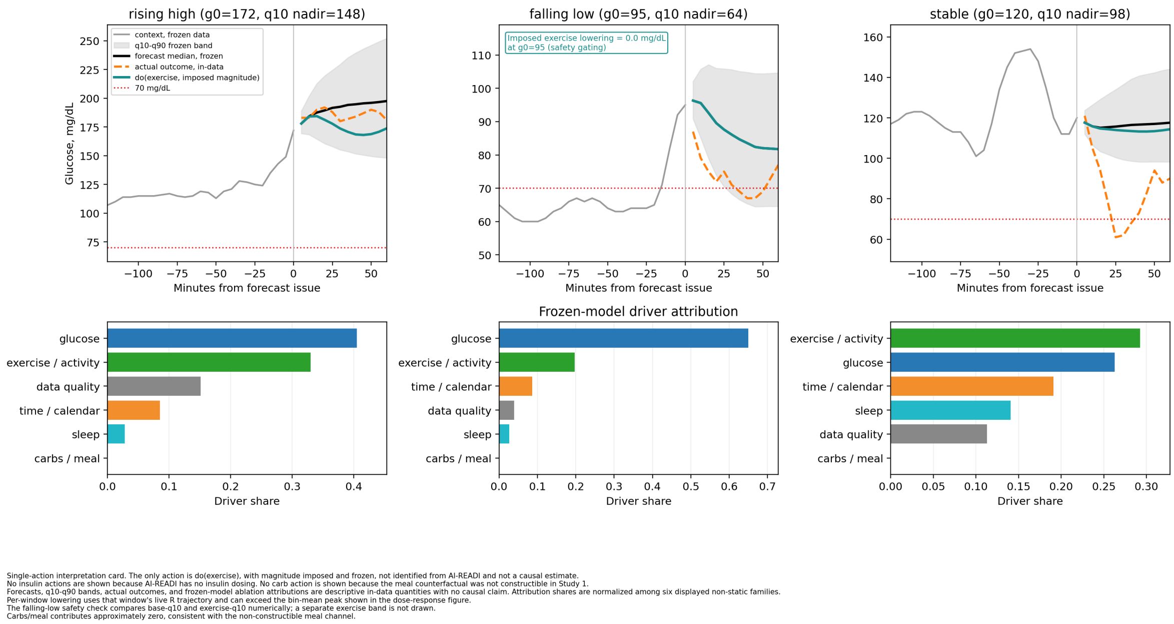


Figure 66: Single action interpretation cards Rising-high, falling-low, and stable states show the frozen forecast, q_{10}–q_{90} band, actual outcome, imposed do(exercise) overlay, and frozen model driver attribution. No insulin action is shown because dosing is unavailable, and no carbohydrate action is shown because the meal counterfactual was not constructible. The falling-low state has $g_0 = 95$ mg/dL and imposed lowering of 0.0 mg/dL, demonstrating safety gating. Per-window lowering uses the window's live R_h and can exceed the bin-mean peak. The cards are descriptive and make no causal claim.

15.4 Identified, deployable exercise aware Hypoglycaemia advisory

In this part we focus on identified predictive advisory built on the frozen model and the measured signed bias trajectory, evaluated against the model's own forecasts and observed outcomes without a control arm. Its scope is 3,813 moderate or vigorous non-insulin windows from 470 episodes, including 105 observed hypoglycaemic outcomes.

Claim 1: ranking. The forecast-only risk-averse q_{10} lowest point ranks observed exercise-associated hypoglycaemia substantially better than current glucose. Pooled AUROC is 0.974 without correction and 0.975 with correction, versus 0.833 for current g_0. Corrected and uncorrected curves overlap because AUROC is invariant to the horizon-specific rank-preserving shift. The ranking gain comes from the streaming encoder reading impending exercise-like dynamics, including rising HR. Median warning lead time is 30 minutes.

Claim 2: calibration at 70 mg/dL. Applying the signed bias correction at the fixed 70 mg/dL threshold raises sensitivity from 0.429 to 0.762 and rescues 35 near miss hypoglycaemic outcomes. The gain is concentrated in vigorous windows, where sensitivity rises from 0.179 to 0.846 and the bias reaches 7.17 mg/dL at 60 minutes; 26 of the 35 rescued outcomes are vigorous. The gain has a clear price: false positives rise from 28 to 119, specificity falls from 0.992 to 0.968, and positive predictive value falls from 0.616 to 0.402. The correction can rescue only cases whose uncorrected nadir lies in the near-miss band between 70 mg/dL and the strain-specific bias-adjusted upper edge, 75.1 mg/dL for moderate and 77.2 mg/dL for vigorous windows. The 25 remaining missed outcomes are the frozen model's residual blind spot.

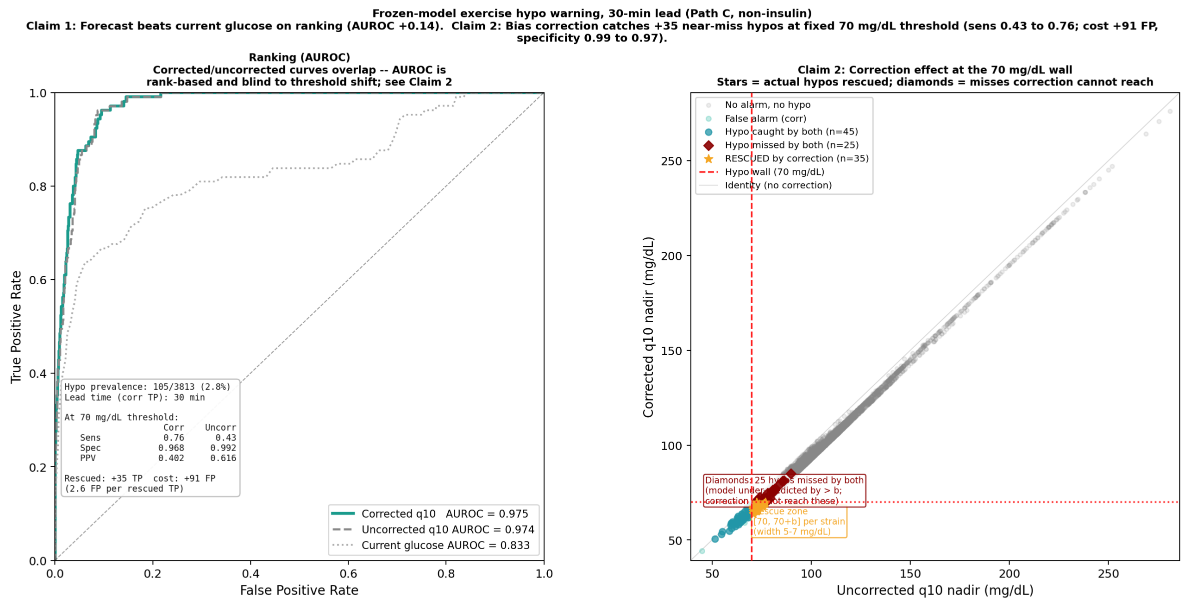


Figure 67: Exercise-aware hypoglycaemia advisory. The ROC panel evaluates Claim 1, that the frozen forecast ranks observed hypoglycaemic outcomes better than current glucose. The rescued-zone panel evaluates Claim 2, that signed-bias correction improves calibration at the fixed 70 mg/dL threshold for near misses, with an accompanying false-alarm cost. This is a bias-corrected predictive warning on the frozen model, with no counterfactual and no causal claim.

15.5 Synthesis and Limitations

The complete exercise part separates into four contributions. First, a wearable only detector identifies ambulatory exercise like episodes without glucose in the detection path. Second, we establishe that the counterfactual magnitude is not identifiable on AI-READI because cross patient matching offers overlap without comparability, while within patient matching offers comparability without positivity. This is a transferable design finding, analogous to the need for participant disjoint evaluation splits. Third, Path 15.3 provides an imposed planning response whose timing, shape, and glucose state gating are data derived but whose magnitude is not estimated. Fourth, Path 15.4 provides an identified advisory with a calibration benefit at a prespecified threshold.

The detector remains wearable defined rather than ground truth labelled and is primarily sensitive to ambulatory activity. Meals, time of day, medication, and selection can influence the descriptive kernel. Path 15.3 inherits the $n = 118$ moderate episode anchor and its imposed magnitude should not be read as a causal estimate. Path 15.4 has 105 hypoglycaemic outcomes, and its correction improves threshold calibration rather than ranking. Insulin dosing is unobserved, so insulin treated participants are excluded from all causal scope.

Table 46: Wearable-defined exercise analysis summary. The detector, residual audit, identifiability result, imposed planning response, and identified advisory are separate claims.

Item	Result	Interpretation
Clean episodes	2,171 across 788 participants	Primary wearable-defined set
Strict recovery	1,015 across 574 participants	Sensitivity subset, not a detection gate
Refined onset	Median 30 min earlier	Backtracks to sustained step rise
Negative controls	HR: 8.01 vs 0.27/−0.46 bpm; steps: 29.44 vs −0.93/−0.23 steps/min	Real onsets separate from shifted/random-awake controls
Cadence	1,859 slow; 223 walk; 81 sub-threshold; 6 brisk; 2 run-like	Slow-walk episodes dominate
Glucose kernel	16.66 mg/dL; nadir 28.31 min	Descriptive-associational anchor only
Residual audit	No MAE gap; bias gap +4.19 vigorous and +4.05 moderate	Provisional residual headroom
Stage 0	Comparability and positivity cannot be achieved together	Magnitude not identifiable
Path B	$S = 0.303636$; $r = -0.9402$; $\Delta\text{MAE} = 0$	Coherent imposed planning response
Path C	AUROC 0.975 vs 0.833 current g_0; 35 outcomes rescued	Identified advisory with a false-alarm cost

16 Environmental Exposure as an External Modality for Overnight Glucose Stability

16.1 Motivation

The previous sections evaluate information streams that are internal to the forecasting task: CGM history, wearable signals, sleep context, static participant descriptors, and derived event states. This section asks whether a more external modality, the bedroom environment, contains additional information about overnight glucose stability. The analysis was intentionally staged. First, environmental exposure was tested as a descriptive candidate signal associated with overnight glycemic outcomes. Second, the only surviving association was tested for forecasting value using a frozen SSM-CGM residual head. This separation matters: an association with a glucose summary does not by itself justify adding a new input channel to the forecasting model.

16.2 Construction of Bedroom-Valid Exposure Nights

Environmental sensors were fixed in the home, so a raw sensor stream is not automatically a personal exposure stream. The primary analysis therefore kept only high-confidence bedroom or bedside placements and excluded living room, kitchen, office, hallway, and unknown placements. Sleep windows were derived from wearable sleep-stage signals, and a valid night required overlapping sleep, CGM, and environmental coverage. This filtering was required because a fixed home sensor outside the bedroom may measure the dwelling environment rather than the participant's sleep exposure.

Table 47: Cohort construction for bedroom environmental exposure analysis.

Filtering step	Count
Selected AI-READI cohort	1591 participants
High-confidence bedroom placement	574 participants
Exposure-valid participants	566 participants
Exposure-valid nights	4419 nights
Median nights per participant	9 nights
Mean nights per participant	7.81 nights

Table 48: Environmental exposure and overnight glucose outcome groups.

Group	Variables
Bedroom environment	`temp_mean` (mean temperature), `temp_sd` (temperature variability), `hum_mean` (mean relative humidity), `pm25_auc` (integrated particulate exposure), `voc_spike_burden`, `nox_spike_burden`, and `night_light_exposure`.
Overnight glucose	Mean glucose, glucose standard deviation, coefficient of variation, glucose range, time below 70 mg/dL, time above 180 mg/dL, wake-to-morning rise, and dawn-associated rise.

16.3 Within-Participant Association Model

Each exposure-outcome pair was fit with a within-between mixed-effects model:

$$Y_{i,n} = \beta_W(x_{i,n} - \bar{x}_i) + \beta_B\bar{x}_i + u_i + \varepsilon_{i,n}, \tag{36}$$

where $Y_{i,n}$ is the glycemic outcome for participant i on night n, $x_{i,n}$ is the environmental exposure on that night, $\bar{x}_i$ is the participant-level mean exposure, $x_{i,n} - \bar{x}_i$ is the within-participant exposure deviation, and u_i is a participant random intercept. The coefficient β_W asks whether the same participant has a different overnight glucose outcome on nights when their bedroom environment differs from their own usual environment. This target is more appropriate than a simple correlation because it reduces confounding by stable participant differences.

Because the grid contained many exposure-outcome pairs, raw p-values were not used as the decision rule. Benjamini-Hochberg FDR correction was applied across the primary exposure-outcome grid, and participant-clustered bootstrap confidence intervals were used because one participant can contribute multiple nights. A result was considered robust only if the FDR-corrected q-value was below 0.05, the participant-bootstrap 95% confidence interval excluded zero, and temporal negative controls did not reproduce the association. Shuffled-night and time-shift controls are the strongest controls in this setting. Non-bedroom and daytime analyses were treated only as specificity checks because personal proximity to the sensor is unknown outside bedroom sleep periods.

16.4 Primary Environmental Association Results

Across 56 primary exposure-outcome pairs, the only pair that cleared the gate was mean bedroom humidity against overnight glucose range:

$$\texttt{hum\_mean} \to \texttt{overnight\_glucose\_range}. \tag{37}$$

The within-participant coefficient was -0.361 mg/dL per 1 percentage-point higher humidity than participant usual (FDR $q = 0.0466$; participant-bootstrap 95% CI $[-0.600, -0.118]$; 566 participants; 4419 nights). In words, on nights when a participant's bedroom humidity was higher than their own average, their overnight glucose range tended to be narrower. The effect size is modest: a 10 percentage-point humidity difference corresponds to approximately 3.6 mg/dL lower overnight glucose range under the linear model. This is a glucose stability association, not evidence for a mean-glucose effect and not a causal estimate.

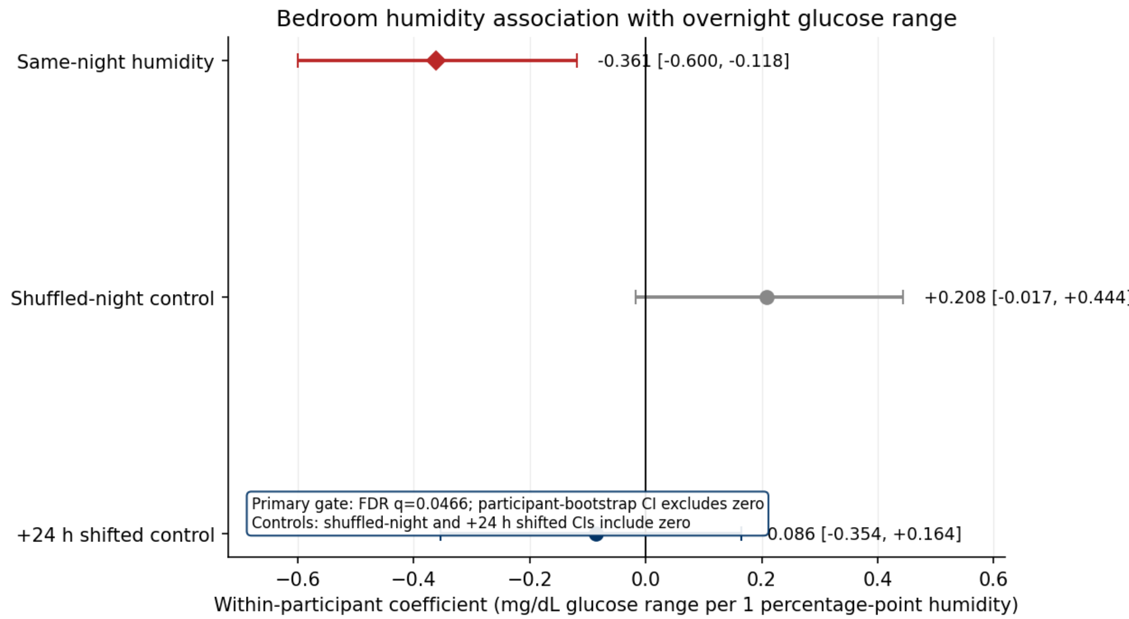


Figure 68: Primary within-participant association between bedroom humidity and overnight glucose range. Higher-than-usual bedroom humidity was associated with a narrower overnight glucose range, and the participant-bootstrap confidence interval excluded zero. Shuffled-night and +24 h shifted exposure assignments attenuated the association and had confidence intervals crossing zero.

16.5 Robustness, Nonlinear, and Heterogeneity Analyses

The humidity finding passed several follow-up checks. A humidity bounds audit found no invalid values outside the plausible 0–100% range. After trimming the most extreme 1% of within-person humidity deviations, the coefficient strengthened from about -0.361 to -0.441, with the bootstrap confidence interval still excluding zero. The real same-night coefficient was not reproduced by shuffled-night exposure assignment and was strongly attenuated under the +24 h shifted exposure control. The 12 h shift is not emphasized because its implementation reproduced the original alignment. Non-bedroom and daytime comparisons were used only as supportive specificity checks, not true negative controls.

Secondary nonlinear models tested whether temperature or humidity had a hidden U-shaped or absolute-deviation relationship:

$$Y \sim (x-\bar{x}) + (x-\bar{x})^2 + \bar{x} + u_i + \varepsilon, \tag{38}$$

and

$$Y \sim |x-\bar{x}| + \bar{x} + u_i + \varepsilon. \tag{39}$$

Across 48 nonlinear tests, zero were FDR-significant and robust. No quadratic or absolute-deviation effect was found for temperature or humidity, supporting the simpler linear within-person model for the main humidity result.

Temperature was then evaluated as a possible heterogeneous exposure rather than a single cohort-wide effect. The subgroup interaction grid contained 204 temperature tests; three model-based FDR-significant signals appeared for `temp_sd` against overnight time above 180 mg/dL, involving insulin use, top BMI quartile, and top sleep-fragmentation quartile. All participant bootstrap intervals included zero, so these were labeled model-based only and not participant-robust. Participant-level temperature slopes were similarly exploratory: among 566 participants, 102 were classified as positive temperature-sensitive candidates, 102 as negative candidates, 303 as not sensitive, and 59 had too few nights. Held-out early/late validation showed 63% sign agreement and near-zero slope correlation across 471 participants. Temperature therefore remains a hypothesis-generating heterogeneous exposure, not a validated personalization signal.

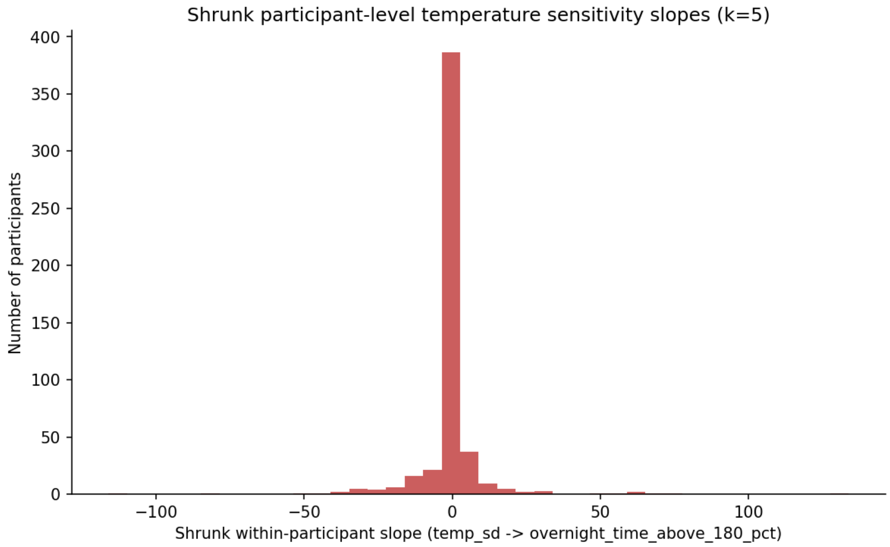


Figure 69: Exploratory participant-level temperature sensitivity estimates. Candidate positive and negative sensitivity groups were observed, but early-late night validation showed weak reproducibility, so these slopes were not treated as stable personalized phenotypes.

The notebook also profiled the candidate groups against clinical and medication context (Figs. 70–71). These plots are descriptive checks on whether the slope-derived groups were clinically interpretable. Positive temperature-sensitive candidates tended to have higher baseline glucose variability and somewhat higher HbA1c and BMI, while the negative and non-sensitive groups were closer together. Medication and study-group composition also differed across candidate groups, especially for insulin use and diabetes-status strata. This pattern makes the candidate labels biologically plausible enough to report, but it also reinforces the main caveat: the labels partly track baseline glycemic instability and cohort composition, and the held-out slope validation was weak. They therefore support hypothesis generation, not a deployed personalization rule.

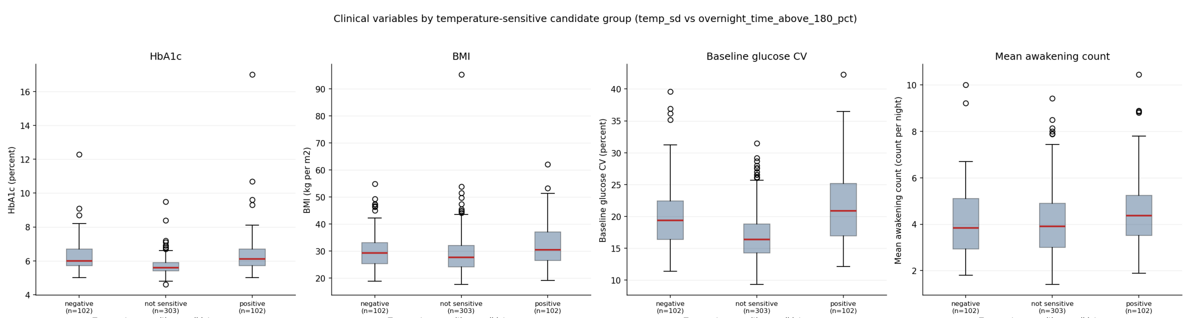


Figure 70: Clinical variables by exploratory temperature-sensitive candidate group for `temp_sd` versus overnight time above 180 mg/dL. The candidate groups show descriptive differences in HbA1c, BMI, baseline glucose variability, and awakening count, but these differences are interpreted as characterization of an unstable slope label rather than validation of a participant phenotype.

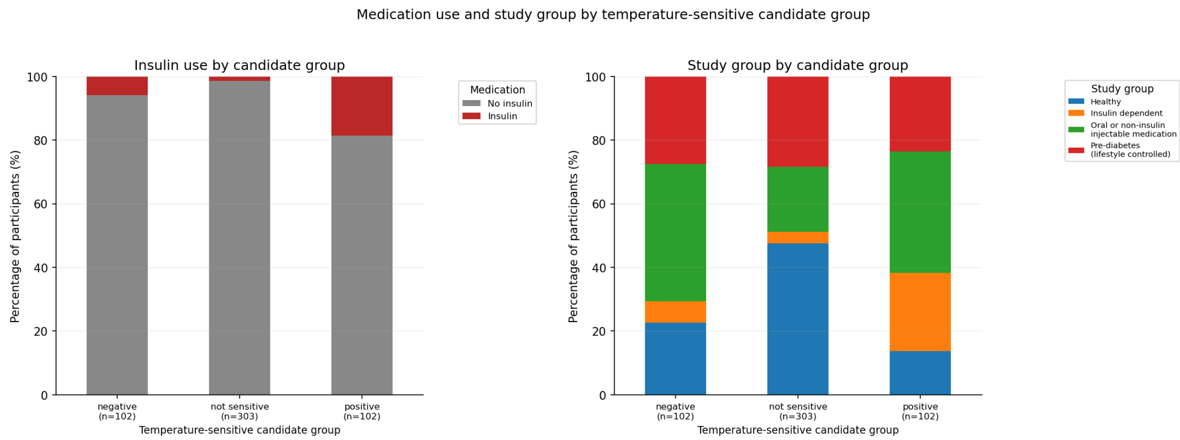


Figure 71: Medication use and study-group composition by exploratory temperature-sensitive candidate group. Insulin use and diabetes-status strata differ across candidate groups, suggesting that apparent temperature sensitivity may partly reflect clinical subgroup structure and baseline glycemic instability.

After the humidity association was identified, sleep architecture was added to test whether the association was simply explained by measured sleep summaries. This was an adjustment and attenuation analysis, not a formal causal mediation analysis:

$$Y_{i,n} = \beta_H H_{i,n}^{\text{within}} + \beta_B \bar{H}_i + \theta^\top S_{i,n} + u_i + \varepsilon_{i,n}, \tag{40}$$

where $S_{i,n}$ contains sleep summaries such as sleep duration and sleep-stage composition. The humidity coefficient did not shrink toward zero: it moved from -0.361 before sleep adjustment to -0.391 after adjustment, and no significant exposure-by-sleep interaction was detected. Thus the association was not explained away by measured sleep architecture, while still remaining a descriptive association rather than proof of independent sleep physiology.

16.6 From Association to Forecasting Value

A statistically robust association is not enough to justify increasing the SSM-CGM input space. The forecasting analysis therefore froze the base SSM-CGM model and tested whether humidity explained residual forecast error. Let the frozen model produce

$$\hat{y}_{i,t+h}^{\text{base}}. \tag{41}$$

The residual is

$$r_{i,t+h} = y_{i,t+h} - \hat{y}_{i,t+h}^{\text{base}}. \tag{42}$$

A lightweight residual head was trained as

$$\Delta\hat{y}_{i,t+h} = g_\phi\left(\hat{y}_{i,t+h}^{\text{base}}, H_{i,n}, S_{i,n}, c_{i,t}\right), \tag{43}$$

and the corrected forecast was

$$\hat{y}_{i,t+h}^{\text{corrected}} = \hat{y}_{i,t+h}^{\text{base}} + \Delta\hat{y}_{i,t+h}. \tag{44}$$

Here $H_{i,n}$ is the night-level humidity feature, $S_{i,n}$ denotes sleep or window context, $c_{i,t}$ denotes available base forecast context, and g_ϕ is a lightweight ridge or histogram-gradient residual model. This design isolates the incremental value of humidity: any gain must come from explaining the frozen base model's remaining error, rather than from full retraining, optimization noise, or representation changes.

Table 49: Forecasting-value comparison for humidity residual correction.

Model	Purpose
Frozen SSM-CGM base	Reference forecast
Base + sleep	Tests sleep-only residual value
Base + humidity	Tests environmental residual value
Base + sleep + humidity	Tests humidity beyond sleep context
Base + shuffled humidity	Negative control for spurious environmental gain

The 30-minute early-wake HGBT residual head was the clearest real-humidity case: `base_plus_env` improved MAE by only 0.046 mg/dL on held-out participants, with a 95% confidence interval crossing zero ($[-0.061, 0.155]$ mg/dL). This is more than six times smaller than the pre-defined 0.3 mg/dL threshold for a meaningful forecasting gain. It also did not beat the corresponding shuffled-night humidity control on the same 30-minute test endpoint; the shuffled control showed a numerically larger 0.066 mg/dL gain with a confidence interval also crossing zero. The notebook's final decision file selected `standalone_attribution_only`: no tested horizon cleared the full candidate gate, and humidity should not be added as an SSM-CGM input channel at this time.

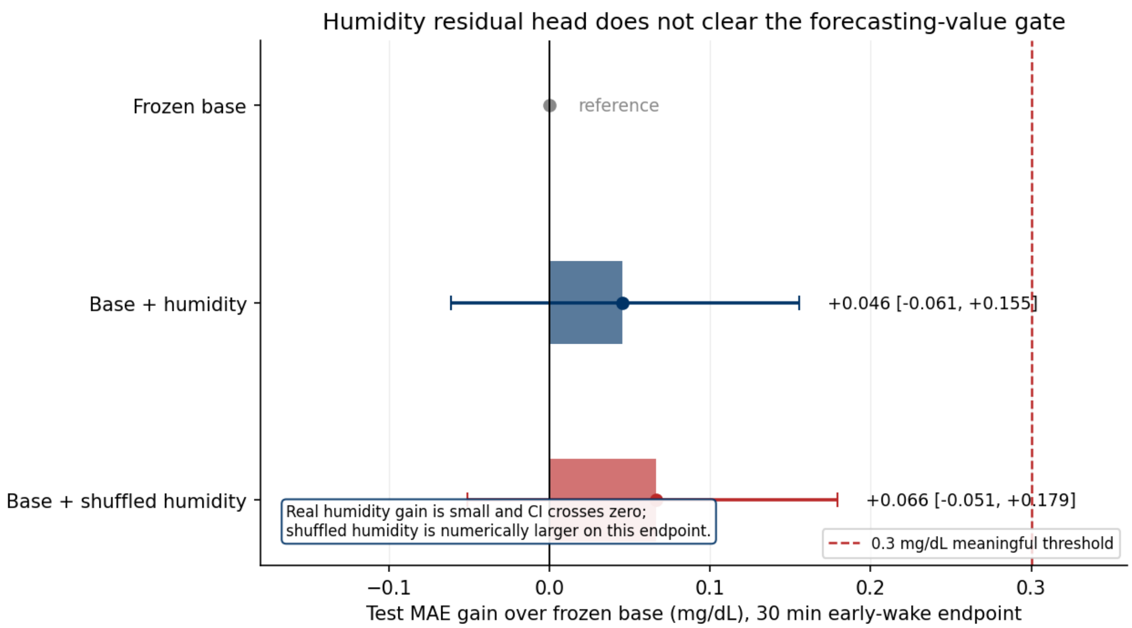


Figure 72: Forecasting-value test for bedroom humidity. Although humidity was associated with overnight glucose range in the association analysis, a residual head using humidity produced only a negligible MAE gain and did not outperform the shuffled-humidity control on the highlighted 30-minute early-wake endpoint.

As a complementary full-model test, environmental features were also added as a seventh streaming group and the model was retrained. The environmental group contained causally aligned temperature and humidity means, their trailing variability, time since the last environmental observation, and an environmental-missingness indicator. This experiment asks a different question from the frozen residual head: whether the recurrent encoder incorporates the environmental stream when allowed to learn a new hidden-state pathway.

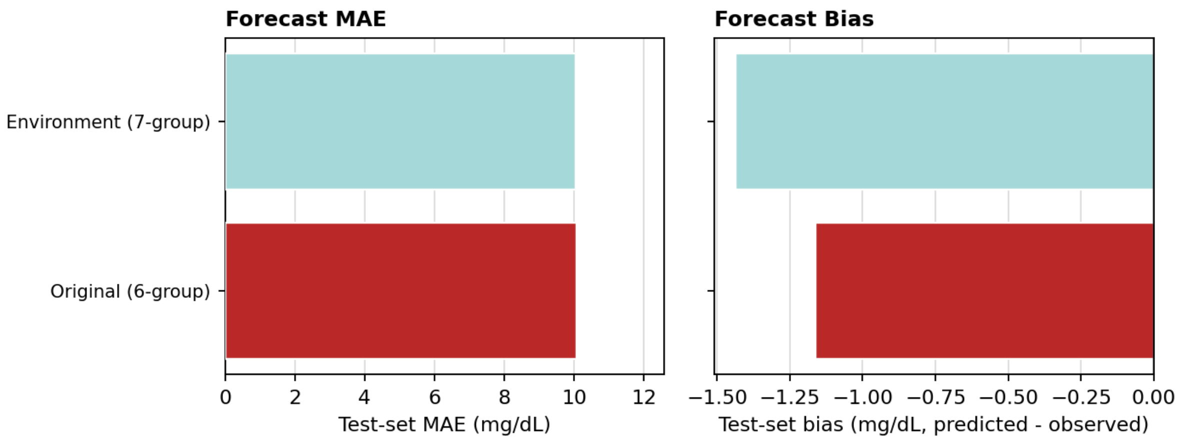


Figure 73: Forecast accuracy after adding environmental exposure to the recurrent model. On the matched test set, the environment-augmented 7-group model obtained an MAE of 10.0565 mg/dL, compared with 10.0718 mg/dL for the original 6-group model: a reduction of only 0.0153 mg/dL (approximately 0.15%). Forecast bias became more negative, changing from -1.1621 to -1.4365 mg/dL (predicted minus observed). Thus the negligible MAE difference is accompanied by 0.2744 mg/dL greater systematic underprediction.

The matched comparison in Fig. 73 reinforces the residual-head result. Although the environment-augmented encoder assigns 12.17% of its rigorous 7-group hidden-state attribution to environmental exposure (Fig. 77), that representation uptake does not translate into a practically meaningful improvement in overall forecast accuracy. The more negative bias also argues against treating the small MAE difference as an unqualified gain. Because the environmental group includes measurement freshness and missingness as well as physical exposure variables, its hidden-state share may partly reflect sensor availability or bedroom-sensor coverage rather than an independent physiological effect of temperature or humidity.

Several implementation limits are relevant to the forecasting-value test. Sections 4–8 of the forecasting notebook used the `experiment_c_split_adapt6h_seed42` split because a validated dense prediction cache was already available, while the association study used the `adapt48h_seed42` split. The decisive comparison between real humidity and shuffled humidity remains internal to the forecasting split. No fresh full inference was run; the existing dense forecast cache was filtered and reshaped. Only short-horizon residual correction was evaluated, and the 30-minute early-wake endpoint was the clearest candidate display. Given the tiny observed gain and the failure against shuffled humidity, these deviations are unlikely to change the decision, but they constrain the claim.

16.7 Interpretation and Decision

Together, these analyses separate four levels of evidence. First, bedroom humidity showed a statistically robust within-participant association with overnight glucose range. Second, this association was not explained away by measured sleep architecture and was robust to key temporal controls. Third, when tested as an input to a residual forecasting head on top of frozen SSM-CGM predictions, humidity produced only a negligible MAE improvement and failed against its shuffled-night control. Fourth, the full environment-augmented model encoded the new group strongly in h_t but reduced overall MAE by only 0.0153 mg/dL while increasing underprediction bias by 0.2744 mg/dL. Therefore, environmental exposure is retained as an interpretable research modality and association, but the current evidence does not support using it as a deployed SSM-CGM input for improved forecast accuracy.

17 Counterfactual Strategy for AI-READI SSMCGM-Stream

17.1 Motivation and scope

The goal of the counterfactual component is to move SSMCGM-Stream from passive forecasting toward controlled, behavior-aware simulation. However, the AI-READI setting is fundamentally different from T1DEXI [1, 9]. T1DEXI contains explicit action streams such as timed insulin boluses, carbohydrate intake, and insulin-on-board, whereas AI-READI provides CGM, wearable signals, static clinical metadata, and a predicted meal flag. Therefore, AI-READI does not support insulin-dose or carbohydrate-dose counterfactuals. The variable `med_insulin` is treated as a static participant descriptor, not as an editable action. All AI-READI counterfactuals are therefore framed as *proxy scenario simulations*: meal-like, activity-like, stress-like, and sleep/rest physiological states.

The current SSMCGM-Stream run already demonstrates a strong participant-held-out forecaster, but its proxy scenario branch is not yet scientifically interpretable. Forecast-only and factual-future metrics are nearly identical, and the measured mean absolute scenario deltas are close to zero. This motivates a conservative strategy: before interpreting any scenario as a plausible counterfactual effect, we must first prove that the model uses the future scenario channel, that the edited scenarios lie on the empirical AI-READI manifold, and that model-implied effects agree with observed natural experiments.

The proposed counterfactual strategy has five linked components:

1. a scenario-aware streaming forecasting model with explicit forecast-only and factual-future pathways;
2. behavior-level, multivariate scenario edits rather than single-variable perturbations;
3. trajectory-level support checks to enforce positivity and avoid out-of-distribution scenario paths;
4. natural-experiment validation against matched observed windows;
5. sensitivity analysis for unmeasured confounding, with additive bias bounds for continuous glucose effects and E-values for binary clinical risks.

This section describes the mathematical and machine-learning framework behind these components.

17.2 Streaming state-space forecasting backbone

Let participant i have static covariates

$$s_i = \{\text{age}, \text{HbA1c}, \text{BMI}, \text{site}, \text{medications}, \ldots\}. \tag{45}$$

These are encoded by a static encoder g_ϕ into an embedding

$$e_i = g_\phi(s_i). \tag{46}$$

At each 5-minute time step t, the dynamic observation vector $x_{i,t}$ contains observed CGM and wearable variables up to the current time, for example CGM, heart rate, steps, stress, and sleep stage. The streaming SSM update is

$$h_{i,t} = f_\theta(h_{i,t-1}, x_{i,t}, e_i), \tag{47}$$

where $h_{i,t}$ is the participant-specific hidden state after observing the history up to time t. In the current SSMCGM-Stream implementation, f_θ is based on selective state-space/Mamba blocks [4, 5]. The practical advantage of this formulation is that the historical context is compressed into a fixed-dimensional state, so multiple future scenarios can be decoded from the same $h_{i,t}$ without re-encoding the full historical window.

For a forecast horizon of H steps, the decoder receives three groups of future inputs:

- $K_{t+1:t+H}$: known future inputs, such as time-of-day and calendar encodings;
- $A_{t+1:t+H}$: future scenario/proxy variables, such as meal proxy, activity footprint, stress, or sleep/rest signals;
- $M_{t+1:t+H}$: masks indicating whether scenario variables are known, unknown, factual, or edited.

The model predicts glucose quantiles through a residual-current target,

$$\hat{y}^{(q)}_{i,t+h} = y_{i,t} + \hat{r}^{(q)}_{i,t,h}, \qquad h = 1, \ldots, H, \tag{48}$$

where $y_{i,t}$ is the current observed glucose and $\hat{r}^{(q)}_{i,t,h}$ is the predicted residual quantile at horizon h. This makes persistence a strong baseline and prevents cold-start forecasts from predicting absolute glucose from scratch.

17.3 Forecast-only and factual-future pathways

The first requirement for counterfactual modeling is to check whether the scenario channel is alive. We therefore evaluate two decoder pathways for the same hidden state $h_{i,t}$.

The forecast-only pathway hides future scenario information:

$$\hat{Y}^{fo}_{i,t+1:t+H} = D_\theta(h_{i,t}, K_{t+1:t+H}, A^0_{t+1:t+H}, M^0_{t+1:t+H}), \tag{49}$$

where A^0 is set to zero or neutral values and M^0 marks the future scenario path as unknown.

The factual-future pathway reveals observed future wearable/proxy variables retrospectively:

$$\hat{Y}^{ff}_{i,t+1:t+H} = D_\theta(h_{i,t}, K_{t+1:t+H}, A^{obs}_{t+1:t+H}, M^{obs}_{t+1:t+H}). \tag{50}$$

This pathway is not deployable, because it uses information from the future. Its role is diagnostic: if the factual-future pathway does not improve prediction on event-rich windows, then the model has not learned to use scenario information.

Using the pinball loss $\rho_q(u) = \max(qu, (q-1)u)$, the multi-horizon quantile losses are

$$\mathscr{L}_{fo} = \sum_{q\in\mathscr{Q}} \sum_{h=1}^{H} \rho_q \left(y_{i,t+h} - \hat{y}^{fo,(q)}_{i,t+h} \right), \tag{51}$$

$$\mathscr{L}_{ff} = \sum_{q\in\mathscr{Q}} \sum_{h=1}^{H} \rho_q \left(y_{i,t+h} - \hat{y}^{ff,(q)}_{i,t+h} \right). \tag{52}$$

The factual-future gain is

$$G_{ff} = \mathscr{L}_{fo} - \mathscr{L}_{ff}. \tag{53}$$

A useful scenario channel should satisfy $G_{ff} > 0$ on event-rich anchors. If $G_{ff} \approx 0$, any inference-time scenario edit will mostly be arbitrary because the decoder was trained to ignore the future scenario path.

17.4 Scenario-aware training objective

To make the scenario path useful, the training objective should include both forecast-only and factual-future losses:

$$\mathscr{L}_{train} = \mathscr{L}_{fo} + \lambda_{ff}\mathscr{L}_{ff} + \lambda_{gain}\mathscr{L}_{gain} + \lambda_{bal}\mathscr{L}_{bal}. \tag{54}$$

The gain term is applied only to event-rich anchors, where future wearable/proxy information is expected to matter:

$$\mathscr{L}_{gain} = \mathbb{1}_{event} \left[m + \mathscr{L}_{ff} - \mathscr{L}_{fo} \right]_+, \tag{55}$$

where $m \geq 0$ is a small margin. This loss penalizes the model when the factual-future path is not at least m better than the forecast-only path. Importantly, we do *not* include any term of the form $-\mathbb{E}|\Delta|$. Such a term would reward large scenario effects and could manufacture fake counterfactuals. The model is encouraged to use real future information when it is predictive, but it is never directly rewarded for making scenario effects large.

The optional balancing term $\mathscr{L}_{bal}$ is inspired by counterfactual recurrent networks [3]. Let B_t denote an observed behavior-proxy label, such as high activity, sleep/rest, stress, or meal-proxy. A behavior classifier c_ψ attempts to predict B_t from the hidden history representation $h_{i,t}$:

$$\hat{B}_t = c_\psi(h_{i,t}). \tag{56}$$

A domain-adversarial objective can be written as

$$\min_\theta \max_\psi \left[\mathscr{L}_{train} - \lambda_{bal}\, CE(B_t, c_\psi(h_{i,t})) \right]. \tag{57}$$

The purpose is to reduce the association between patient history and behavior assignment, which is a source of time-dependent confounding. This term should be introduced only after simpler two-pathway training and event oversampling have been tested, because excessive balancing can degrade forecasting performance.

17.5 Behavior-level scenarios rather than symptom edits

A central design choice is to define scenarios at the level of behaviors, not individual wearable variables. Directly setting heart rate to a high value is not a valid intervention on exercise: heart rate is a physiological consequence of many possible causes, including activity, stress, illness, caffeine, and glucose dynamics. Therefore, AI-READI scenarios are represented by behavior labels

$$B \in \{\text{meal\_proxy}, \text{mild\_activity}, \text{high\_activity}, \text{sleep\_rest}, \text{sedentary\_stress}\}. \tag{58}$$

Each behavior is realized as a multivariate wearable footprint:

$$B \longrightarrow (\text{steps}, \text{heart rate}, \text{stress}, \text{sleep stage}, \text{awake}, \ldots). \tag{59}$$

This avoids impossible single-variable counterfactuals and is closer to a target-trial emulation view: the intervention is a behavioral strategy, and the wearable variables are its observed proxy signature.

17.6 On-manifold empirical scenario path generation

A first attempt could set scenario variables using independent percentiles, for example steps at the participant's 75th percentile and heart rate at the 65th percentile. This is useful for prototyping, but it can create impossible combinations because marginally plausible values are not necessarily jointly plausible. The preferred approach is empirical conditional path sampling, related in spirit to controllable sequence editing for counterfactual generation [8].

For each anchor (i,t), define the matching context

$$C_{i,t} = [y_{i,t-L:t}, \Delta y_{i,t-L:t}, HR_{i,t-L:t}, steps_{i,t-L:t}, stress_{i,t-L:t}, sleep_{i,t-L:t}, hour(t), s_i]\,. \tag{60}$$

For a behavior B, the future scenario path is sampled from observed training windows with similar context:

$$A^B_{i,t+1:t+H} \sim p_{train}\left(A_{u+1:u+H} \mid B_u = B, C_u \approx C_{i,t}\right). \tag{61}$$

In practice, this can be implemented by nearest-neighbor retrieval, cluster prototypes, or weighted averaging among observed analogue windows. This makes the generated scenario path lie closer to the empirical AI-READI manifold than independently constructed percentiles.

The actual edit is localized in time and scope. For a behavior B starting at horizon step τ and lasting d steps, the scoped edit operator is

$$(A^B, M^B, S^B) = \mathscr{E}_{B,\tau,d}(A, M, S; C_{i,t}), \tag{62}$$

where S is a source/type variable distinguishing unknown future values, factual future values, planned proxy values, and edited proxy scenarios. For affected variables $j \in \mathscr{V}_B$,

$$A^B_{t+h,j} = \begin{cases} \tilde{A}^B_{h,j}, & h \in [\tau, \tau+d],\ j \in \mathscr{V}_B, \\ A_{t+h,j}, & \text{otherwise}, \end{cases} \tag{63}$$

where $\tilde{A}^B$ is the empirical analogue path. The mask is set to one for edited variables during the intervention interval, and the source/type label marks them as edited proxy values.

17.7 Scenario predictions and model-implied effects

Given the same hidden state $h_{i,t}$, the forecast-only prediction is

$$\hat{Y}^{fo}_{i,t+1:t+H} = D_\theta(h_{i,t}, K, A^0, M^0, S^0), \tag{64}$$

whereas the behavior-scenario prediction is

$$\hat{Y}^{B}_{i,t+1:t+H} = D_\theta(h_{i,t}, K, A^B, M^B, S^B). \tag{65}$$

The model-implied proxy scenario effect at horizon h is

$$\Delta_\theta(B,i,t,h) = \hat{y}^{B}_{i,t+h} - \hat{y}^{fo}_{i,t+h}. \tag{66}$$

This is not automatically a causal effect. It is the change in the model's prediction when the future proxy path is changed while holding the observed history and current state fixed. It becomes scientifically interpretable only if the scenario channel is used, the scenario has empirical support, and the model-implied effect agrees with observed natural experiments.

A useful decomposition is

$$\hat{y}^{B}_{i,t+h} = \hat{y}^{base}_{i,t+h} + \Delta_\theta(B,i,t,h), \tag{67}$$

where $\hat{y}^{base}$ captures the forecast from history and known future time variables, and Δ_θ captures the additional scenario-dependent component. This decomposition should be used diagnostically: if $\mathbb{E}|\Delta_\theta| \approx 0$, the scenario branch is inactive or weakly used.

17.8 Trajectory-level support and positivity

Longitudinal counterfactual identification requires positivity: each scenario should have nonzero probability among participants with comparable histories. In this application, this means the edited future wearable path must resemble observed AI-READI paths. Support should therefore be assessed over the whole trajectory rather than at a single time point.

Let $\Psi(A_{t+1:t+H}, C_t)$ summarize a future path and its context, including total steps, maximum heart rate, mean stress, sleep fraction, time of onset, glucose level, glucose slope, time of day, site, HbA1c quartile, and study group. Define the path support distance

$$d_{path}(B,i,t) = \min_{u \in \mathcal{D}^{B}_{train}} \left\| \Psi(A^{B}_{i,t+1:t+H}, C_{i,t}) - \Psi(A^{obs}_{u+1:u+H}, C_u) \right\|_{\Sigma^{-1}}. \tag{68}$$

The support class is then

$$support(B,i,t) = \begin{cases} \text{supported}, & d_{path} < c_1 \text{ and } n_{neighbors} > n_{min}, \\ \text{weak}, & d_{path} < c_2, \\ \text{unsupported}, & \text{otherwise.} \end{cases} \tag{69}$$

Only supported scenarios are used in the main counterfactual analysis. Weakly supported scenarios are reported in the appendix. Unsupported scenarios are excluded from effect interpretation.

17.9 Natural-experiment validation

Because AI-READI is observational, model-implied effects must be compared against real observed windows that resemble the proposed scenario. For each behavior B, define a treatment indicator

$$T^{B}_{i,t} = 1 \tag{70}$$

when the observed future window contains the behavior footprint, and $T^{B}_{i,t} = 0$ for matched control windows. For example, an activity treated window may require high steps, elevated heart rate, and awake state; a control window should have low steps or normal heart rate while matching on pre-onset glucose and clinical context.

Matching should use rich pre-onset histories, not only current glucose and one slope. The matching vector is

$$H^{match}_{i,t} = [y_{i,t-L:t}, \Delta y_{i,t-L:t}, HR_{i,t-L:t}, steps_{i,t-L:t}, stress_{i,t-L:t}, sleep_{i,t-L:t}, hour(t), s_i]. \tag{71}$$

The observed natural-experiment effect is

$$\Delta_{obs}(B,h) = \mathbb{E}\left[y_{i,t+h} - y_{i,t} \mid T^{B}_{i,t} = 1,\ matched\right] \tag{72}$$

$$- \mathbb{E}\left[y_{i,t+h} - y_{i,t} \mid T^{B}_{i,t} = 0,\ matched\right]. \tag{73}$$

The model-implied effect is averaged over the same supported anchors:

$$\Delta_\theta(B,h) = \mathbb{E}_{(i,t)\in\mathscr{S}_B}\left[\hat{y}^{B}_{i,t+h} - \hat{y}^{fo}_{i,t+h}\right], \tag{74}$$

where $\mathscr{S}_B$ is the set of supported anchors for behavior B.

Validation metrics include effect MAE,

$$EffectMAE(B) = \frac{1}{H}\sum_{h=1}^{H}\left|\Delta_\theta(B,h) - \Delta_{obs}(B,h)\right|, \tag{75}$$

and sign agreement,

$$SignAgreement(B) = \frac{1}{H}\sum_{h=1}^{H}\mathbb{1}\left[\operatorname{sign}(\Delta_\theta(B,h)) = \operatorname{sign}(\Delta_{obs}(B,h))\right]. \tag{76}$$

The primary validation report should include treated-window counts, control-window counts, support coverage, sign agreement, and effect MAE. Calibration slope is useful as a secondary metric, but it should not replace interpretable diagnostics.

17.10 Sensitivity analysis for unmeasured confounding

Sequential ignorability cannot be verified in AI-READI. Behavior may be confounded by unobserved intent, symptoms, meals, insulin, illness, or other latent factors. Sensitivity analysis is therefore a core component of the counterfactual framework rather than an optional add-on [6].

For continuous glucose effects in mg/dL, we use an additive bias bound. Let δ denote a hypothetical unmeasured confounding bias. The adjusted observed effect is

$$\Delta^{adj}_{obs}(B,h;\delta) = \Delta_{obs}(B,h) - \delta. \tag{77}$$

We evaluate a grid such as

$$\delta \in \{-20,-15,-10,-5,0,5,10,15,20\}\ \text{mg/dL}. \tag{78}$$

The sign-flip threshold is

$$\delta_{flip}(B,h) = \min_{\delta}\left\{\operatorname{sign}(\Delta^{adj}_{obs}(B,h;\delta)) \neq \operatorname{sign}(\Delta_{obs}(B,h))\right\}. \tag{79}$$

This gives a direct interpretation: how large an unmeasured glucose-scale bias would need to be to reverse the estimated effect.

For binary clinical outcomes, such as hypoglycemia within H minutes,

$$Y^{hypo}_{i,t,H} = \mathbb{1}\left[\min_{1\le h\le H} y_{i,t+h} < 70\right], \tag{80}$$

or hyperglycemia,

$$Y^{hyper}_{i,t,H} = \mathbb{1}\left[\max_{1\le h\le H} y_{i,t+h} > 180\right], \tag{81}$$

we estimate risk ratios comparing treated and matched controls and report E-values [10]. E-values are appropriate for risk-ratio-scale sensitivity analysis; they should not be used as the main sensitivity tool for continuous mg/dL effects unless those effects are converted into binary clinical risks.

17.11 Attribution and channel-use diagnostics

Attribution is useful only after the behavioral tests show that the scenario channel matters. Raw gradients such as

$$\left\|\frac{\partial \hat{y}_{t+h}}{\partial A_{t+1:t+H}}\right\| \tag{82}$$

can be unstable in gated SSMs. Therefore, the primary channel-use diagnostic remains the factual-future gain G_{ff} on event windows. Attribution should be used as a secondary explanation tool.

When applied, attribution should answer three questions:

1. Does the decoder place non-negligible attribution on the scenario path rather than only on current glucose?
2. Does attribution increase after the planned scenario onset τ?
3. Does attribution vary by scenario and horizon in a physiologically plausible way?

The hidden-attention interpretation of Mamba models can support this analysis by converting the selective SSM recurrence into an attention-like view of which past or scenario tokens influence the forecast [2].

17.12 Counterfactual horizon design

The current forecasting headline uses a 60-minute horizon, but some behavior effects may unfold over longer periods. Activity, stress, and sleep/rest effects can be delayed or sustained, and meal-like effects may peak beyond the first hour depending on timing. Therefore, the counterfactual validation should be run at several horizons:

$$H \in \{12, 24, 48, 72\} \tag{83}$$

corresponding to 1, 2, 4, and 6 hours at 5-minute resolution. The 1-hour horizon remains the deployment-oriented forecast benchmark, while 2–6 hour horizons test whether behavioral scenario effects are hidden by the short forecasting window.

17.13 Experimental roadmap

The proposed implementation is divided into four phases.

Phase 0: audit the current model without retraining. Compute factual-future gain G_{ff} and scenario deltas by all anchors and by event-rich anchors: meal-proxy windows, high-activity windows, sleep transitions, stress windows, and large glucose-change windows. If $G_{ff} \approx 0$, the current scenario decoder should not be interpreted.

Phase 1: train a scenario-aware stream model. Train variants with the two-pathway objective in Equation (54), event-enriched sampling, and optional source/type embeddings. The primary metric is not large scenario delta; it is positive factual-future gain on event windows.

Phase 2: replace hand-set scenarios with empirical on-manifold scenario paths. Compare independent percentile scenarios, nearest-neighbor empirical paths, and cluster-prototype paths. Report support coverage, path distance, effect stability, and scenario effect by horizon.

Phase 3: validate against natural experiments. For each behavior, compare model-implied effects with observed matched effects using sign agreement and effect MAE. Add continuous sensitivity bounds for mg/dL effects and E-values for binary hypo/hyper risk outcomes.

Phase 4: optional CRN-style balancing. If natural-experiment validation suggests substantial time-dependent confounding, add adversarial behavior balancing. This should be treated as an ablation because too much balancing may reduce forecast accuracy.

17.14 Phase 0 Results: Wearable Counterfactual Readiness Audit

We ran the five-block readiness evaluation for the `wearable_plus_shape` detector on the non-insulin cohort (9,667 unique meal onsets, 1,591 participants). The purpose is not to claim causal validity, but to document precisely where the current system stands relative to the four-phase roadmap.

Table 50: Matched natural-experiment effect $\Delta_{obs}(h)$ for the `wearable_plus_shape` detector on the non-insulin cohort. Bootstrap 95% CI from participant resampling ($n_{\text{boot}} = 300$). Δ_θ is not available because the scenario channel is dead.

Horizon h	Δ_{obs} (mg/dL)	95% CI	n pairs
60 min	+22.2	[21.6, 22.8]	7,480
120 min	+24.2	[23.5, 25.0]	7,432
240 min	+3.6	[2.8, 4.5]	7,355
360 min	+0.3	[−0.5, 1.0]	7,302

Table 51: Sensitivity analysis for the `wearable_plus_shape` natural experiment. Sign-flip threshold is the smallest additive bias that reverses the sign of Δ_{obs}. E-values use the VanderWeele-Ding formula [10] for the binary hyperglycemia risk ratio; they are reported only for the binary endpoint, not for the continuous glucose effect.

Horizon	Δ_{obs}	Sign-flip δ_{flip}	E-value (hyper)	RR (hyper)
60 min	+22.2 mg/dL	22.2 mg/dL	3.80	2.19
120 min	+24.2 mg/dL	24.2 mg/dL	3.33	1.96
240 min	+3.6 mg/dL	3.6 mg/dL	1.84	1.26
360 min	+0.3 mg/dL	0.3 mg/dL	—	—

Block A: channel gate. The factual-future gain is $G_{ff} = 1.8 \times 10^{-5}$ overall, and the `meal_proxy_event` event-window evaluation has zero anchors (`predmeal_flag` future mask coverage is zero). The scenario channel is therefore *dead*: the decoder was trained without a factual-future loss term on meal windows, so any inference-time wearable scenario edit produces near-zero prediction deltas regardless of detector quality. Scenario-aware retraining (Phase 1) is required before the channel can be used.

Block B: matched natural experiment. Despite the dead channel, the matched natural experiment recovers a clear physiological signal. We matched 7,825 treated episodes (`wearable_plus_shape` detected, non-insulin) against 7,538 controls from the same participant in the same two-hour clock bin, with pre-onset glucose within 25 mg/dL and no nearby meal. The observed effect $\Delta_{obs}(h)$ is shown in Table 50.

The effect profile follows classic postprandial glucose kinetics: rapid rise peaking at 120 minutes, returning to baseline by 360 minutes. At 360 minutes the confidence interval crosses zero, consistent with full resolution. The detector is capturing real meal events, not noise.

Block C: trajectory support. Of 7,825 treated anchors, 7,814 (99.9%) are classified as supported; 11 are unsupported due to missing CGM context. The positivity condition is essentially satisfied: the detector fires in states well-represented in training data, so future counterfactual comparisons will not extrapolate into uncharted CGM territory.

Block D: sensitivity to unmeasured confounding. Table 51 reports the additive-bias sign-flip threshold $\delta_{flip}(h) = |\Delta_{obs}(h)|$ and the VanderWeele-Ding E-value for the binary hyperglycemia endpoint.

At 60 and 120 minutes the effect is robust: an unmeasured confounder would need to shift glucose by more than 22 mg/dL (or be associated with both wearable state and hyperglycemia risk by a factor of 3.3–3.8) to explain away the finding. At 240 minutes the sign-flip threshold drops to 3.6 mg/dL, making the effect fragile to modest unmeasured confounding. At 360 minutes Δ_{obs} is +0.3 mg/dL with a CI crossing zero; no E-value is reported because the continuous effect is indistinguishable from zero.

Block E: detection quality linkage. The `wearable_plus_shape` detector provides the most and best-timed treated episodes (0-minute median latency, 46.8% pre-rise) compared to all other setups and CGM-only baselines (35–40 minute post-hoc detection). The higher false-positive rate (11.5 FP/day) dilutes the natural-experiment effect toward zero, making the +22 mg/dL finding conservative rather than inflated.

Phase 0 conclusion. The wearable detector captures real pre-meal glucose events with near-perfect trajectory support and robust sensitivity at the 1–2 hour horizons. It is ready to serve as the treated indicator in a natural experiment and as the input to a scenario-aware decoder once the channel is live. The single limiting factor is the dead scenario channel. Phase 1 (scenario-aware retraining) is the required next step; improving the detector further is not.

17.15 Expected outputs

The counterfactual pipeline should produce the following report-ready outputs:

- `scenario_channel_audit.csv`: factual-future gain, event-window gain, and scenario delta by mode;
- `scenario_support.csv`: support class, path distance, and neighbor counts by scenario;
- `scenario_effects_by_horizon.csv`: $\Delta_\theta(B,h)$ by scenario and horizon;
- `natural_experiment_effects.csv`: treated/control counts, Δ_{obs}, Δ_θ, sign agreement, and effect MAE;
- `sensitivity_bounds.csv`: additive bias bounds for continuous effects and E-values for binary clinical risks;
- figures for factual-future gain, support coverage, model-vs-observed effects, and sensitivity thresholds.

17.16 Interpretation rules

The final interpretation should follow strict rules:

1. If $G_{ff} \leq 0$ on event windows, the scenario channel is considered inactive; do not interpret scenario deltas.
2. If a scenario is unsupported by the trajectory-level support check, exclude it from the main counterfactual results.
3. If model-implied effects disagree with observed natural-experiment effects, report the scenario as a failed or unvalidated proxy.
4. If continuous effects are sensitive to small additive bias, state that the result is fragile to unmeasured confounding.
5. E-values are reported only for binary clinical risk endpoints, not as the main sensitivity metric for continuous glucose effects.
6. AI-READI counterfactuals should never be described as insulin-dose or carbohydrate-dose counterfactuals.

Under these rules, the contribution is not a claim that AI-READI already supports definitive causal recommendations. The contribution is a validated framework for deciding which proxy behavioral scenarios are meaningful, which are unsupported, and which require additional action labels or model supervision.

18 Clinical Safety

Median hypoglycemia alarms are conservative: precision is high but recall is low. The q0.1 risk rule substantially increases hypoglycemia recall at the cost of lower precision, which is the expected clinical safety trade-off. Hyperglycemia is more predictable than hypoglycemia, but the q0.9 risk rule similarly trades precision for sensitivity.

Tail errors remain clinically important despite good average MAE. The forecast-only p90, p95, and p99 absolute errors are 24.25, 34.05, and 59.69 mg/dL, respectively. True TIR is 89.7% and predicted TIR is 91.6%, giving a 1.86 percentage-point overprediction of time in range.

Table 52: Hypoglycemia and hyperglycemia event detection using median and risk-quantile alarm rules.

Event	Horizon	Rule	Prec.	Recall	Spec.	F1
Hypo < 70	overall	median_q50_lt_70	0.734	0.231	1.000	0.352
Hypo < 70	30 min	median_q50_lt_70	0.651	0.147	1.000	0.239
Hypo < 70	60 min	median_q50_lt_70	0.610	0.038	1.000	0.072
Hypo < 70	overall	risk_q10_lt_70	0.314	0.585	0.995	0.409
Hypo < 70	30 min	risk_q10_lt_70	0.291	0.571	0.994	0.385
Hypo < 70	60 min	risk_q10_lt_70	0.224	0.349	0.995	0.273
Hyper > 180	overall	median_q50_gt_180	0.886	0.781	0.988	0.830
Hyper > 180	30 min	median_q50_gt_180	0.879	0.777	0.987	0.825
Hyper > 180	60 min	median_q50_gt_180	0.833	0.642	0.984	0.726
Hyper > 180	overall	risk_q90_gt_180	0.610	0.935	0.926	0.738
Hyper > 180	30 min	risk_q90_gt_180	0.614	0.935	0.927	0.741
Hyper > 180	60 min	risk_q90_gt_180	0.484	0.887	0.883	0.626

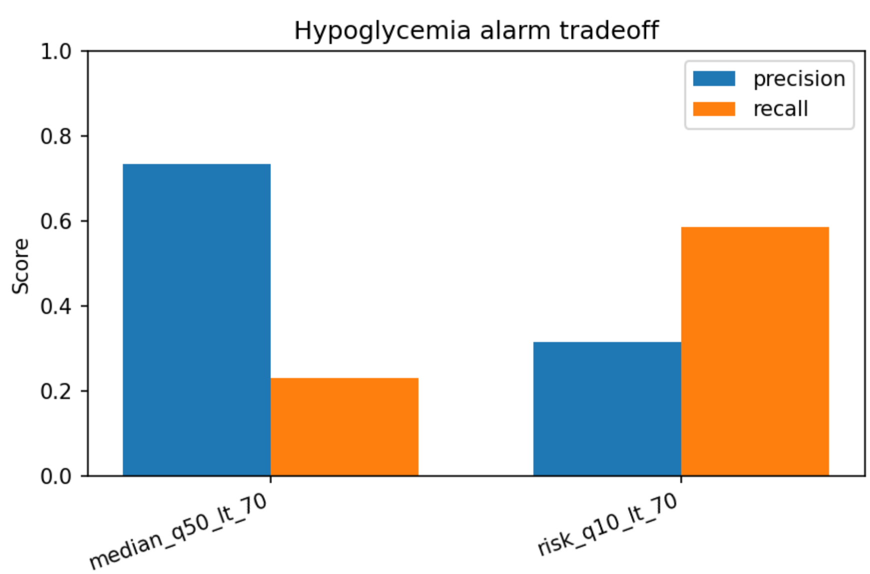


Figure 74: Clinical alarm trade-off. Median forecasts are conservative for hypoglycemia, while the q0.1 risk rule increases recall at the cost of lower precision.

19 Interpretability

19.1 Question and Scope

In this section we are looking at *what does the trained AI-READI forecasting model rely on?* The answer is reported through two views: The first view is *output level reliance*: how much the 12 step forecast changes when a modality is ablated. The second view is *state construction reliance*: how much each modality contributes to the recurrent Mamba hidden state h_t before the decoder turns that state into a forecast.

All results in this section use the full held out AI-READI test split: 153,591 anchors sampled at a 15 minute stride across 381 participant/segment streams from 221 test participants. The analysis is not restricted by insulin status metadata.

19.2 Interpretability based on two directions

Interpretability view 1: output level reliance. Leave-one-modality-out (LOMO) ablation measures the mean absolute change in the final 12step forecast when one modality is replaced by its population mean. This view is defined for all 11 modality groups because every group reaches the forecast decoder: glucose history, static clinical metadata, site/cohort, time/calendar, data quality/missingness, heart rate/physiology, medication metadata, sleep, wearable activity/exercise, stress, and the meal-proxy diagnostic channel. LOMO therefore answers a direct deployment question: if this information were removed or made uninformative, how much would the forecast move?

Interpretability view 2: state construction reliance. The joint group norm measures how much a modality contributes to the Mamba encoder hidden state h_t. For each group g, per feature contribution vectors u_f are summed first and then normed, $\|\sum_{f\in g} u_f\|$. Summing before taking the norm matters because many features are collinear: heart-rate mean/min/max/std and multiple activity-stage indicators can point in similar directions in representation space. The sum of individual norms would have double counted those shared directions.

In the original model, h_t interpretability is available for the 6 groups that pass through the streaming encoder's grouped linear fusion into the recurrent state: glucose history, heart rate/physiology, wearable activity/exercise, sleep, stress, and data quality/missingness. A separate environment-augmented model adds environmental exposure as a seventh h_t-reachable group; its state attribution and matched accuracy comparison are reported below. Time/calendar and meal proxy feed the horizon decoder directly, while static clinical metadata, medication metadata, and site/cohort modulate the decoder through the static FiLM context. Those groups are therefore interpreted through LOMO interpretability.

19.3 View 1: Output-Level Reliance

Table 53 and Figure 75 report LOMO attribution on the full held-out test split. The dominant output-level driver is **glucose history**: ablating it changes the forecast by 5.87 mg/dL on average and accounts for 36.7% of the normalised output-level attribution. This is the expected result for a residual-current CGM forecasting model: recent glucose contains the strongest information about the next hour.

The next largest output-level group is **static clinical metadata** (14.6%, 2.33 mg/dL). This does not mean that static variables cause short-term glucose movement; it means the decoder uses phenotype and enrolment context to calibrate the mapping from recent dynamics to future glucose. **Site/cohort** (9.0%) and **data quality/missingness** (8.4%) together account for 17.4% of output-level reliance. That is useful but also cautionary: part of the forecast depends on collection structure, recruitment site, or missingness patterns rather than purely physiological variation.

Time/calendar contributes 8.5%, and **heart rate/physiology** contributes 7.7%. Sleep, wearable activity/exercise, stress, and medication metadata have smaller but nonzero output-level effects. The **meal-proxy diagnostic channel** contributes 0.0%, which is the important negative control: under the available AI-READI representation, this channel does not move the forecast and should not be interpreted as a meal counterfactual mechanism.

Table 53: Output-level modality reliance on the full held-out test split. LOMO ablation measures the mean absolute change in the 12-step forecast when the named modality is replaced by its population mean. Fractions sum to 100% across the 11 output-visible groups.

Modality	Mean abs. change (mg/dL)	Fraction (%)
Glucose history	5.87	36.7
Static clinical	2.33	14.6
Site / cohort	1.45	9.0
Time / calendar	1.35	8.5
Data quality / missingness	1.34	8.4
Heart rate / physiology	1.23	7.7
Medication metadata	0.70	4.4
Sleep	0.65	4.0
Wearable activity / exercise	0.61	3.8
Stress	0.48	3.0
Meal proxy (diagnostic only)	0.00	0.0

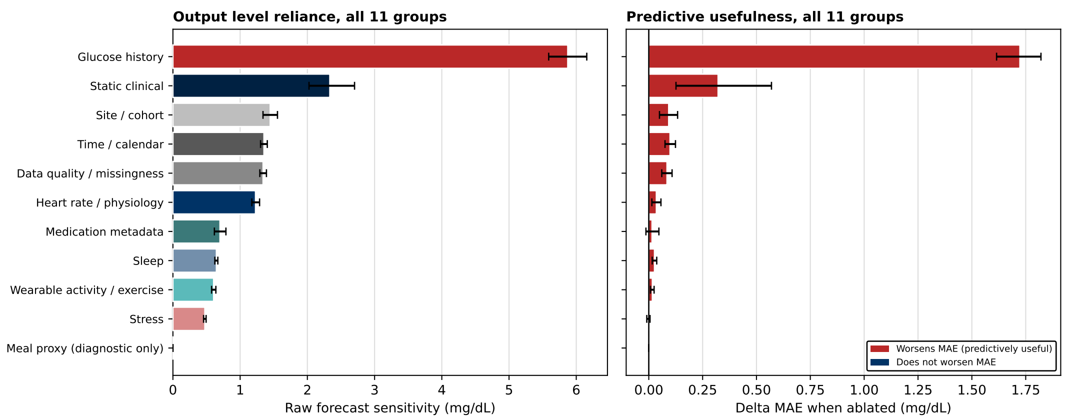


Figure 75: Output-level LOMO attribution on the full held-out test split (*n* = 153,591 anchors). Bars show the mean absolute forecast change after replacing each modality by its population mean. Glucose history is the largest output-moving input group, followed by static clinical context, site/cohort, time/calendar, data quality, and heart-rate/physiology. The meal-proxy diagnostic channel has no measurable output-level effect.

Fig. 75 places physiological and non-physiological inputs on the same scale. The important interpretation is not only that glucose dominates, but also that site/cohort and data-quality signals remain visible in the output. Those groups may improve forecast accuracy, but they also indicate possible reliance on recruitment and measurement structure.

19.4 View 2: State-Construction Reliance

The h_t view changes the interpretation. Among the 6 groups that actually enter the streaming recurrent state, glucose history is still important but no longer overwhelming. The rigorous joint-group norm assigns 21.9% of state-construction reliance to **glucose history**, 19.4% to **heart rate/physiology**, and 18.9% to **data quality/missingness**. **Sleep** (14.9%), **wearable activity/exercise** (13.4%), and **stress** (11.6%) make up the remainder.

This means the recurrent state is not just a compressed glucose trace. It also stores information about physiology, missingness, sleep/activity context, and stress-related channels. The decoder then decides how much of that internal state should affect the final forecast. That distinction explains why wearable activity and sleep can matter much more for h_t construction than for output-level LOMO: they help build the internal state, but the final forecast is still dominated by recent glucose and static calibration.

Table 54: State-construction reliance in the recurrent hidden state h_t. Rigorous attribution uses $\|\sum_{f\in g} u_f\|$, summing each group's contribution vectors before taking the norm. Post-hoc sum uses $\sum_{f\in g}\|u_f\|$ and is shown only to illustrate collinearity-driven overcounting. The LOMO output column renormalises output-level reliance over the same 6 h_t-reachable groups.

Modality (h_t-reachable)	Rigorous (%)	Post-hoc sum (%)	Ratio	LOMO output, renorm. (%)
Glucose history	21.9	9.7	0.44	57.8
Heart rate / physiology	19.4	24.2	1.25	12.1
Data quality / missingness	18.9	22.9	1.21	13.2
Sleep	14.9	10.1	0.68	6.4
Wearable activity / exercise	13.4	20.5	1.54	6.0
Stress	11.6	12.5	1.08	4.7

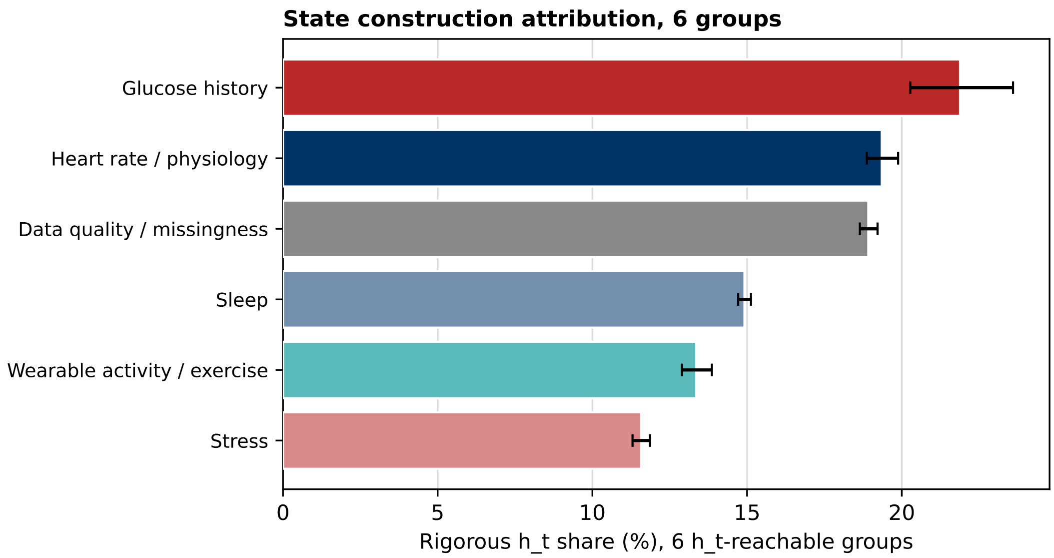


Figure 76: Grouped state-construction importance (AI-READI v2). Bars show each h_t-reachable group's rigorous joint-group-norm share. Glucose, heart-rate/physiology, and data-quality/missingness carry similar state-construction weight, while sleep, activity, and stress provide additional context that is less visible in output-level ablation.

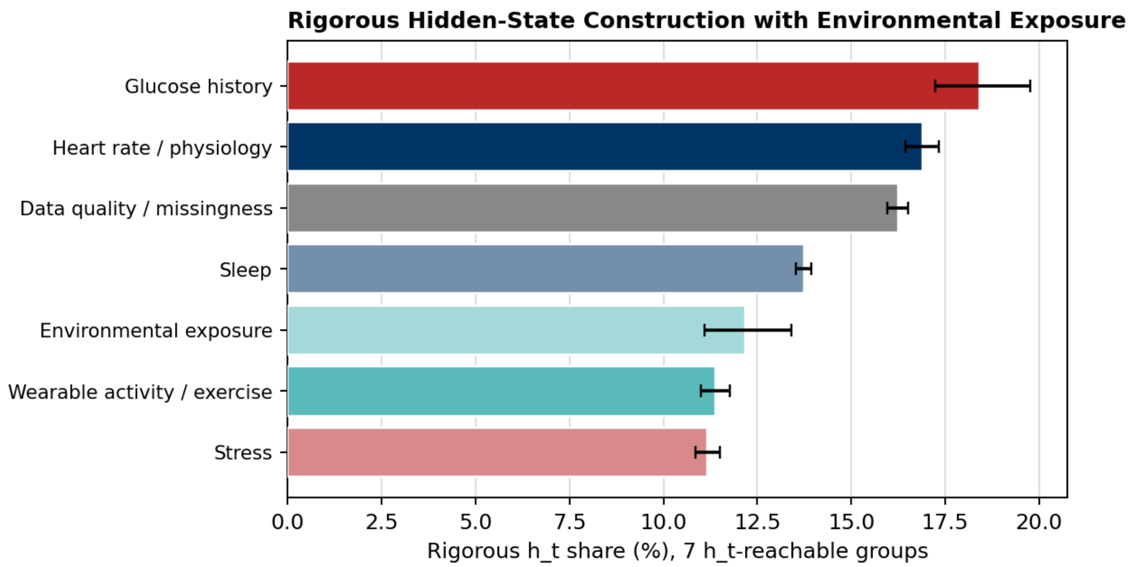


Figure 77: Global state-construction reliance after adding environmental exposure. Rigorous joint-group norms are normalised over the 7 groups that reach h_t in the environment-augmented model; error bars show the corresponding uncertainty intervals. Environmental exposure accounts for 12.17% of hidden-state construction, behind glucose history (18.41%), heart-rate/physiology (16.89%), data-quality/missingness (16.24%), and sleep (13.73%), but ahead of wearable activity/exercise (11.38%) and stress (11.17%). These values measure representation uptake, not causal influence or incremental forecast utility.

The environment-augmented attribution in Fig. 77 shows that the recurrent state is distributed rather than glucose-only. Environmental exposure has a nontrivial 12.17% joint-group share, so the encoder demonstrably stores the temperature, humidity, trailing variability, measurement-freshness, and missingness features supplied through this pathway. The wider uncertainty interval for the environmental group is consistent with its more limited and heterogeneous sensor availability. Importantly, a large hidden-state norm does not establish that the information improves the decoded forecast: it only establishes that the trained encoder uses representational capacity for that group.

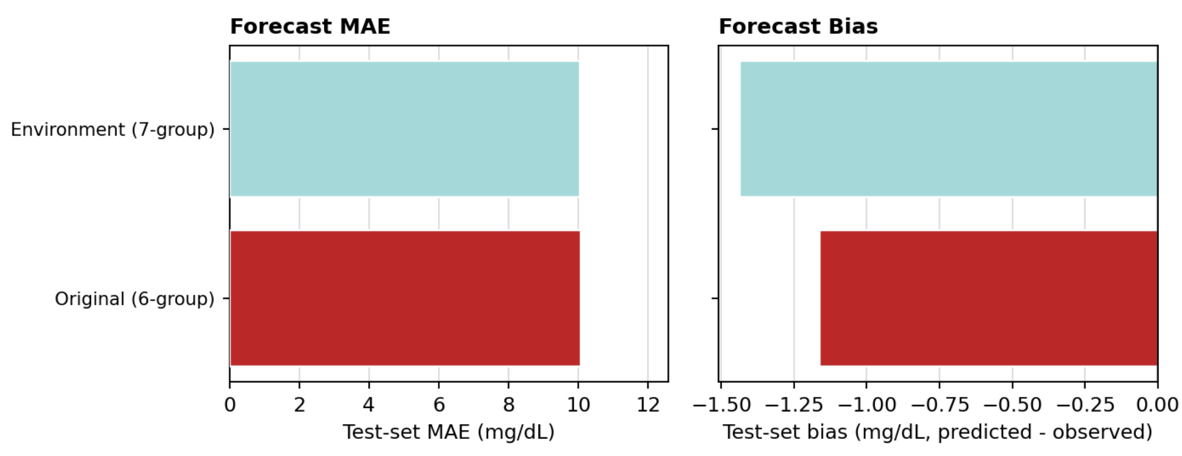


Figure 78: Matched test-set accuracy and bias with and without environmental exposure. The 7-group environment model has an MAE of 10.0565 mg/dL versus 10.0718 mg/dL for the original 6-group model, an absolute reduction of only 0.0153 mg/dL (approximately 0.15%). Its bias is more negative (-1.4365 versus -1.1621 mg/dL, predicted minus observed), increasing systematic underprediction by 0.2744 mg/dL. Environmental exposure is therefore encoded in h_t, but it provides no practically meaningful overall accuracy gain and worsens calibration bias.

Figures 77 and 78 together illustrate why representation importance and predictive value must be separated. The environment group occupies a substantial direction in the hidden state, yet the matched forecasting comparison is essentially unchanged in MAE and less favorable in bias. The most defensible interpretation is that the encoder can absorb environmental and environmental-availability context, not that those inputs add clinically meaningful forecasting information.

19.5 Reconciling the Two Views

Fig. 79 places the two reliance views next to each other. The central finding is an asymmetry between *representation* and *forecast movement*. Glucose history dominates forecast movement, but the hidden state distributes its capacity more evenly across physiological, behavioral, and data-quality channels. Wearable activity/exercise and sleep are the clearest examples: their state-construction shares are roughly 13–15%, but their output-level LOMO shares are only about 4%. The model appears to encode these channels as contextual state information; the decoder only lets a smaller part of that context move the final forecast.

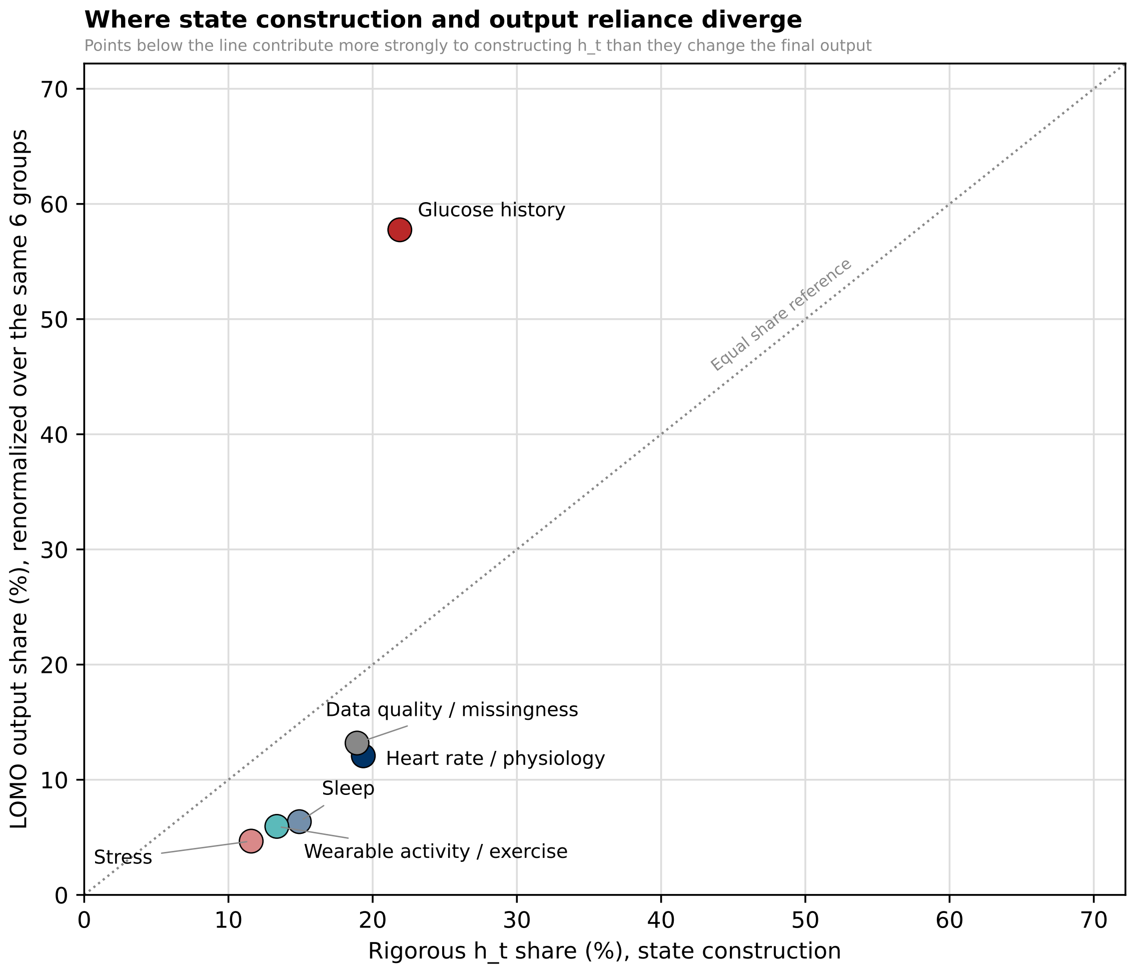


Figure 79: State-construction reliance versus output-level reliance. The same h_t-reachable groups are compared under the rigorous state norm and output-level LOMO attribution. The figure shows that representation importance and forecast sensitivity are not interchangeable: glucose drives the forecast most strongly, while wearable and sleep channels contribute more to the internal state than to immediate output movement.

The post-hoc comparison in Fig. 80 explains why the rigorous group norm is necessary. For collinear groups such as heart-rate/physiology and wearable activity/exercise, summing individual feature norms after the fact inflates the apparent group share. The rigorous method avoids this by summing contribution vectors before taking the norm.

This is why glucose history's post-hoc share looks artificially small: its absolute contribution is not reduced, but the denominator is inflated by other collinear groups.

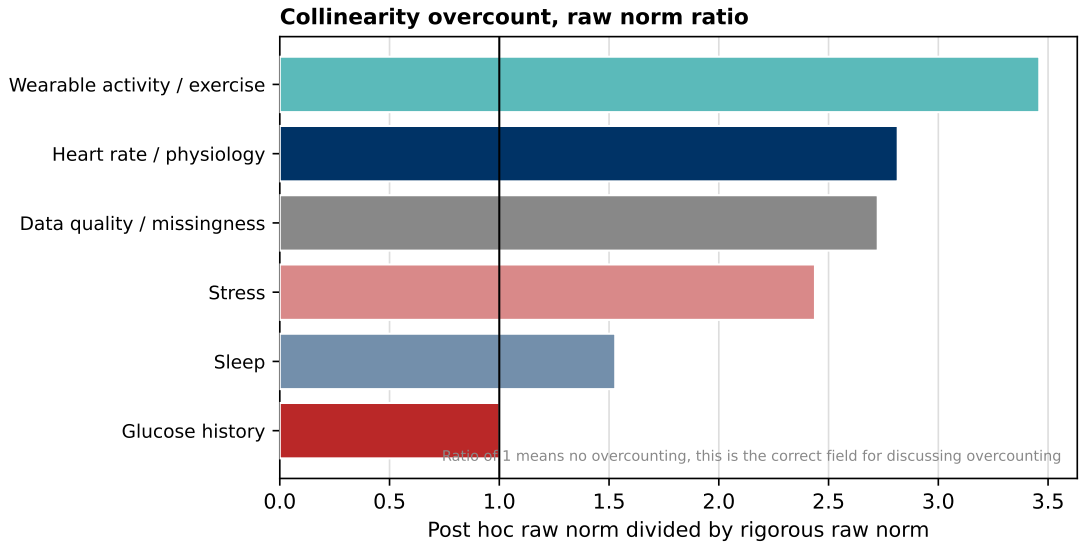


Figure 80: Why the rigorous joint-group norm is used. Ratios above 1 indicate groups whose post-hoc sum-of-norms attribution is inflated by collinearity. The rigorous group norm is the appropriate state-construction measure because it counts shared directions once rather than once per correlated feature.

19.6 Static Clinical Interpretation

Static covariates describe participant-level characteristics that remain fixed over the monitored time series, including glycaemic status, baseline clinical measurements, demographics, medication metadata, and clinical site. Their role differs from that of streaming variables. Historical glucose and wearable measurements describe the participant's current state, whereas static variables provide a participant-specific context for interpreting those measurements.

The static analysis addresses three separate questions:

1. How much does each static domain or feature change the final forecast?
2. Does this dependence improve held-out predictive performance?
3. Through which architectural pathway does the static information affect the model?

These questions are evaluated separately because a feature can substantially change the forecast without improving its accuracy.

19.6.1 Output Sensitivity and Predictive Usefulness

For a static feature or domain g, output sensitivity is measured using leave-one-group-out reference ablation. Let $\widehat{\mathbf{y}}_i$ denote the original forecast at anchor i and $\widehat{\mathbf{y}}_i^{(-g)}$ the forecast obtained after replacing g by its training-derived reference representation. The raw sensitivity is

$$A_g = \frac{1}{NH} \sum_{i=1}^{N} \sum_{h=1}^{H} \left| \widehat{y}_{i,h} - \widehat{y}_{i,h}^{(-g)} \right|, \tag{84}$$

where A_g is expressed in mg/dL. A large value indicates that the trained model changes its forecasts when that information is removed. It does not by itself indicate that the feature improves accuracy.

Predictive usefulness is evaluated using

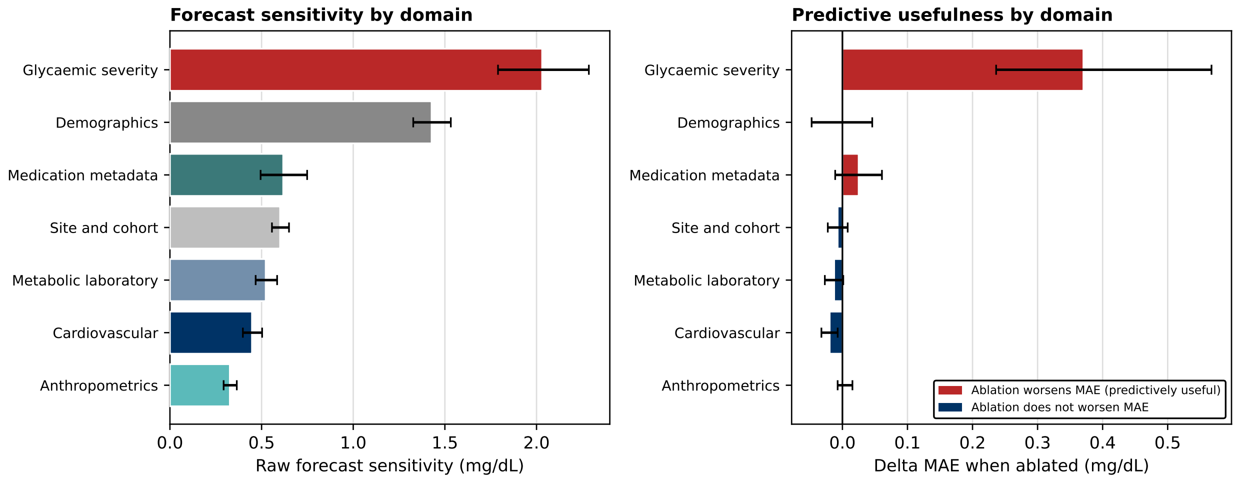


Figure 81: Static-domain output sensitivity. Bars show the mean absolute change in the 12-step forecast after replacing one static domain by its training-derived reference representation. Error bars show participant-clustered bootstrap confidence intervals. Glycaemic severity and demographics produce the largest output changes. The values measure model reliance in mg/dL and do not represent causal clinical effects or accuracy improvements.

$$\Delta \mathrm{MAE}_g = \mathrm{MAE}_{-g} - \mathrm{MAE}_{\mathrm{full}}. \tag{85}$$

A positive value means that removing the feature worsens held-out MAE and that the feature is predictively useful in the frozen model. A value close to zero indicates limited incremental predictive value, whereas a negative value means that the ablated model performs better. Negative frozen-model ablation results do not automatically justify removing a feature, since replacing a feature after training is not equivalent to retraining the architecture without it.

At the domain level, glycaemic-severity information produces the largest forecast change, followed by demographic information. Medication metadata, site/cohort, metabolic laboratory measurements, cardiovascular variables, and anthropometric measurements have smaller effects. Domain-level values should be interpreted cautiously because the domains contain different numbers of features and different degrees of internal correlation.

The feature-level comparison provides a more precise interpretation of the aggregated static-clinical result. Study group, baseline HbA1c, and baseline serum glucose are both influential and predictively useful: their removal changes the forecast and increases held-out MAE. These variables therefore provide participant-level information that is not fully recoverable from the recent CGM trajectory alone.

Study group is the strongest individual static feature. This variable summarizes broad differences in glycaemic phenotype, including normoglycaemia, prediabetes, non-insulin-treated diabetes, and insulin-treated diabetes. However, its attribution should not be interpreted as an isolated clinical effect because it is correlated with HbA1c, glucose variability, medication status, and other participant characteristics.

HbA1c is the most influential continuous clinical measurement. Its positive predictive usefulness indicates that long-term glycaemic control helps the model adapt near-term forecasts across participants. Baseline serum glucose provides an additional, smaller participant-level glycaemic prior.

In contrast, several demographic variables substantially change the forecast but provide little or uncertain improvement in MAE. Marital status is a particularly clear example of high sensitivity with limited predictive usefulness. Race, ethnicity, sex, and site variables also require caution. Their influence may reflect population structure, socioeconomic or behavioural proxies, clinical-site differences, missingness patterns, or shortcut learning rather than robust physiological information.

The feature-level results can be summarized into four categories:

- **Strongly used and useful:** study group, HbA1c, and baseline serum glucose.
- **Strongly used but weakly useful:** marital status and selected demographic variables.

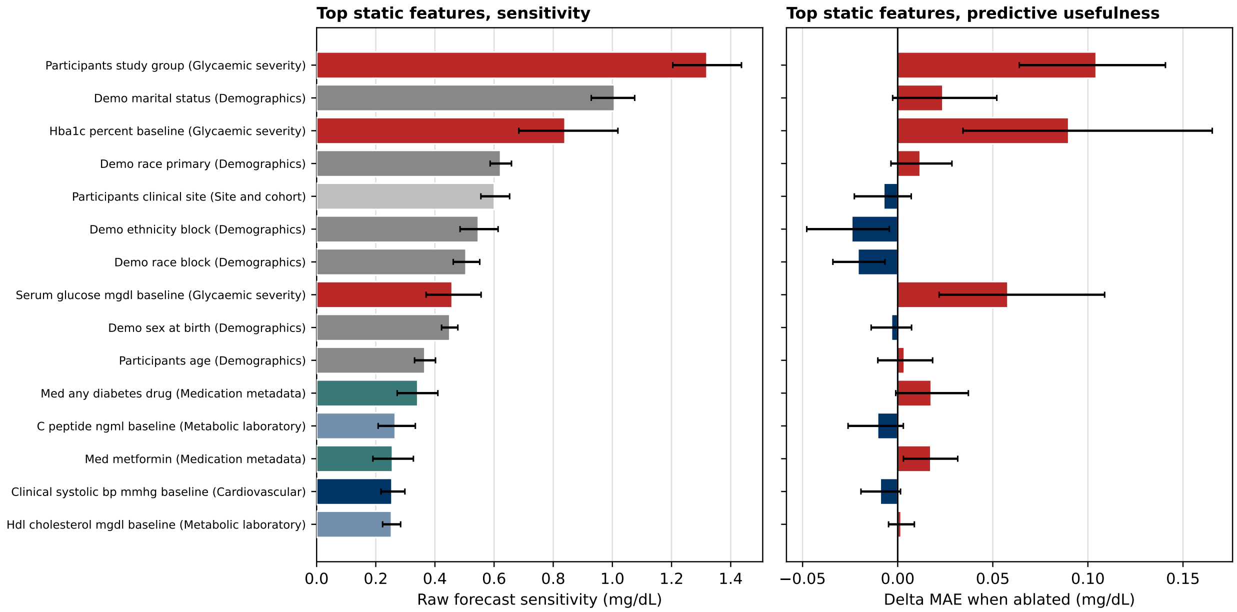


Figure 82: Static-feature reliance and predictive usefulness. Left: raw mean absolute forecast change after reference ablation of each feature or categorical feature block. Right: $\Delta\mathrm{MAE} = \mathrm{MAE}_{-g} - \mathrm{MAE}_{\mathrm{full}}$. Positive values on the right indicate that removing the feature worsens prediction and that the feature is useful in the frozen model. Features may therefore be strongly used without providing a corresponding accuracy benefit. Categorical variables represented by multiple one-hot columns are ablated jointly as conceptual blocks.

- **Neutral or potentially harmful under frozen-model ablation:** selected race/ethnicity blocks, clinical site, C-peptide, and other variables whose confidence intervals include or fall below $\Delta\mathrm{MAE} = 0$.
- **Secondary static variables:** anthropometric, cardiovascular, and lipid measurements with relatively small forecast effects.

These findings motivate retrained ablation experiments rather than immediate feature deletion. Protected demographic and site variables should additionally be evaluated using conditional permutation or matched ablation within study group, HbA1c range, and clinical site.

19.6.2 Static Conditioning Pathways

Output-level ablation establishes that static information changes the forecast, but it does not identify where that dependence enters the architecture. Static covariates influence the model through several pathways. The static representation initializes the participant-specific recurrent state,

$$h_0 = \psi(e_S), \tag{86}$$

and conditions the historical representation using feature-wise linear modulation,

$$u_t^{\mathrm{cond}} = \gamma(e_S) \odot u_t + \beta(e_S), \tag{87}$$

where $\gamma(e_S)$ multiplicatively rescales the dynamic representation and $\beta(e_S)$ applies an additive shift.

For each static domain, the pathway analysis replaces that domain by its training-derived reference representation and measures the resulting change in h_0, γ, β, and the final forecast. The first three quantities are representation-space L_2 norms, whereas output sensitivity is expressed in mg/dL. Their absolute magnitudes are therefore not directly comparable between panels.

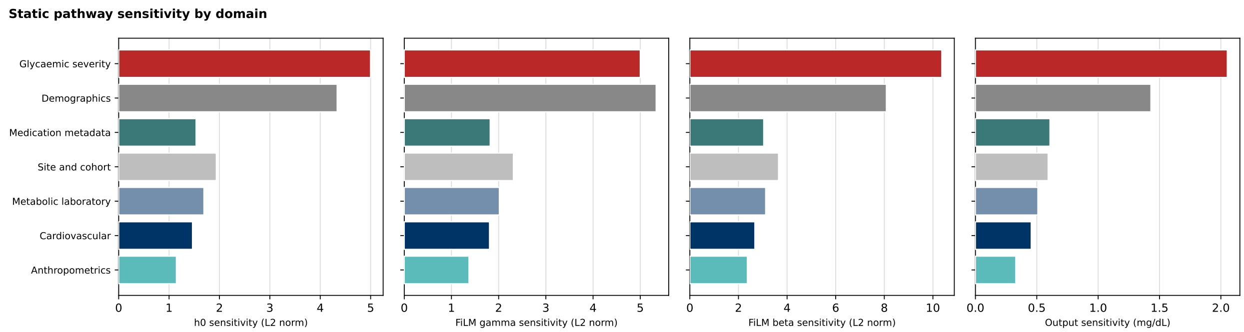


Figure 83: Static-domain sensitivity across model-conditioning pathways. Each panel measures the change produced by replacing one static domain with its training-derived reference representation. From left to right, the panels show sensitivity of the participant-specific initial state h_0, the FiLM multiplicative scale $\gamma(S)$, the FiLM additive shift $\beta(S)$, and the final forecast output. Glycaemic-severity and demographic variables have the largest influence across the static-conditioning pathways, indicating that they affect both participant initialization and the processing of subsequent dynamic inputs. The first three panels report representation-space L_2 norms, whereas the final panel reports mean absolute forecast change in mg/dL. Magnitudes should therefore be compared across domains within each panel, not directly between panels. These quantities measure model sensitivity and do not represent causal effects.

The pathway ranking is consistent with the output-level analysis. Glycaemic severity has the largest effect on the initial state, the additive FiLM shift, and the final output. This suggests that glycaemic phenotype establishes a participant-specific metabolic baseline that persists throughout the streaming forecast.

Demographic information also strongly affects all three static-conditioning pathways and is particularly influential in the FiLM scaling pathway. This means that demographic variables can change not only the initial participant representation but also how subsequent dynamic glucose and wearable signals are scaled by the model. This result is important for model auditing because it creates a possible mechanism through which protected attributes or correlated population structure can influence every later streaming update.

Medication, site/cohort, laboratory, cardiovascular, and anthropometric domains have smaller but non-zero pathway sensitivities. Their effects are more consistent with secondary participant-level adjustments than with dominant short-horizon forecast drivers.

19.6.3 Sensitivity Across Clinical Values and Forecast Horizons

For selected continuous static variables, the original value is replaced by training-distribution quantiles while all other inputs are held fixed. For feature j, quantile q, and horizon h, the magnitude of the resulting forecast change is

$$M_{j,q,h} = \frac{1}{N}\sum_{i=1}^{N}\left|\widehat{y}_{i,h} - \widehat{y}_{i,h}\left(S_{i,j} = Q_{j,q}\right)\right|. \tag{88}$$

This experiment evaluates model sensitivity across the observed training distribution. It is not a clinical intervention and should not be interpreted as the effect of changing HbA1c, BMI, age, or another participant characteristic.

The horizon dependence is consistent across the selected variables: participant-level static information has limited influence at the shortest horizons, where the latest CGM reading dominates, but becomes more influential towards 60 minutes as the decoder relies more heavily on participant-specific priors.

HbA1c produces the largest sensitivity across the tested horizons, reinforcing its role as the most important continuous static clinical variable. Baseline serum glucose also modifies the longer-horizon prediction, while BMI, age, and resting heart rate have smaller effects. The approximately U-shaped magnitude curves indicate that replacing a participant's value with an extreme population quantile changes the forecast more than replacing it with a typical value. Since absolute changes are plotted, this pattern does not establish whether low or high values raise the forecast.

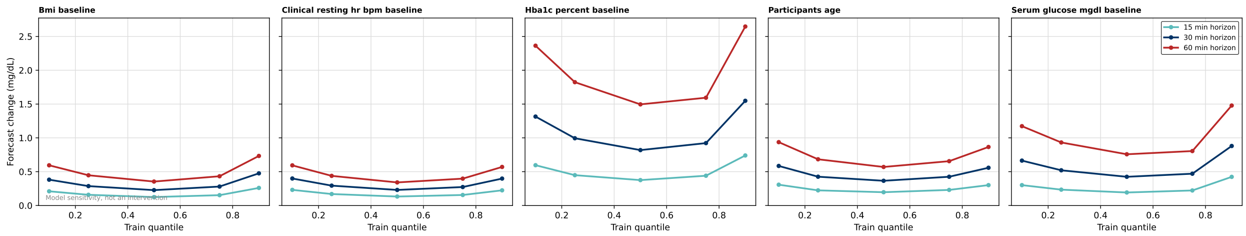


Figure 84: Static-feature sensitivity across training-distribution quantiles. Selected continuous variables are replaced by their training-split quantiles, and the resulting mean absolute forecast change is shown at 15-, 30-, and 60-minute horizons. Static influence generally increases with forecast horizon. HbA1c has the largest continuous sensitivity, followed by baseline serum glucose, whereas BMI, age, and resting heart rate produce smaller changes. Because the plotted quantity is absolute, the curves show magnitude but not whether a given value raises or lowers the forecast. These are model-sensitivity analyses, not interventions.

19.6.4 Interpretation and Clinical Cautions

The static analysis supports three principal conclusions.

First, static clinical information is not a single homogeneous input block. Most of its useful predictive contribution is concentrated in variables that describe glycaemic phenotype, particularly study group, HbA1c, and baseline serum glucose.

Second, glycaemic-severity variables influence several architectural pathways. They affect participant initialization, FiLM conditioning, and the final forecast. Their high output reliance is therefore not solely an output-layer artefact.

Third, the strong sensitivity to demographic and site variables requires additional auditing. High reliance does not guarantee robust predictive value, and a variable that improves performance on the current held-out split may still reduce fairness or transportability under a new population, site, or data-collection protocol.

The results therefore motivate three follow-up analyses:

1. retraining the model without weak or potentially harmful static features;
2. conditional demographic and site ablations within matched clinical strata;
3. subgroup calibration and error analysis across protected and clinical categories.

Overall, the model uses static clinical information primarily to construct a participant-specific glycaemic prior. This improves forecasting when the static variable captures stable metabolic phenotype, but it also creates a pathway through which demographic, site, and measurement-process structure can influence the prediction.

19.7 Interpreting Non-Glucose Reliance

The non-glucose reliance results are important because they separate useful context from potential shortcuts. Heart-rate/physiology, sleep, activity, stress, and data-quality channels all help construct h_t. That does not imply that each channel has a large independent causal effect on next-hour glucose. It means the model has learned to encode these channels as part of the current state from which the forecast is decoded.

The wearable subgroup figure shows that the activity/sleep block is not a single opaque signal. The model distributes state reliance across activity intensity, sleep-stage, and data-quality-related channels. This supports the interpretation that the recurrent state stores behavioral context, not just CGM history. At the same time, the data-quality detail figure shows why this must be read cautiously: missingness and quality indicators are useful to the model but may reflect device wear, site workflow, or participant adherence rather than physiology.

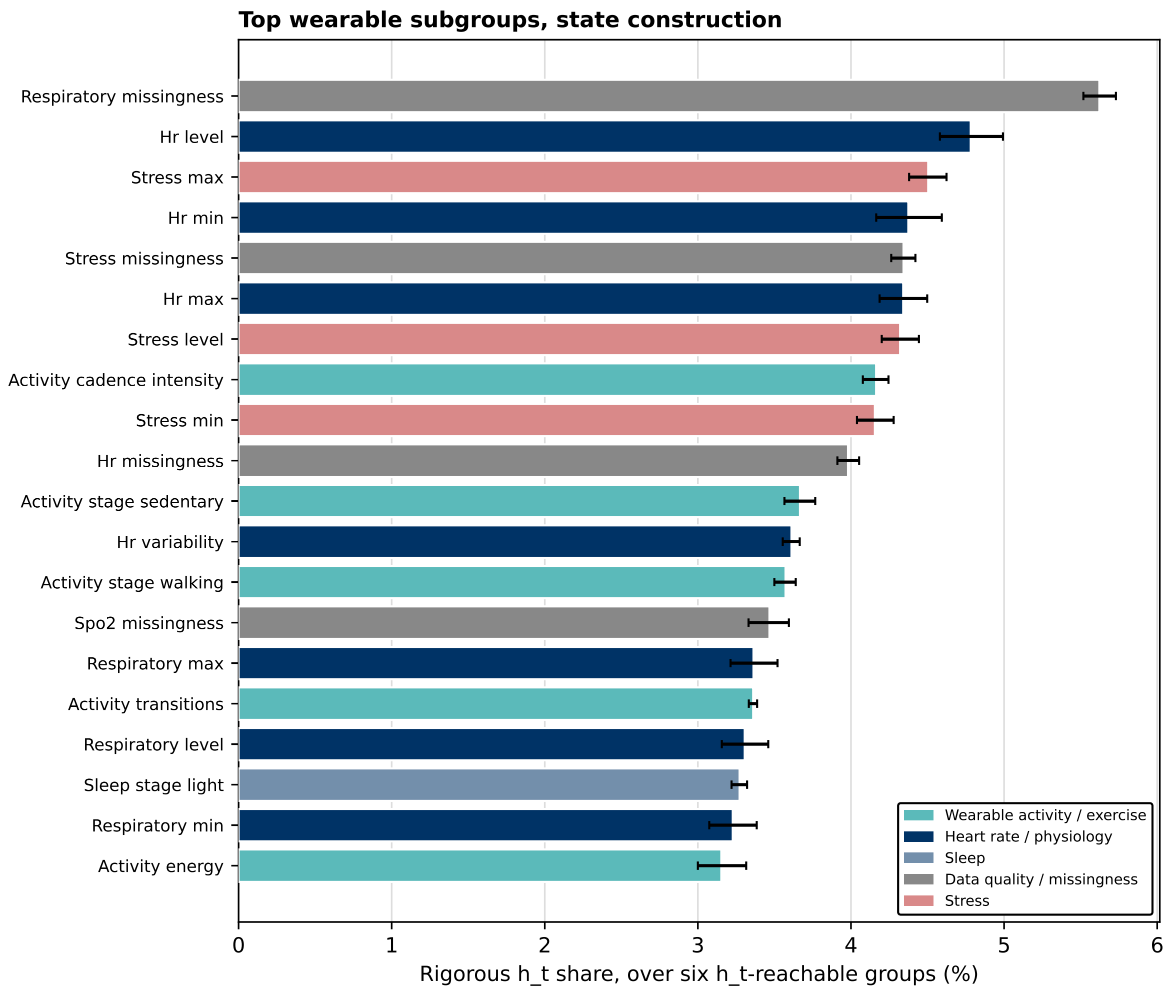


Figure 85: Wearable and data-quality subgroup contributions to h_t. The recurrent state uses several behavioral and quality-related channels rather than a single activity proxy. This is evidence of contextual state encoding, not by itself evidence of a causal activity effect.

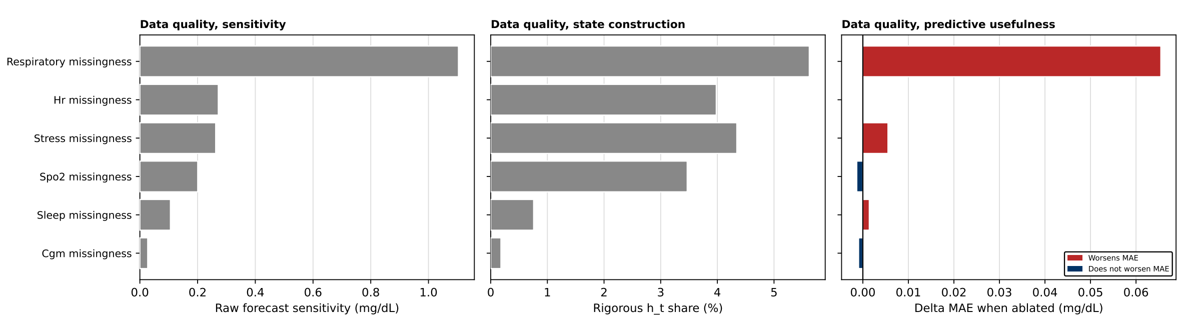


Figure 86: Data-quality and missingness detail. Data-quality channels contribute to both state construction and output behavior. This improves prediction but also flags a deployment concern: the model may partially exploit measurement process, device coverage, or site-specific missingness structure.

19.8 Hidden-State Sanity Checks

The probe figures ask whether h_t contains clinically meaningful state information after the streaming encoder has fused the input groups. The regression probes show that the hidden state carries strong information about current glucose and short-term glucose dynamics. The classification probes show separability for categorical state targets, but these should be interpreted as representation diagnostics rather than as standalone clinical classifiers. The PCA view provides a simple geometric check: the first principal component of h_t tracks current glucose strongly (correlation 0.821), indicating that current glycemic state is a dominant axis of the learned representation.

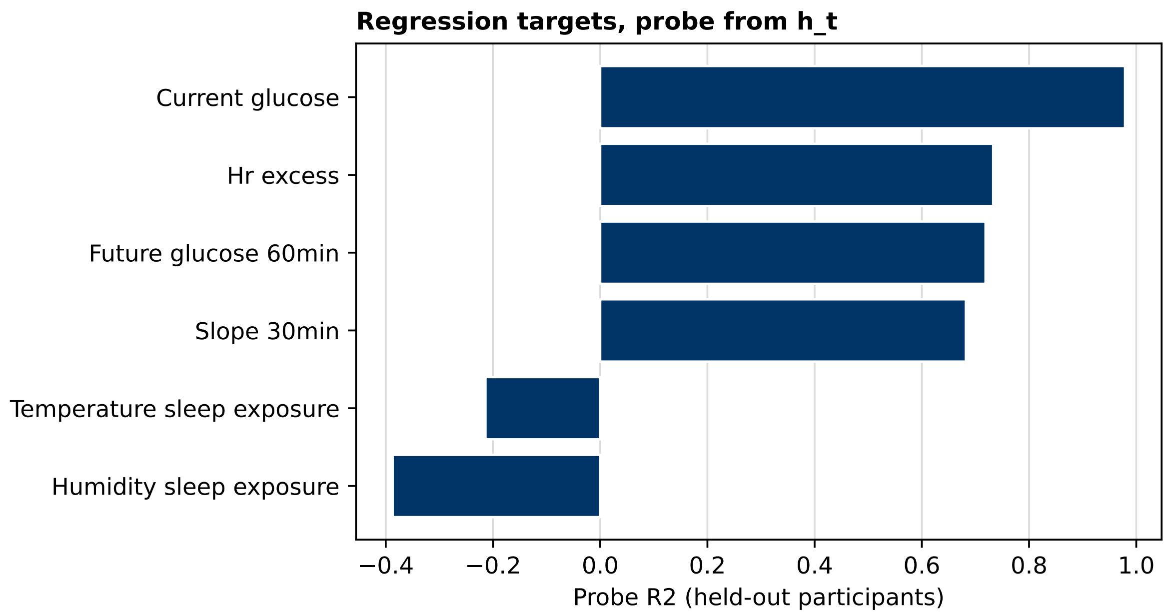


Figure 87: Hidden-state regression probes. Probe performance indicates that h_t contains recoverable information about current glucose and short-term dynamic state, consistent with the reliance results above.

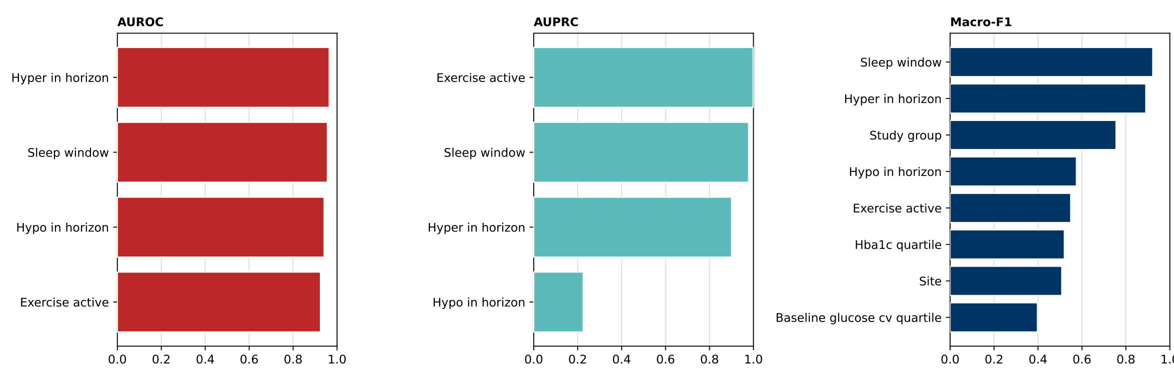


Figure 88: Hidden-state classification probes. Classification probes test whether categorical state information is encoded in h_t. These are representation checks, not deployment classifiers.

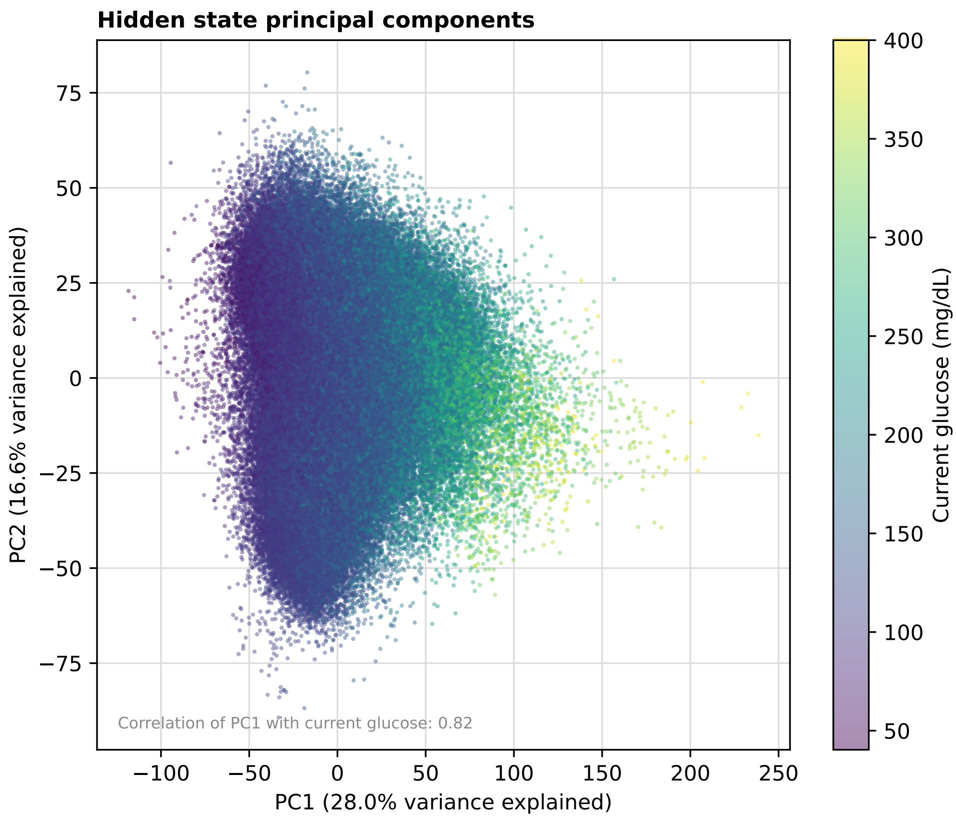


Figure 89: PCA projection of h_t colored by current glucose. The first principal component is strongly aligned with current glucose (correlation 0.821; PC1 variance 28.0%, PC2 variance 16.6%), confirming that glycemic state is a major organizing direction of the hidden state.

19.9 Local Case Studies: From Global Reliance to Individual Forecast Behaviour

Global attribution summarizes average model reliance across the held-out cohort, but it does not show how that reliance is expressed in individual forecasts. We therefore selected representative local cases to connect the forecast trajectory with two complementary interpretability views. The left panel evaluates factual forecast behaviour, the centre panel shows which historical groups contributed to constructing the streaming state h_t, and the right panel shows which input groups most strongly affected the final output under leave-one-modality-out ablation. These examples are descriptive analyses of the trained model and are not estimates of causal physiological effects.

The left panel in each case shows historical CGM, the forecast median, the 10th–90th percentile interval, the observed future trajectory, the forecast anchor, and any context annotation such as a sleep window or exercise-detector interval. The centre panel reports rigorous hidden-state construction attribution, normalized over the six directly decomposable historical groups: glucose history, heart rate/physiology, wearable activity/exercise, sleep, stress, and data quality/missingness. The right panel reports output-level LOMO reliance, normalized over all eleven input groups visible to the forecast output. These two percentage scales have different denominators and should not be compared as numerical ratios. Their rankings and qualitative patterns can be compared: hidden-state attribution shows what builds the representation, whereas LOMO shows what changes the output. Neither view proves causal importance or establishes that a modality improved the forecast in a specific case.

For each case, we ask three questions at the same forecast anchor: whether the future glucose trajectory was predicted correctly, which historical modalities contributed to constructing h_t, and which input groups most strongly changed the final forecast output. The selected anchors are fixed artifacts from the case-study generation pipeline rather than newly selected examples: participant 1302, segment 0, anchor 540 for the exercise case; participant 7347, segment 1, anchor 1305 for the stable sleep case; and participant 4162, segment 0, anchor 651 for the reversal case.

19.9.1 Exercise-Annotated Forecast-Only Case

Fig. 90 shows an exercise-annotated factual forecast. The Study 2 exercise detector was external to the base forecasting model. It was used to identify and annotate the walking episode, but the detector-active state was not a dedicated input to the displayed forecast. Historical steps, heart rate, and related wearable measurements were available to the model through the standard wearable and physiology channels. This case therefore evaluates the forecast-only pathway, not the exercise counterfactual branch. The hatched interval marks an independently detected 70-minute walking bout with moderate strain and is displayed for interpretation; it is not a forecast input under the checkpoint inspection summarized in `external_candidate_check_v2.csv`.

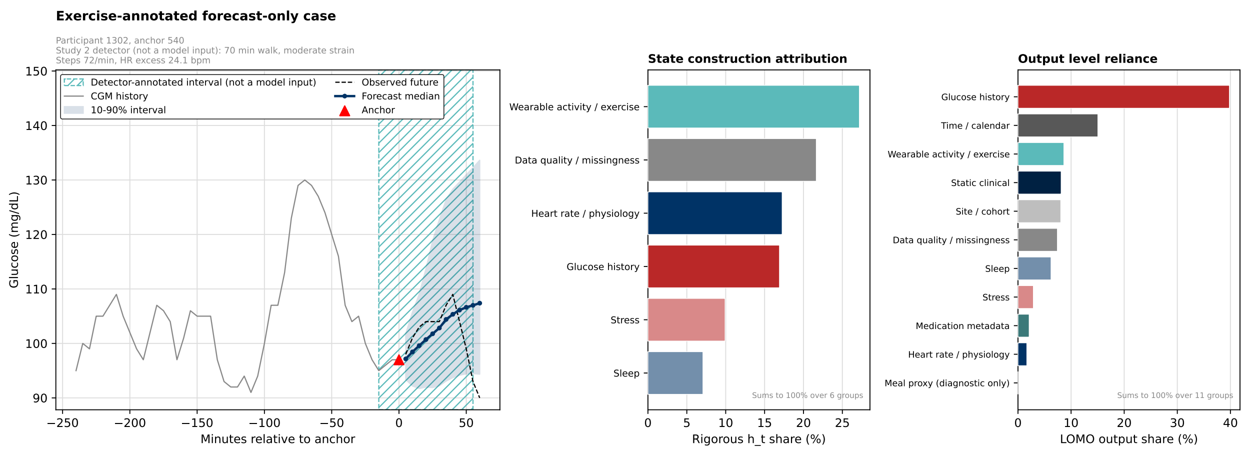


Figure 90: Exercise-annotated forecast-only case. The episode was independently identified by the Study 2 detector, but the detector state, cadence and event phase were not supplied as dedicated inputs to the displayed forecast. Left: historical CGM, forecast median, 10th to 90th percentile interval, observed future and detector-annotated exercise interval. Centre: rigorous hidden-state construction attribution normalized over the six directly decomposable historical groups. Right: output-level LOMO reliance normalized over all eleven input groups. Strong activity representation in h_t does not imply that an explicit exercise-response function was available to the forecast decoder.

The forecast in Fig. 90 captures the initial increase in glucose, and the observed future initially follows a similar direction. The later decline is missed: by the end of the 60-minute horizon the absolute error is 17.39 mg/dL, interval coverage is 83.3%, the first interval exit occurs at 55 minutes, and the maximum distance outside the interval is 4.35 mg/dL. This is therefore a late-horizon point-forecast error and an uncertainty-calibration error.

The state-construction view shows that wearable activity/exercise is the largest contributor to h_t in this case, accounting for 27.2% of the six-group hidden-state attribution. Data quality/missingness contributes 21.6%, heart rate/physiology 17.3%, and glucose history 16.9%. This shows that raw wearable activity information reached the streaming representation. It does not prove that the model constructed an explicit semantic state for exercise onset, duration, cadence, recovery phase, or time since exercise. In the output-level view, the final forecast remains primarily driven by glucose history (39.8%), followed by time/calendar (15.0%). Wearable activity/exercise contributes 8.6% of the eleven-group LOMO attribution, while heart rate/physiology contributes 1.7%. Thus, activity influences the output less strongly than it contributes to the hidden state, and heart rate has little direct output reliance in this example.

The episode was independently identified by the Study 2 detector as a walking bout. Activity-related historical inputs were the largest contributors to construction of the streaming state at the forecast anchor, showing that raw wearable information was strongly represented. However, the displayed trajectory was produced in forecast-only mode, without providing the detector state, exercise onset, cadence class, recovery phase or time since exercise as dedicated decoder inputs. The final forecast remained primarily driven by recent glucose history and calendar context. It captured the initial glucose increase but missed the later decline, which fell below the prediction interval. This case therefore does not show failure of an explicitly exercise-aware forecasting architecture. Instead, it shows the limitation of expecting a general forecaster to infer a delayed exercise-response function solely from historical wearable measurements.

Because the external detector uses steps and heart rate, elevated activity attribution in a detector-selected exercise episode is partly expected by construction. The informative result is the mismatch between strong activity representation and incomplete prediction of the subsequent glucose trajectory. This local case evaluates factual forecasting and representation. The separate exercise-counterfactual analysis evaluates how the decoder output changes when an activity scenario is explicitly asserted.

19.9.2 Stable Overnight Sleep Forecast Case

Fig. 91 is included as a successful comparison against the two failure cases. Glucose is stable at the anchor: the current glucose is 108 mg/dL and the 30-minute pre-anchor slope is 0.033 mg/dL/min. The model predicts a nearly flat trajectory, the observed future declines only slightly, and the observed trajectory remains inside the prediction interval. The 60-minute absolute error is 3.28 mg/dL, the horizon-average MAE is 1.95 mg/dL, interval coverage is 100.0%, and the 60-minute interval width is 8.51 mg/dL. This is a small-error, well-calibrated forecast.

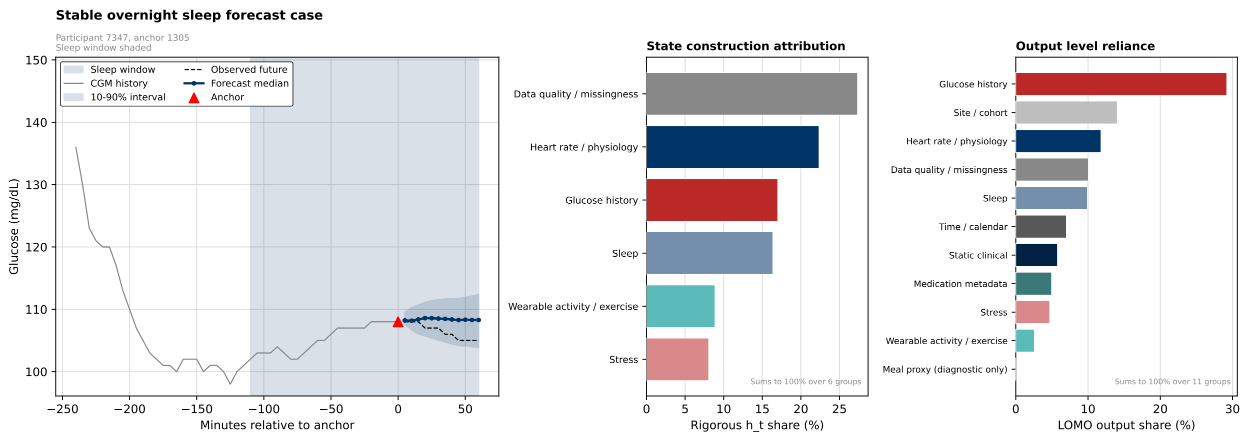


Figure 91: Stable overnight sleep forecast case. Left: sleep window, historical CGM, forecast median, prediction interval and observed future. Centre: hidden-state construction attribution over the six historical groups. Right: output-level LOMO reliance over all eleven groups. The balanced multimodal state is translated into a stable forecast whose observed future remains inside the prediction interval.

The hidden-state attribution in Fig. 91 is multimodal rather than CGM-only. Data quality/missingness is the largest state-construction group at 27.3%, followed by heart rate/physiology at 22.4%, glucose history at 17.0%, and sleep at 16.4%. The output-level LOMO view remains led by glucose history at 29.2%, but site/cohort (14.1%), heart rate/physiology (11.8%), data quality/missingness (10.1%), and sleep (9.9%) also influence the output. Compared with the exercise case, sleep and physiology are more visibly translated into the final forecast.

The stable overnight case illustrates a context in which multimodal state information is translated into an accurate and calibrated forecast. Data quality, heart-rate physiology, glucose history and sleep all contribute to construction of the hidden state, while the final output remains primarily glucose-driven. The observed future remains within the prediction interval and differs only slightly from the forecast median. In this relatively stable context, the model successfully combines recent glucose with sleep and physiological information.

The large contribution of data-quality variables should not be interpreted as a physiological mechanism. It may reflect legitimate reliability conditioning, but it may also encode sleep state, wearable non-wear, participant identity, site workflow or device-specific missingness.

19.9.3 Post-Decline Reversal Failure Case

The reversal example in Fig. 92 uses the `fast_down` case rendered as `fig_case_reversal_v2`. The case-quality audit confirms that the future CGM values are observed, no invalid CGM gap crosses the forecast horizon, no participant or segment boundary is crossed, timestamps are correctly aligned, no inverse-scaling error exists, values remain inside the accepted CGM sensor range, and the trajectory is not generated by a preprocessing artifact. The corresponding audit row has `all_checks_pass=True`.

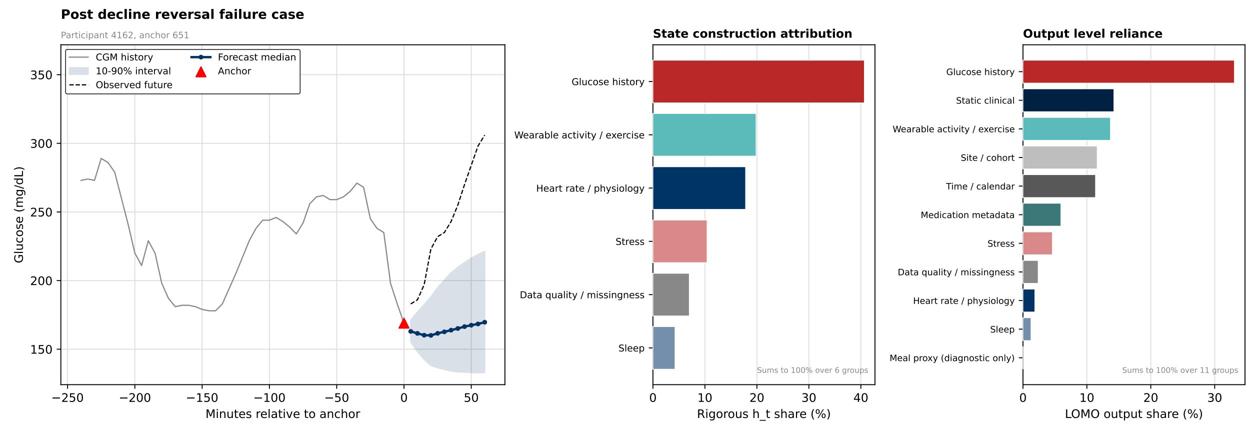


Figure 92: Post-decline reversal failure. The model extrapolates the recent downward glucose tendency, whereas observed glucose reverses sharply and exceeds the upper prediction interval. Glucose history dominates both state construction and output reliance. The example illustrates directional, magnitude and uncertainty failure under an abrupt trajectory change.

Glucose declines strongly before the forecast anchor, with a 30-minute pre-anchor slope of -3.300 mg/dL/min and current glucose of 169 mg/dL. The model predicts a small further decline followed by a relatively stable level, whereas the observed future reverses sharply and rises above 300 mg/dL. The observed trajectory exceeds the upper prediction interval rapidly: interval coverage is 0.0%, the first interval exit occurs at 5 minutes, the maximum distance outside the interval is 84.76 mg/dL, and the 60-minute absolute error is 136.33 mg/dL.

This failure has three components. It is a directional failure because the forecast continues the recent decline rather than predicting the reversal. It is a magnitude failure because the later increase is strongly underestimated. It is an uncertainty failure because the observed trajectory lies outside the prediction interval throughout the horizon.

The centre panel shows that glucose history is the largest contributor to h_t, accounting for 40.7% of the six-group hidden-state attribution. Activity is also represented at 19.8%, and heart rate/physiology at 17.8%. The recent declining glucose trajectory is therefore the dominant state-construction signal. In the right panel, glucose history also dominates final-output reliance at 33.1%. Static clinical information contributes 14.2%, wearable activity/exercise 13.7%, site/cohort 11.6%, and time/calendar 11.4%. Heart rate/physiology is represented in h_t but has little direct output influence in this case, contributing 1.9% of the eleven-group LOMO attribution.

The reversal case exposes a major limitation of the forecast-only model. The sharply declining pre-anchor glucose trajectory dominates both hidden state construction and final-output reliance, and the decoder continues the recent downward tendency. The observed future instead reverses rapidly and rises well above the upper prediction interval. Activity and heart-rate information are present in the state, but they are not translated into a prediction of the reversal. This example therefore represents a directional, magnitude and uncertainty-calibration failure.

The reversal may depend on information that cannot be identified from the recent observed trajectory, such as an unrecorded meal, treatment response or another physiological disturbance. No specific cause should be assigned without a corresponding logged event.

19.9.4 What the Local Cases Reveal

The three cases separate representation from response modelling. In the exercise episode, wearable activity strongly contributes to the hidden state, but the decoder does not fully predict the later glucose decline. In the stable overnight case, multimodal state information is translated into an accurate and calibrated forecast. In the post-decline reversal, recent glucose history dominates both the state and output, and the decoder continues the recent trend while missing an abrupt change in direction. These cases suggest that the encoder captures meaningful multimodal context, while the main remaining limitation is the decoder's ability to convert that context into event-specific future glucose trajectories.

The exercise case motivates explicit event-aware modelling rather than relying only on raw wearable history. A future extension could condition a residual head or scenario decoder on exercise-active state, time since onset, time since end, duration, cadence, strain and participant phenotype. The current case study motivates this extension but does not evaluate it.

The local examples are selected illustrations of model behaviour. They support interpretation of the cohort-level analyses but do not establish the frequency of each failure mode. Population-level conclusions must rely on systematic stratified evaluation across all eligible anchors.

19.10 Interpretation and Limitations

The two-view analysis supports four conclusions. First, the forecast is primarily driven by glucose history, as it should be for short-horizon CGM prediction. Second, the hidden state is more distributed than the output attribution: it stores physiology, activity/sleep context, stress, and data-quality structure in addition to glucose history. Third, Data-quality and site/cohort variables improve prediction on the current held-out cohort, but their contribution may reflect sensor availability, collection procedures, or population differences rather than direct physiological information. Their robustness should therefore be evaluated under unseen sites, devices, and missingness patterns before deployment. Fourth, the meal-proxy diagnostic channel has no measurable output-level influence in this run, so the AI-READI model should not be described as having learned an interpretable meal counterfactual.

The main limitation is that these are reliance measures, not causal effect estimates. LOMO ablation asks how the trained model's output changes under a replacement operation; it does not identify what would happen biologically if a modality changed in the real world. The h_t joint-group norm asks how the encoder constructs its internal state; it does not say which input would independently change the forecast under an intervention. Together, the two views give a defensible account of model reliance: what moves the output, what builds the hidden state, and where predictive performance may be borrowing from non-physiological structure.

20 Ablation and Runtime

The stateful Mamba run is feasible on a single GPU. The best validation checkpoint occurs at epoch 5; later epochs improve training loss but do not improve validation pinball, so the best checkpoint rather than the final epoch is used for evaluation. The ablation launcher prepares short probes, but full ablations remain a follow-up experiment.

Table 55: Runtime, memory, and training-throughput summary from the selected AI-READI stream run.

Quantity	Value
Best validation epoch	5
Best validation pinball	3.286 mg/dL
Max training throughput	4868 anchors/s
Peak training GPU memory	897 MB
Evaluation GPU memory	307 MB
Evaluation CPU RSS	8699 MB
Median update latency	3.76 ms
Mean 1h forecast latency	0.0037 ms

Table 56: Ablation status. Some ablations are configured as probes but have not yet been fully run.

Ablation	Status	Interpretation
Full residual-current model	configured	available as reference
Absolute-target / no residual-current	prepared	not run by current pipeline
No static h0/FiLM	probe supported	run with launch_ablation_probes.sh
No scenario masks	prepared	current model lacks safe switch
No scenario decomposition	probe supported	run with launch_ablation_probes.sh

[TODO: No runtime backend figure was found; runtime is summarized in Table 55.]

The current ablation table is a status table, not a result table. The full model, no-static-FiLM, and no-decomposition probes can be launched with the provided script. Absolute-target and no-scenario-mask variants require additional implementation before their numbers can be interpreted.

21 Discussion and Limitations

21.1 Summary of Findings Across Three Studies

The AI-READI port produces a viable participant-held-out streaming forecaster and improves over persistence on matched anchors. The improvement over persistence is real but not large, which is the expected limitation of residual-current CGM forecasting: current glucose already carries most of the short-horizon predictive signal.

Study 1 (Section 11) establishes that AI-READI cannot support a meal counterfactual with any constructible label. The glucose-defined arm is sign-inverted ($\Delta\theta_{60} \approx -35$ mg/dL versus observed $+22.2$ mg/dL) because the label selects precisely the anchors where the forecast-only path already anticipates a rise. The wearable arm is a near-zero null ($\Delta\theta_{60} \approx -0.21$ mg/dL), and a dose-matched sweep to model-space 6.25 yields -0.77 mg/dL. The three convergent findings are not independent causal proofs, but they rule out the main alternative explanations (dead channels, dose magnitude, encoding artefacts). The required next step is exogenous meal annotation from a dataset with prospective logging (T1DEXI or CGMacros).

Study 2 (Section 12) takes a different approach: rather than attempting a meal counterfactual, it asks whether deployable model-surprise signals can correct forecasts *after* the base model's error becomes observable. The rich correction head (Ridge full, split up/down; 19 causal trailing features) beats the frozen base by 0.175 mg/dL on test triggered windows (95% CI $[-0.259, -0.094]$, CI excludes zero) and beats the best non-learning baseline (decaying shift) by 0.172 mg/dL (CI $[-0.238, -0.109]$). The practical contribution is an interpretable, online correction mechanism: it fires on approximately 9.3% of deployment anchors (triggered rate at $\tau = 0.2$), requires no meal labels or future covariates, and can be retrained without touching the frozen base. The improvement is modest in absolute terms. Direction-specific heads (separate up/down) outperform a combined head; Ridge and HGBT are statistically tied; the richer causal feature set beats a simple z_t-only correction but the margin is interpretable, not surprising.

Study 3 (future work) asks whether the scenario-aware channel supports a causal meal counterfactual on a dataset with prospective meal logging such as T1DEXI or CGMacros. AI-READI does not serve this role: no constructible label carries pre-rise causal information orthogonal to the CGM slope.

21.2 Interpretability

Section 19 reports two complementary views of frozen-base model reliance on the full held-out test split. At the output level, recent glucose history is the dominant driver (36.7% of LOMO-attributed influence), with secondary contributions from static clinical metadata, site/cohort, time/calendar, data-quality indicators, and heart-rate/physiology. Data quality and site/cohort together account for 17.4% of output-level influence, which is predictive but also cautionary: part of the model's signal comes from collection structure rather than purely physiological variation. The meal-proxy diagnostic channel registers 0.0% output-level influence, consistent with Study 1 and arguing against any AI-READI meal-counterfactual interpretation.

The hidden-state view gives a different but compatible picture. Among the six groups that build the recurrent Mamba state h_t, glucose history, heart-rate/physiology, and data-quality/missingness carry comparable state-construction weight, while sleep, wearable activity/exercise, and stress also contribute. Thus the model's internal state is richer than a glucose-only trace, even though the final forecast remains most sensitive to glucose history.

21.3 Limitations

Dataset. AI-READI lacks timed insulin doses, insulin-on-board records, and carbohydrate quantity logs. The variable `med_insulin` is static therapy metadata, not an action covariate. The predicted meal flag is noisy and potentially retrospective. No confirmed meal timestamps are available; the event-enriched proxy used in Study 2 is a glucose-level proxy and is not distribution-matched to its control group.

Forecasting target. The residual-current target can hide weak dynamic learning if persistence is not separately reported. The model does beat persistence, but the margin is modest at every horizon (Section 6).

Personalisation. Matched-anchor personalisation is flat in the current artifacts; offset correction worsens MAE. Participant-level averages are necessary because anchor-weighted metrics overrepresent participants with longer clean segments.

Study 2 scope. The correction head is evaluated on triggered windows only ($\approx 9.3\%$ of deployment anchors). The event-vs-control comparison in Section 12.6 is descriptive and structurally confounded: the event definition selects on glucose level, so the two groups differ by construction. The correction head is slightly harmful for extreme downward residuals ($r_t < -20$ mg/dL, 77 test anchors) and for narrow prediction-interval windows (Q1); these cases should be handled with explicit abstention in deployment.

Interpretability scope. The output-level modality attribution is based on 153,591 dense anchors across 381 test streams (221 participants), the full held-out test split. The h_t joint-group norm applies only to the six modality groups that enter the streaming encoder state; static/context and direct-decoder inputs are interpreted through output-level LOMO instead. Both analyses are model-reliance audits, not causal effect estimates.

Site and phenotype effects. Site and phenotype subgroup effects require follow-up audits before being interpreted as biological effects rather than cohort composition or domain shift. The insulin-dependent subgroup in Study 2 failure-mode analysis comprises only 8 participants (1,498 anchors); results in that cell should not be generalised.

21.4 What a True Meal Counterfactual Requires

A scientifically valid "what if I eat now?" counterfactual requires exogenous meal information not derivable from the CGM signal: prospective meal logs, carbohydrate quantities, timed insulin boluses, or a standardised meal-challenge dataset. Study 1 demonstrates that approximating this signal from glucose-shape or wearable-activity features fails sign agreement with the observed natural experiment. Study 2 provides a deployable alternative that avoids this requirement entirely: it corrects forecasts using only causal residual and shape features available at trigger time, without claiming to model the meal causal mechanism. Future work connecting Study 2 event-aware triggers to prospective meal logs (e.g., from CGMacros) would enable a direct comparison of the two approaches on the same participants.

A Compute Placement: GPU vs. CPU

CUDA accelerates training through the Mamba/Triton scan backend. The streaming recurrence and decoder are still compatible with CPU execution for portability, but the reported full-cohort training runs used one NVIDIA A100 GPU.

B Generated Appendix Tables and Figures

Table 57: Full training history for the 10-epoch resumed stateful Mamba run.

Epoch	Train	Val	Anchors/s	Seconds
0	3.563	3.423	3089	241.8
1	3.394	3.330	4714	158.4
2	3.339	3.298	4709	158.6
3	3.304	3.325	4693	159.1
4	3.280	3.316	4720	158.2
5	3.263	3.286	3241	230.4
6	3.243	3.321	4829	154.7
7	3.225	3.306	4831	154.6
8	3.213	3.300	4868	153.4
9	3.197	3.319	4843	154.2

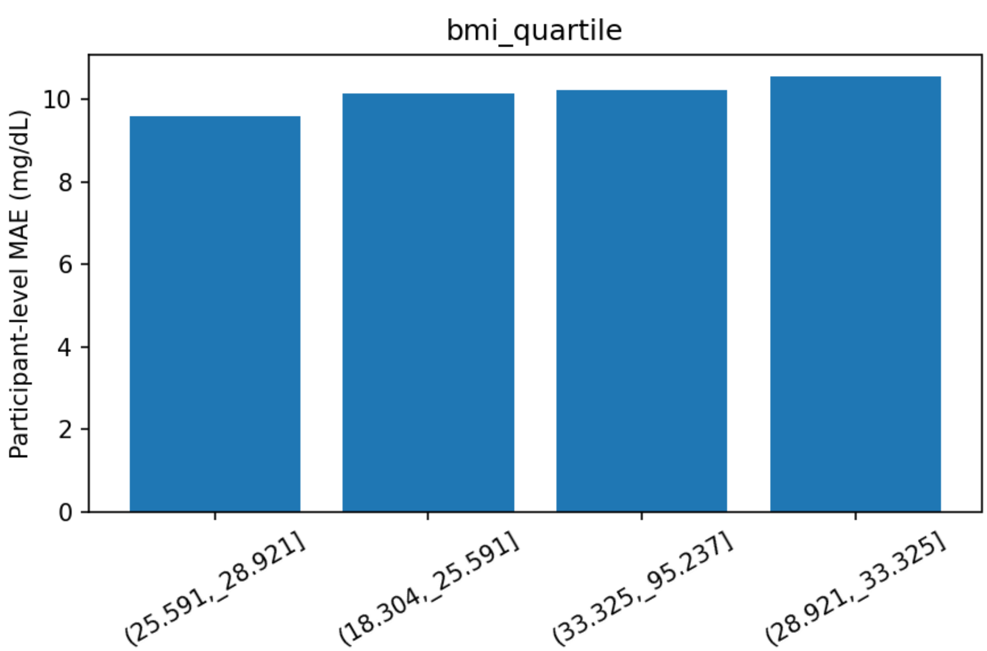


Figure 93: Appendix subgroup plot: participant-level MAE by BMI quartile.

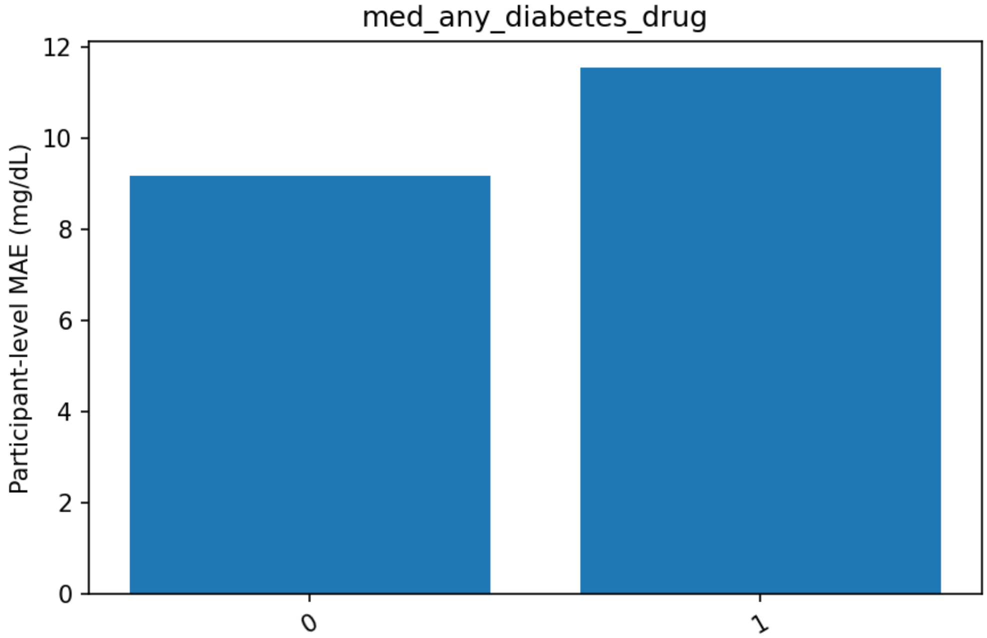


Figure 94: Appendix subgroup plot: participant-level MAE by any diabetes-drug metadata flag.

C Interpretability: Supplementary Figures

These figures support Section 19 but were moved out of the main text to keep the chapter to one figure per scientific question. All source CSVs and captions match the corresponding main-text discussion.

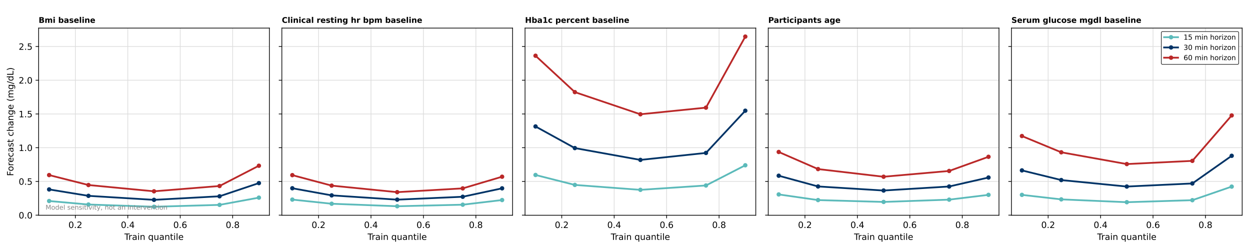


Figure 95: Static sensitivity curves: HbA1c, BMI, age, baseline serum glucose, resting heart rate. Model-sensitivity magnitude at observed train-split quantiles Q10/Q25/Q50/Q75/Q90, by forecast horizon. Magnitude only; no directional claim. Source: `static_sensitivity_curves_v2.csv`.

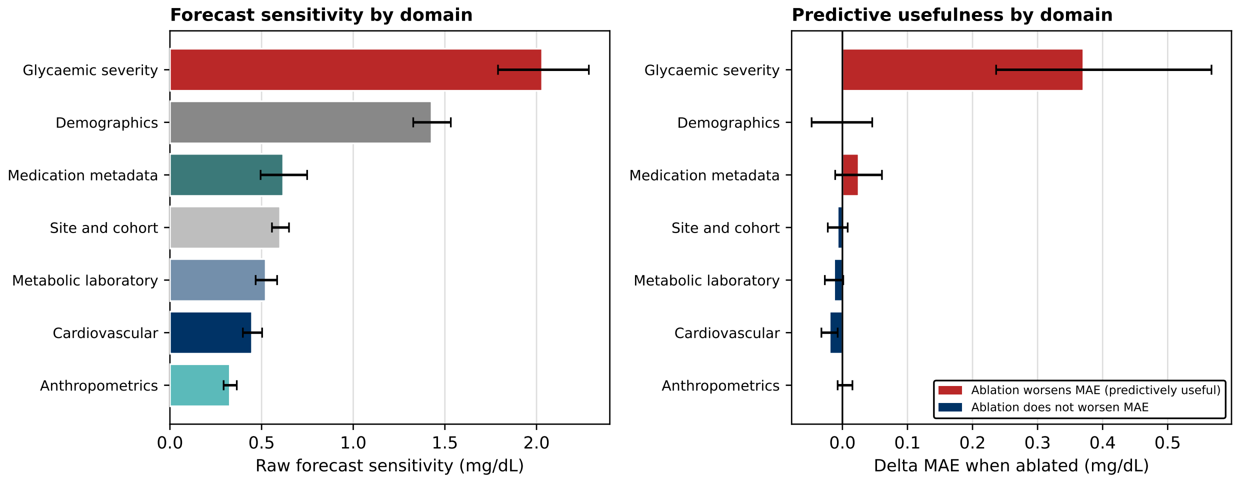


Figure 96: Static domain LOMO ablation, full test split. Source: `static_domain_lomo_v2.csv`.

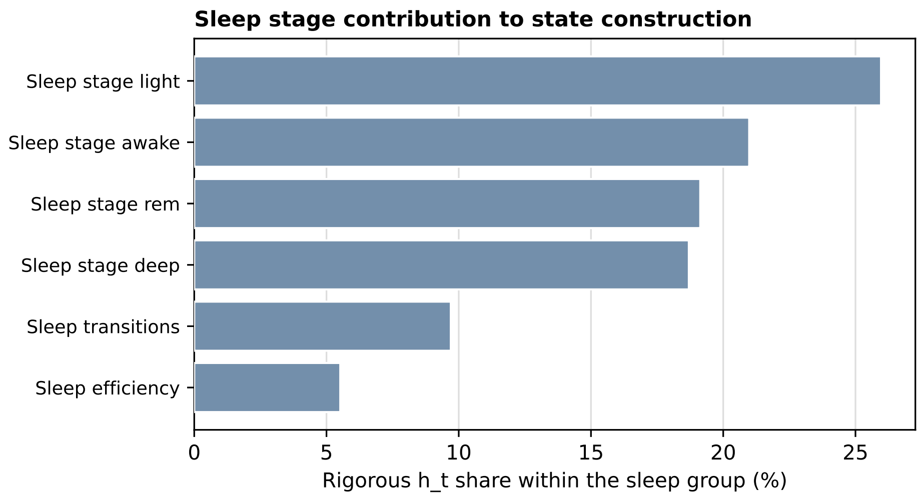


Figure 97: Sleep-stage contribution to state construction, within the sleep group. Source: `wearable_subgroup_ht_v2.csv`.

D Generated Results Snapshot

Scalar result macros are loaded from `sections/generated_results_summary.tex`. The file is retained with the report source for provenance, while the main appendix keeps the generated tables focused on model results rather than macro bookkeeping.

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